

MAGNETOM

Replacement of Parts System

Host / Imager

Symphony, A Tim System

Document Version

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1.1 General

1.1.1 Safety information



Adhere to ESD regulations and use the appropriate tool set: anti-static wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. ground pin of the mains plug) and the protective earth wire connected to the "main component" (e.g. screw of the housing).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

1.1.2 Handling of patient data

The data protection guidelines specify the mandatory procedures for handling patient data.

Data carriers, storage devices, or other media (such as the hard disk inside the host PC) must not leave the customer site if there is a chance they still contain patient data.



The service report needs to list the activities performed to ensure the secure handling of patient data.

As a first step, the PC needs to be analyzed. Once a defect in any part of the PC hardware is identified as the cause of the problem, the part replacement follows.

For PCs there are two cases:

1. The PC does not contain any patient data. - The individual product documentation needs to list hard disks without patient data. (→ No patient data on hard disks / Page 6)

2. The PC contains patient data.

The disks need to be returned without patient data.

Please decide how to proceed for your individual case. The recommendation:

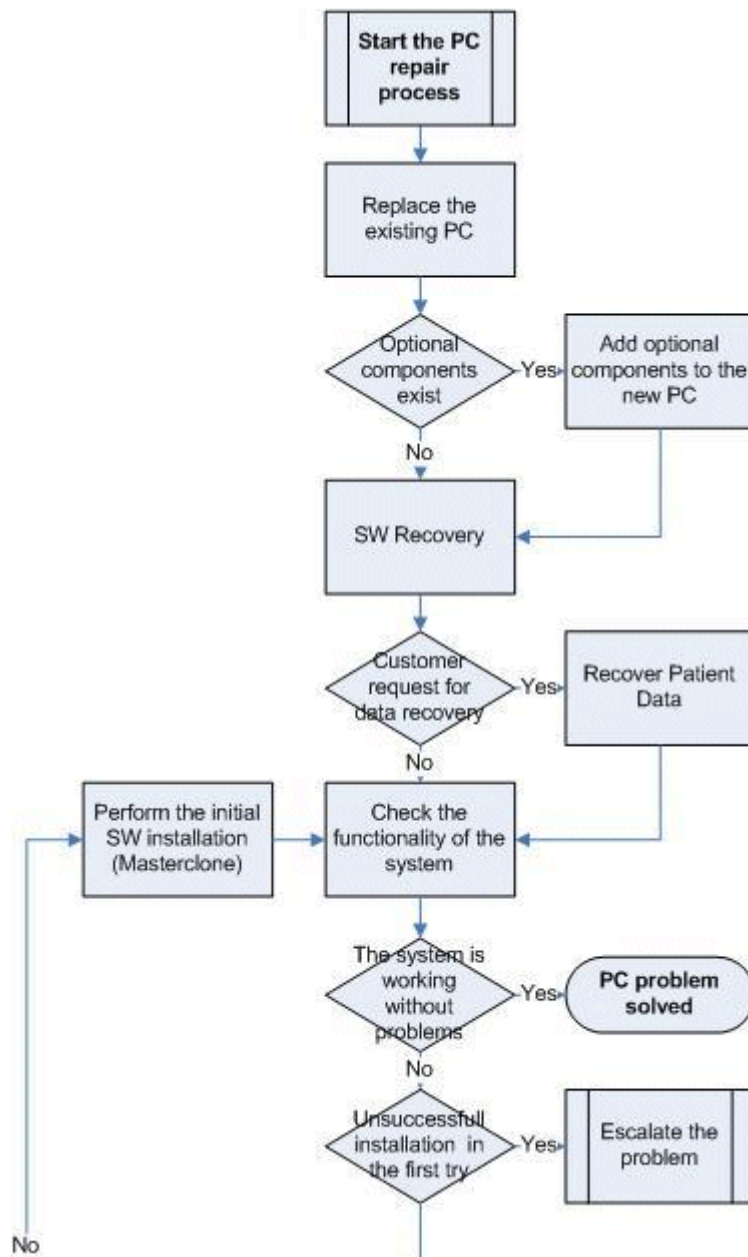
1. Case: Valid if the new PC has the same material number as the defective PC being replaced. The existing disk has to be removed from the host computer that is going to be returned. This disk must be installed in the replacement host computer. At the same time, the new disk that was just removed has to be installed in the defective host PC that is being returned. (→ Swap the disks / Page 7)
2. Delete the patient data with the "blancco" tool. (→ Deleting Patient Data / Page 10)
3. Case: All hard disks potentially containing patient data are removed from the defective PC and remain at the customer site.

1.1.2.1 No patient data on hard disks



Only continue with the parts replacement procedure outlined in the flow chart below if you have confirmation from the product-specific documentation that the PC contains no patient data, and that no patient information is contained in any logfiles on the PC.

Fig. 1: Replacing a PC without patient data



1.1.2.2 Swap the disks

The preferred case is replacing the hard disk: This case entails additional activities and this will surely take some time. - This time span may be compensated for by the time required for the restore to be performed. CSML confirms that such an action as opening the housing of the host computer and performing such activities will not negatively influence any warranty rights or claims.

Such a course of action, however, may provide the following benefits:

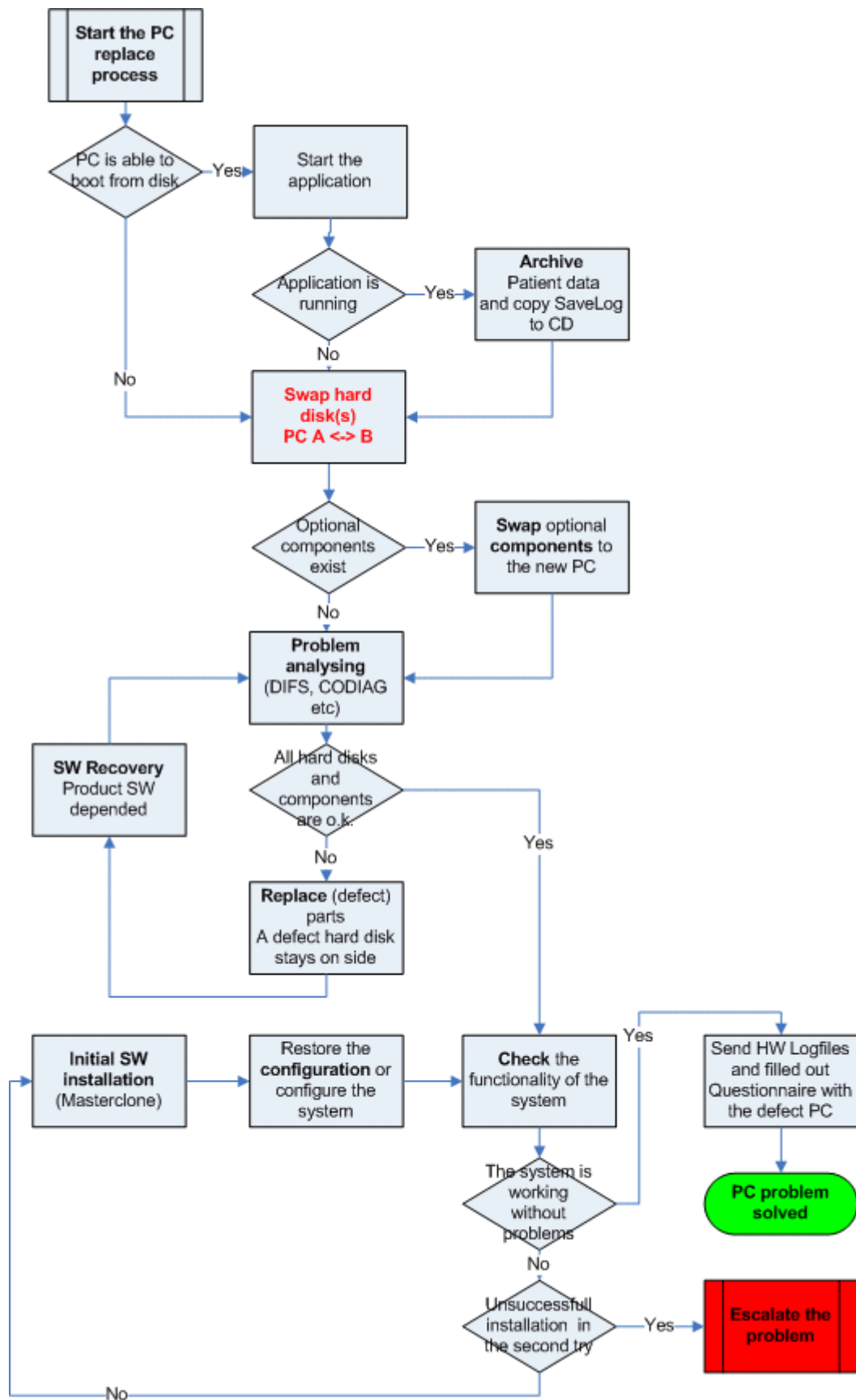
- The returned host computer will not contain any patient data.

- The disk swapped to the new PC will run without need of the rebuild/restore function.
- The customer may even have the benefit of reduced downtime.
- No reconstruction or loading of images from the archive will be necessary (all patient data will be available).

The decision in such cases will be made by the attending CSE or the responsible service manager.

The work steps are shown in the following flowcharts:

Fig. 2: Replacing a PC with patient data



1.1.2.3 Deleting Patient Data

Since the end of 2013, there is a tool to delete data from hard disk. The tool needs to be used in cases where replacement of the disk from the defective to the new PC is not possible.

The principles of "Return Goods Management" require that all storage devices containing personal patient data, which are to be removed from the site (i.e. spare parts returns), shall have all data wiped clean (securely deleted) or the storage device shall be handed over to the customer when deletion is not possible.

The tool is called "blancco". The tool can be downloaded from the Siemens Intranet.

The certified blancco tool permanently removes data from IT assets such as PCs, servers and data center equipment.

Data erased using blancco cannot be recovered with any existing technology. This helps to have all data wiped clean (securely deleted) and achieve compliance with existing "patient data handling" regulations.

SIEMENS Employees: The tool is available as an iso image from the CS Home page

<https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>

CS Home page -> Tools -> blancco - Erasure Software

For others see e.g.: <http://blancco.com/en/products/>

The time required by the procedure depends on the size of the hard disk.

Warning: before you start using the blancco software, make sure that all files, folders, software applications or any other information that you want to save for later use are backed up on an appropriate media device other than the original hard drive. Data that has been erased from the hard disk by this program cannot be recovered by any existing method.



Selecting the wrong hard disk can result in loss of data with the use of blancco. Be sure that the selected hard disk for deletion is the correct one.

Creating the blancco CD

An ISO image of the blancco CD is available for download from: CS Home page -> Tools -> blancco (CS Home page, <https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>)

1. Download the ISO image.
2. Label the blank CD as "blancco - Erasure Software"
3. Insert the CD into the drive of your service computer.
4. Use a standard burn program (such as Nero) to open the file and burn the CD image to the CD.
5. Remove the CD from the drive.

Using the blancco CD

1. Booting with blancco

1. Insert the "blancco - Erasure Software" CD into the PC with the drive(s) to be deleted

2. Switch off the PC.



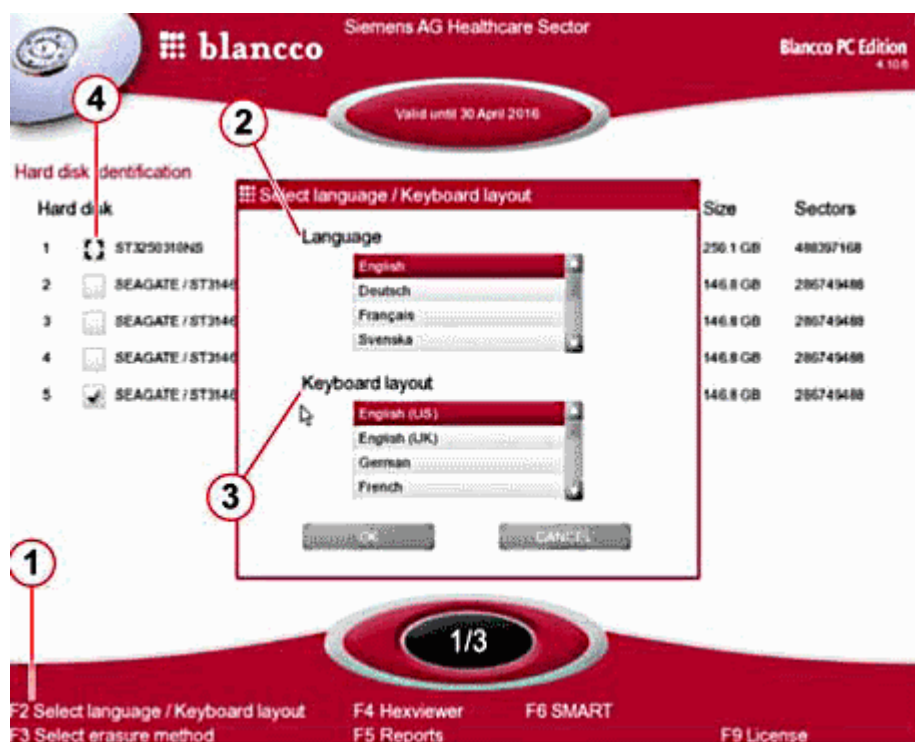
If this is an IAS-A PC, disconnect the service backup drive to prevent accidental loss of data.

3. Switch on the computer power to boot the system from the CD (if necessary, change the BIOS boot priority). The blanco Home screen is visible with all identified hard disks of the PC



If necessary, press F2 to change the language and the keyboard layout.

Fig. 3: Blanco language/keyboard layout



- [1] F2 to open menu
- [2] Language
- [3] Keyboard layout
- [4] Checkboxes to select hard disks

Deletion with blanco

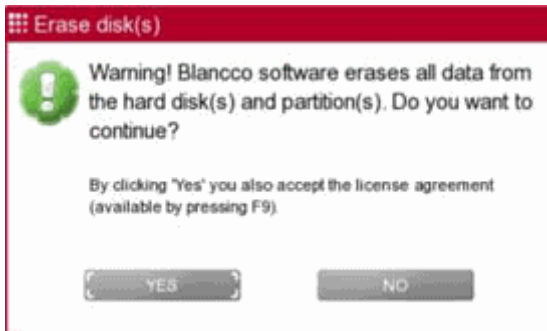
1. Select the hard disks to be deleted.



blanco can erase up to 16 hard disks simultaneously.

2. Press Erase, a warning message appears.

Fig. 4: Warning message



3. Press Yes to start the erasure process. The erasure process starts and the progress and duration is visible.
4. Click on OK when the process is finished and the Erase successful message appears.
5. Insert a memory stick to save the report.
6. 6. Press Save or F10. The report is saved to the memory stick.
7. Press Ok. The process is finished and the PC can be turned off.

1.2 FRU overview

FRU	Time	Instructions	Final checks	Tools
DVD-RW	45 min	(→ DVD-RW drive / Page 14)	Archive a patient images on CD-R / DVD media with the new DVD-RW drive	Field service grounding kit (ESD), set of standard tools
Host	180 min	(→ Host / Page 16)	Perform a measurement, view the images produced. Check with the customer that all options are available	Field service grounding kit (ESD), set of standard tools
Dongle (USB)	60 min	(→ Dongle / Page 17)	Check with the customer that all options are available Perform a measurement and view the images produced.	Service license/Application licenses. The Service/Applications licenses generated in relation to the flexid of the new dongle
Fans	45 min	(→ Fans, ventilation duct, crosspiece / Page 19)	Ensure that the new fan(s) works.	Set of standard tools
Power supply	45 min	(→ Power supply (R640 / R650) / Page 30)	Perform a measurement and view the images produced.	Set of standard tools
Battery	60 min	(→ Lithium Battery / Page 33)	Perform BIOS setup according to TDOC Installation Software. Perform a measurement and view the images produced.	Field service grounding kit (ESD), set of standard tools
In-room MRC	120 min	(→ In-Room MRC / Page 35) Safety-related tests	Run the auto configuration program and perform a visual evaluation according to the start-up of the In-Room MRC Measure protective conductor resistance for the "In-room MRC"	Set of standard tools and TDOC start-up of the In-room MRC

1.3 DVD-RW drive

1.3.1 Required time and tools

Time

- 45 min

Tools

- Set of standard tools

1.3.2 Preliminary

- Shutdown and switch off the system.

1.3.3 Disassembly

Disconnect the main power plug, network drop cable(s), mouse, and keyboard cables. Remove the right side cover (as seen from back), remove the 2 screws for the side cover.

Slide the side cover 15 mm back, and remove the cover laterally.

 Adhere to ESD regulations and use the appropriate tool set: anti-static wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

Fig. 5: Host "Celsius R610", front cover



- Remove the plastic front cover to access the CD-RW drive. Gently bend the plastic front cover clips, and pull the cover forward

Fig. 6: Host, disconnect IDE and power cable



- Disconnect the IDE and power cables in back of the CD-RW drive

Press the metallic clips located along the sides at the front of the DVD-RW drive. Gently pull out the DVD-RW drive.

Only valid for IDE devices; the new DVD-RW drive must be jumpered as "SLAVE" for the IDE bus. Refer to the label on the DVD-RW drive. (Since R650 S-ATA DVD-RW drives are used)

1.3.4 Assembling

- Reassemble the "host" with the new DVD-RW drive

1.3.5 Final checks

- Switch on the system, start the MR application
- Press **Ctrl - Esc**; select **Start, Advanced User**; enter the advanced user password; click **OK**
- Check that drive letters for the CD writer and DVD-drive are correct with the "windows file explorer" (CD/DVD-burner drive letter "**S**" and DVD/CD-ROM drive letter "**R**"). (In case of wrong drive letters, change the drive letter with the disk manager as follows: Simultaneously press keys **Ctrl - ESC** select "Run..." **cmd** ↵ in the cmd tool enter the following command: **runas /user:administrator cmd** ↵ in the new cmd window with administrative rights enter the following command: **diskmgmt.msc** ↵ ; **right click** the drive to be corrected; select **Change Driveletters and Path...** select proper drive letter in the pull-down menu, confirm the pop-up with **Yes**, repeat this sequence until you active the correct drive letter assignment)
- Archive patient images on CD-R / DVD media with the new DVD-RW drive

1.4 Host

1.4.1 Required time and tools

Time

- 180 min

Tools

- Set of standard tools

1.4.2 Preliminary

- If the condition of the "Host" allows access to the "Service platform" (Service Software, e.g., as user administrator), select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master Backup**
- Shutdown and switch off the system.

1.4.3 Disassembly



Remember to move the dongle from the old "host" to the new "host"

- Disconnect the main power plug, network drop cable(s), mouse, and keyboard cables.

1.4.4 Assembling

- Plug the HW dongle to the new Host.
- Plug the network drop cable(s), mouse, keyboard cables, and main power plug.

1.4.5 Final checks

- Install software on the new "Host" according to the "Install Software" technical documentation (TDOC) valid for your system.
- Restore "site specific data" according to TDOC "Install Software", valid for your system.
- Start the Local Service, select **Quality Assurance**, Perform the "Quality Assurance" step Image Orientation

1.5 Dongle

1.5.1 Required time and tools

Time

- 60 min

Tools

- Licence file with all customer licenses for the new DONGLE
- Valid service license for the new DONGLE

1.5.2 Preliminary

If the service license and all customer options are invalid, the dongle is most likely defective.



Place dongle orders only through TDHSC1 in Erlangen, a dongle defect must be directly escalated to the TDHSC1 in Erlangen, phone number:+49 9191-188080

1.5.3 Disassembly

Fig. 7: Host, USB dongle



- Ensure that the "Host" is switched off; replace the dongle in back of the "Host".

1.5.4 Assembling

- Switch on the system

1.5.5 Final checks

- Login as user **administrator**, start the Internet Explorer, start Local Service, select **Configuration**, Service, **Licensing**, Click radio button **Browse to select the license file**, navigate to the media with your new license file, click **Save**.

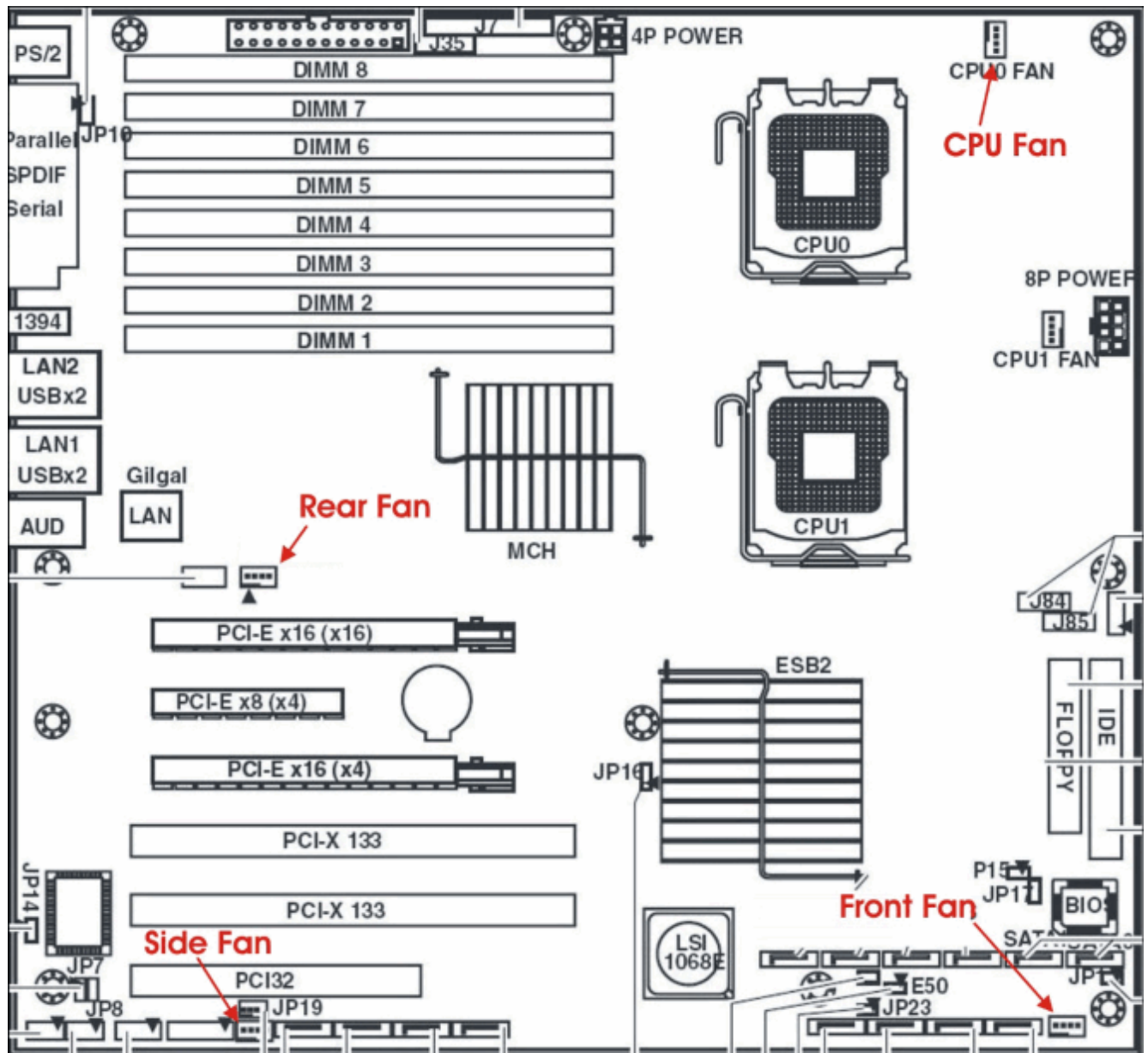
- Re-boot the system, start the MR application
- Check with the customer that all options are available
- Start the Local service, select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master Backup**

1.6 Fans, ventilation duct, crosspiece (R640 / R650)

1.6.1 Front fan

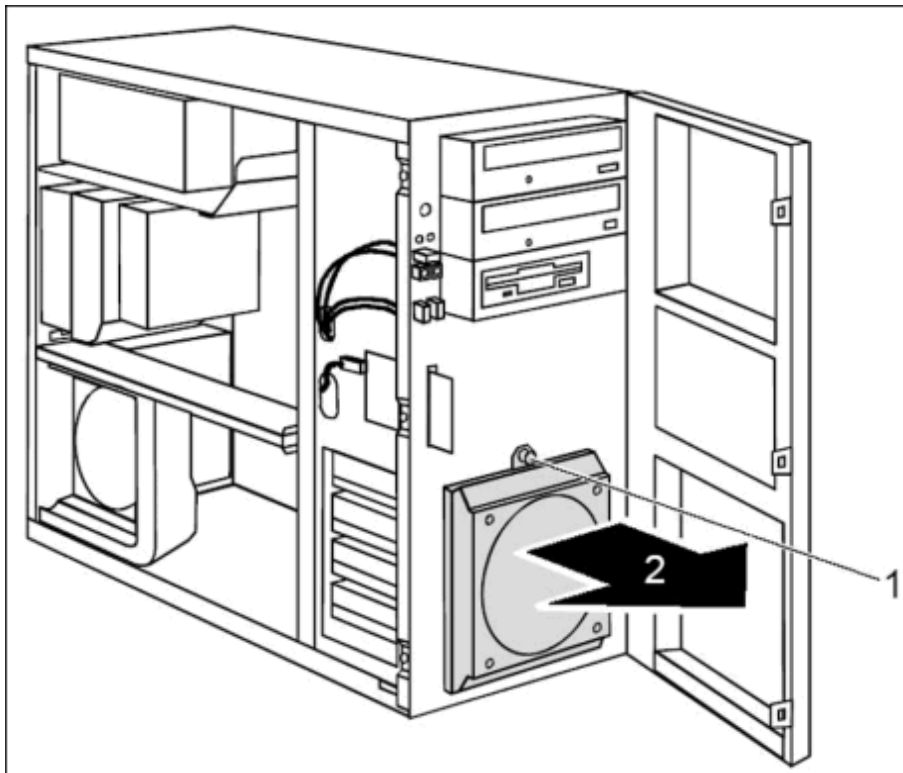
- Open the housing.
- Open the front panel.

Fig. 8: Fan connectors on the mainboard



1.6.1.1 Removing the front fan

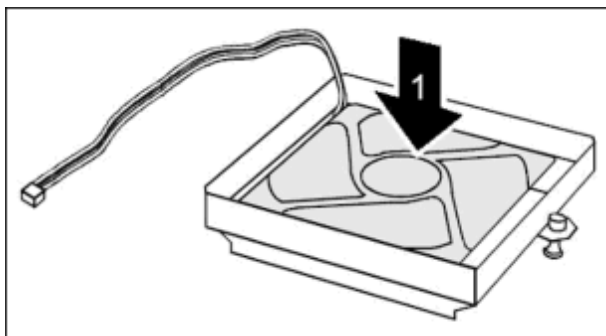
Fig. 9: Removing front fan



- Detach the green expansion rivet (1).
- Ensure that the green expansion rivet (1) is not pulled off the fan.
- Tilt the fan toward the front in the direction of the arrow (2) and remove it from the housing.
- Disconnect the fan cable (refer to (→ Fig. 8 Page 19), "Front fan").
- Pull the fan cable out of the housing.
- Remove the fan.

1.6.1.2 Securing the front fan on the fan bracket

Fig. 10: Fan bracket

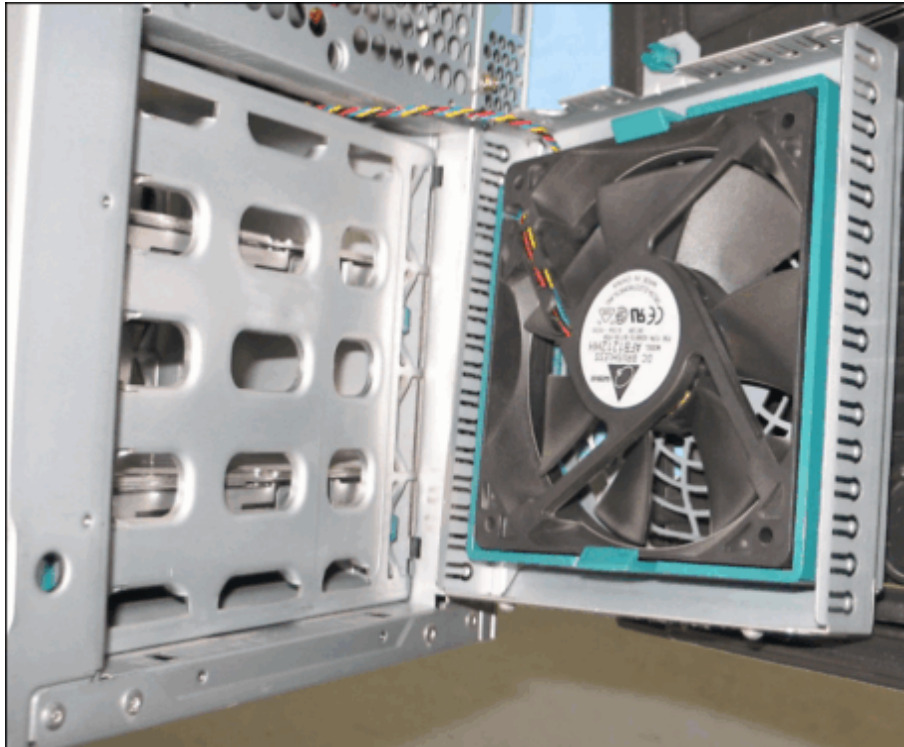


- Lay the fan onto the fan bracket in the direction of the arrow (1).



When doing so, ensure the proper rotation and airflow direction.
Look around the label of the fan as shown in (→ Fig. 11 Page 21).

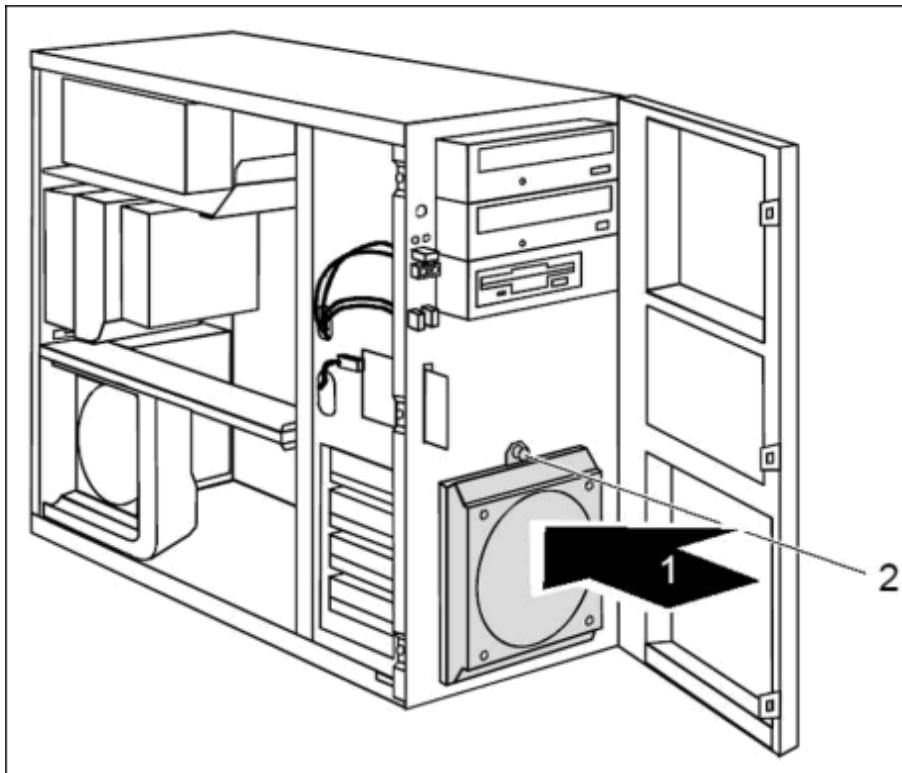
Fig. 11: Air flow of the front fan



Press the fan firmly into the fan bracket until it is felt to engage.

1.6.1.3 Installing the front fan

Fig. 12: Installing the front fan



- Connect the fan cable (refer to (→ Fig. 8 Page 19), "Front fan").
- Insert the fan into the housing in the direction of the arrow (1) so that the lower edge of the fan is flush with the bottom of the device.



When doing so, make sure not to pinch the power supply cable.

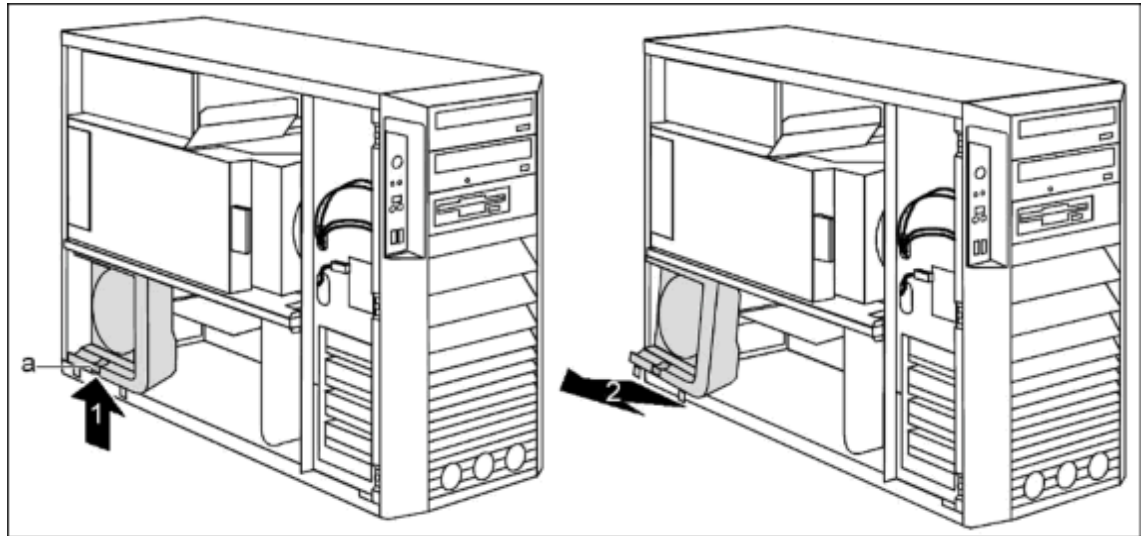
- Slightly pull back the green expansion rivet (2) until the tip of the expansion rivet is no longer spread.
- Press the green expansion rivet (2) into the hole on the housing.
- Secure the fan by pressing the green expansion rivet (2) into the housing until the tip of the expansion rivet is spread again.
- Close the front panel.
- Close the housing.

1.6.2 Side fan

You will need to remove the side fan inside the housing in order to install boards or upgrade the main memory.

1.6.2.1 Removing the side fan

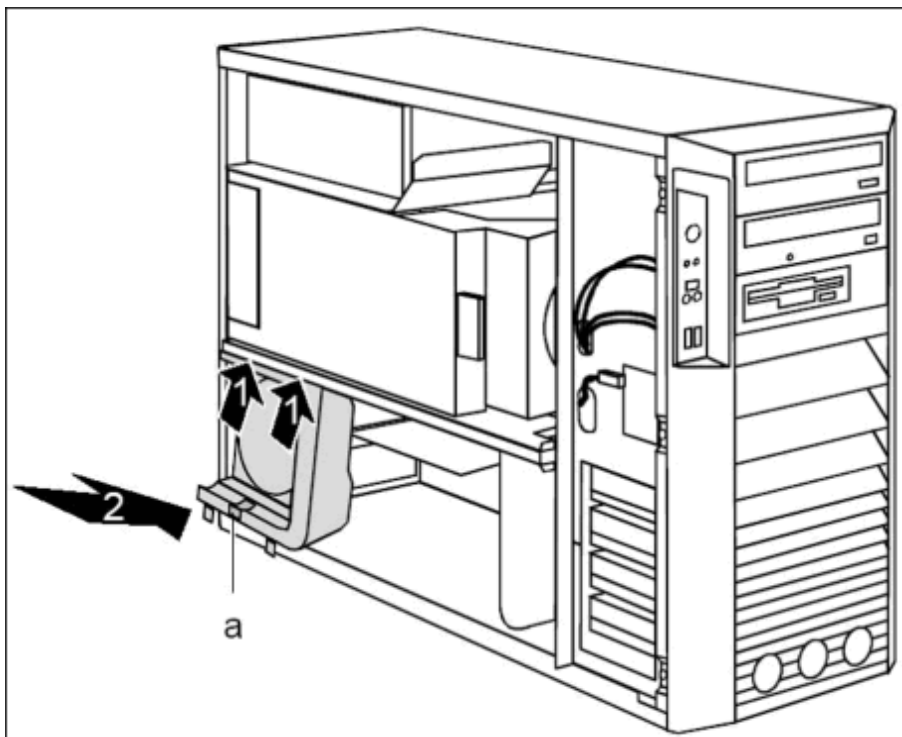
Fig. 13: Removing the side fan



- Open the housing (for better access remove (→ Crosspiece / Page 28)).
- Release the side fan by pressing the locking hook (a) in the direction of the arrow (1).
- Pull the bottom edge of the side fan in the direction of the arrow (2) out of the housing.
- Disconnect the cable of the fan from the mainboard (refer to (→ Fig. 8 Page 19), "Side fan").
- Remove the side fan from the housing.

1.6.2.2 Installing the side fan

Fig. 14: Installing the side fan



- Connect the fan cable to the corresponding plug contact on the mainboard (refer to (→ Fig. 8 Page 19), "Side fan").
- Position the side fan at a slight angle in the direction of the arrow (1) and hook the fan lugs into the notches on the crosspiece.



When doing so, ensure the proper rotation and airflow direction.
Look around the label of the fan as shown in (→ Fig. 15 Page 24).

- Push the side fan into the housing in the direction of the arrow (2).

Fig. 15: Air flow of the side fan



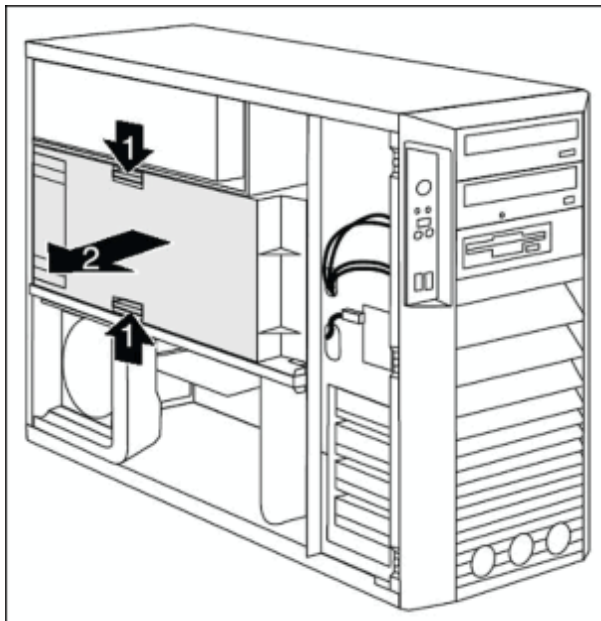
- Lock the side fan by pressing the locking hook (a) downward.
- Close the housing.

1.6.3 Ventilation duct, CPU fan and rear fan

1.6.3.1 Removing the ventilation duct

Before you can upgrade the main memory or the processor, or before you can replace the processor or the lithium battery, you must remove the ventilation duct.

Fig. 16: Removing the ventilation duct



- Open the housing.
- Release the ventilation duct by pressing the locking hooks in the direction of the arrow (1).
- Grasp the ventilation duct by the cutouts and pull it in the direction of the arrow (2) out from the housing.



When removing the ventilation duct, be careful not to damage the processor cooler(s) on the mainboard. If necessary, disconnect the cable to the mainboard.

1.6.3.2 Removing the CPU fan

- Disconnect the cable of the fan from the mainboard and take note of where it was plugged in (refer to (→ Fig. 8 Page 19), "CPU fan").
- Release the fan by pressing the locking hook. Remove the fan.

1.6.3.3 Removing the rear fan

- Disconnect the cable of the fan from the mainboard and take note of where it was plugged in (refer to (→ Fig. 8 Page 19), "Rear fan").

Fig. 17: Locking hook of the rear fan



Release the fan by pressing the locking hook. Remove the fan.

1.6.3.4 Installing the CPU fan

- Connect the fan cable to the mainboard at the connection point you noted before (refer to (→ Fig. 8 Page 19), "CPU fan").
- Mount the fan in the ventilation duct.

1.6.3.5 Installing the rear fan

- Connect the fan cable to the mainboard at the connection point you noted before (refer to (→ Fig. 8 Page 19), "Rear fan").

- Mount the fan in the housing.

Fig. 18: Air flow of the rear fan



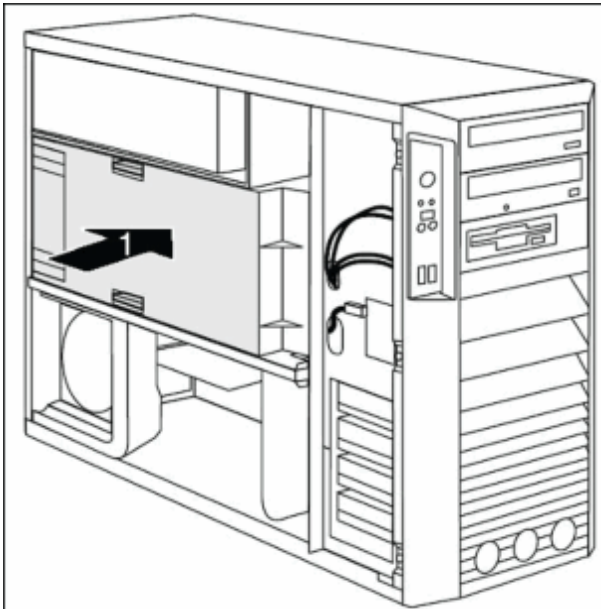
1.6.3.6 Installing the ventilation duct



When routing the fan cable, make sure that it is not kinked or pinched.

When fitting the ventilation duct, be careful not to damage the processor cooler(s) on the mainboard.

Fig. 19: Installing the ventilation duct



- Slide the ventilation duct in the direction of the arrow (1) into the housing until the locking hooks noticeably engage.

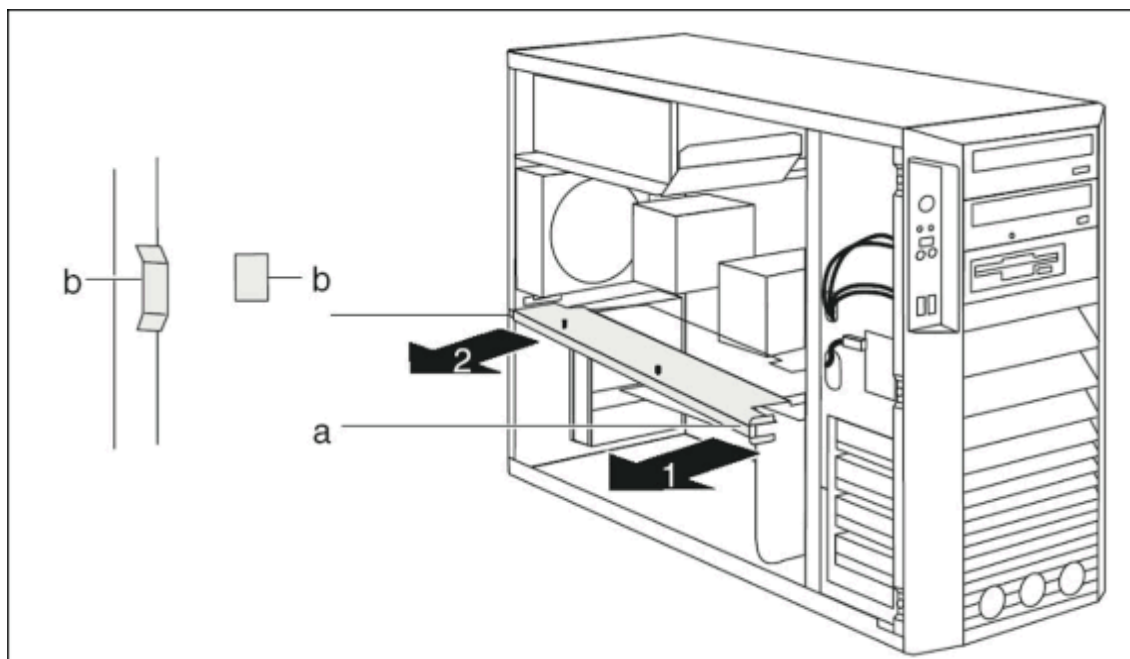
- Close the housing.

1.6.4 Crosspiece

1.6.4.1 Removing the crosspiece

Before replacing the main memory, you will need to remove the crosspiece.

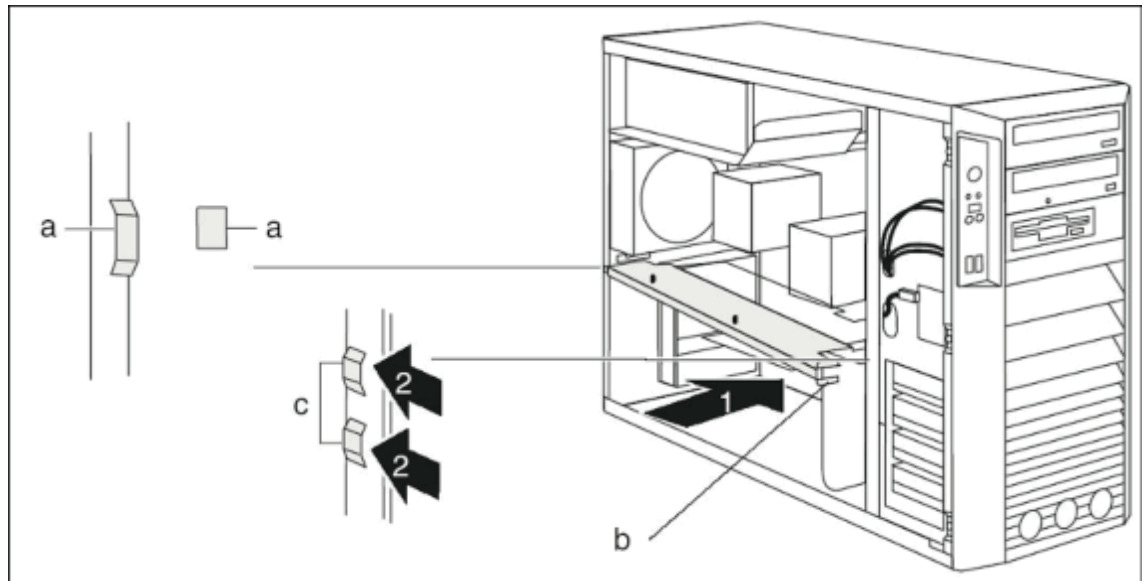
Fig. 20: Removing the crosspiece



- Open the housing.
- Remove the ventilation duct.
- Remove the side fan.
- Grasp the crosspiece by the lug (a) and unhook it from the brackets in the housing.
- Pull the crosspiece slightly out of the housing in the direction of the arrow (1).
- Detach the crosspiece from the brackets (b) on the back of the housing.
- Remove the crosspiece from the housing in the direction of the arrow (2).

1.6.4.2 Installing the crosspiece

Fig. 21: Installing the crosspiece



- Insert the crosspiece into its brackets (a) on the back of the housing at a slight angle.



Make sure that the Plexiglas plate of the crosspiece is not pinched between the memory slots.

- Press the crosspiece into the housing in the direction of the arrow (1) until the lugs (b) of the crosspiece hook into the brackets (c) in the housing.
- Install the side fan.
- Install the ventilation duct.
- Close the housing.

1.7 Power supply (R640 / R650)

1.7.1 Preparatory work steps

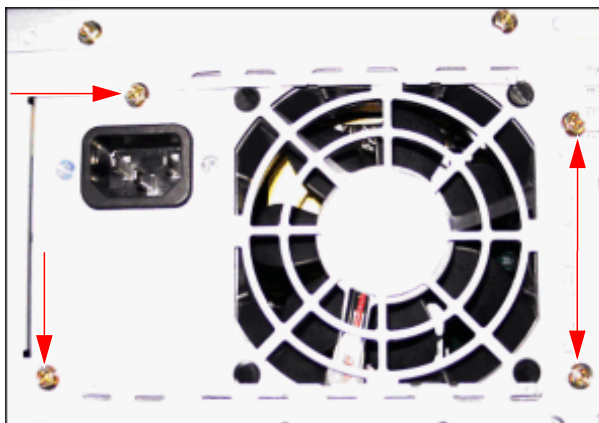
- Open the housing.
- Remove the ventilation duct.
- Remove the crosspiece.



The power supply unit "PC/WE 1000W" only fits mechanically to the R650 host.

1.7.2 Changing the power supply

Fig. 22: Power Supply (1).jpg



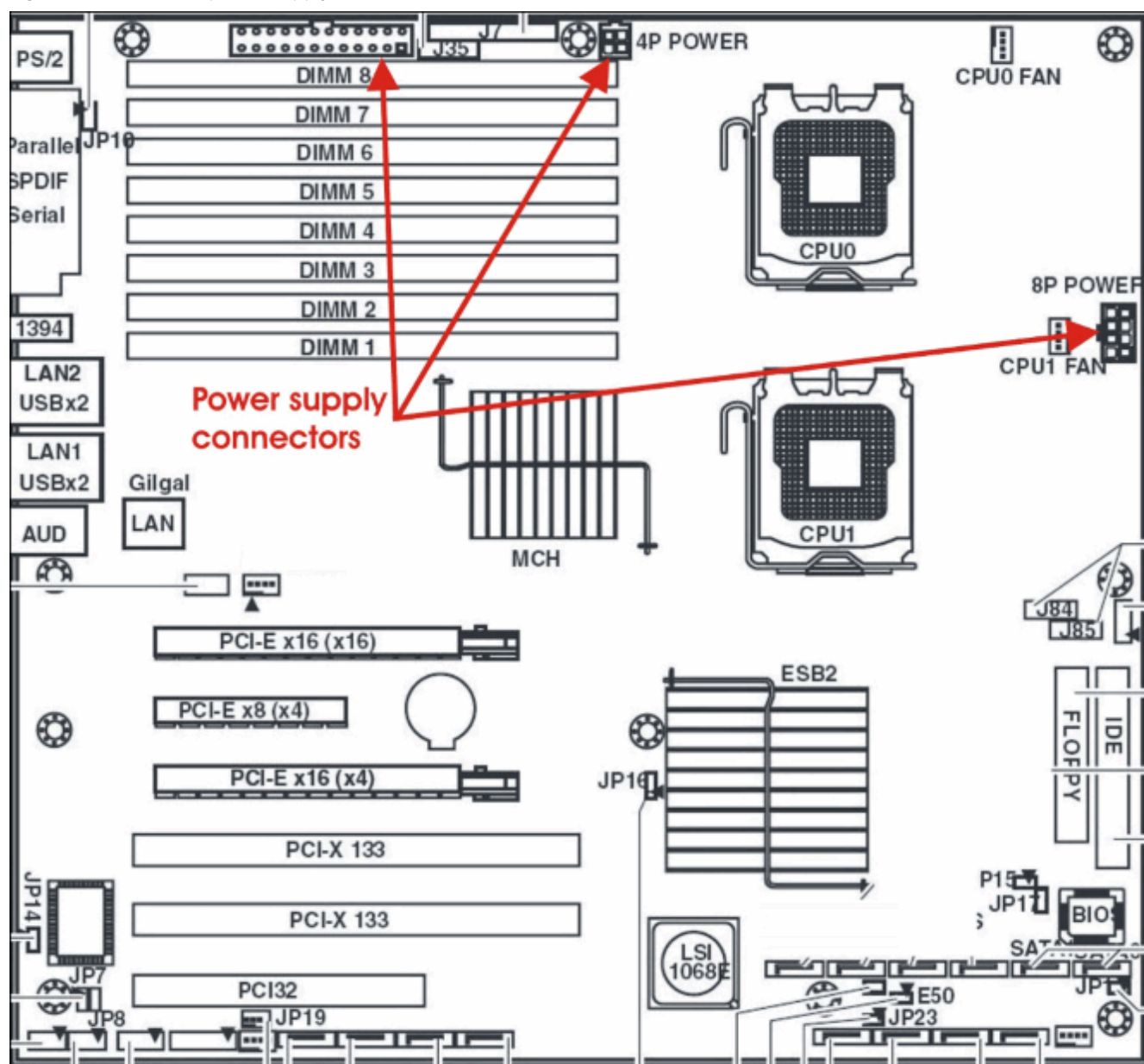
- Make notes about the power connectors (mainboard [refer to (→ Fig. 24 Page 32)], drives and graphics board) and the ducting of the power supply, and disconnect all of them.
- Remove the 4 screws (arrows) on the back of the power supply.
- (R650 remove the screw at the side from the power supply).
- Pull the power supply out from the case laterally.

Fig. 23: Pillar



- (R640 Remove the pillar (2 screws) and mount it on the new power supply).
- Move the new power supply into the case and attach it with the 4 screws on the back.
- (R650 attach the screw at the side to the power supply).
- Connect all power connectors and attach the cables as they were before.

Fig. 24: Mainboard / power supply



1.7.3 Final work steps

- Install the crosspiece.
- Install the ventilation duct.
- Close the housing.

1.8 Lithium Battery



CAUTION

The BIOS update may result in a loss of patient data.

- » Check the specific system documentation for details concerning the BIOS installation.
- » If necessary, save patient data before starting the BIOS installation.
- » On some systems, a BIOS installation requires a complete SW installation

1.8.1 Changing the lithium battery

In order to permanently save the system information, a lithium battery is installed to provide the CMOS memory with a current. A corresponding error message notifies the user when the charge is too low or the battery is dead. The lithium battery must then be replaced.



CAUTION

Incorrect replacement of the lithium battery may lead to a risk of explosion.

- » The lithium battery may be replaced only with an identical battery or with a type recommended by the manufacturer, following the instructions below.



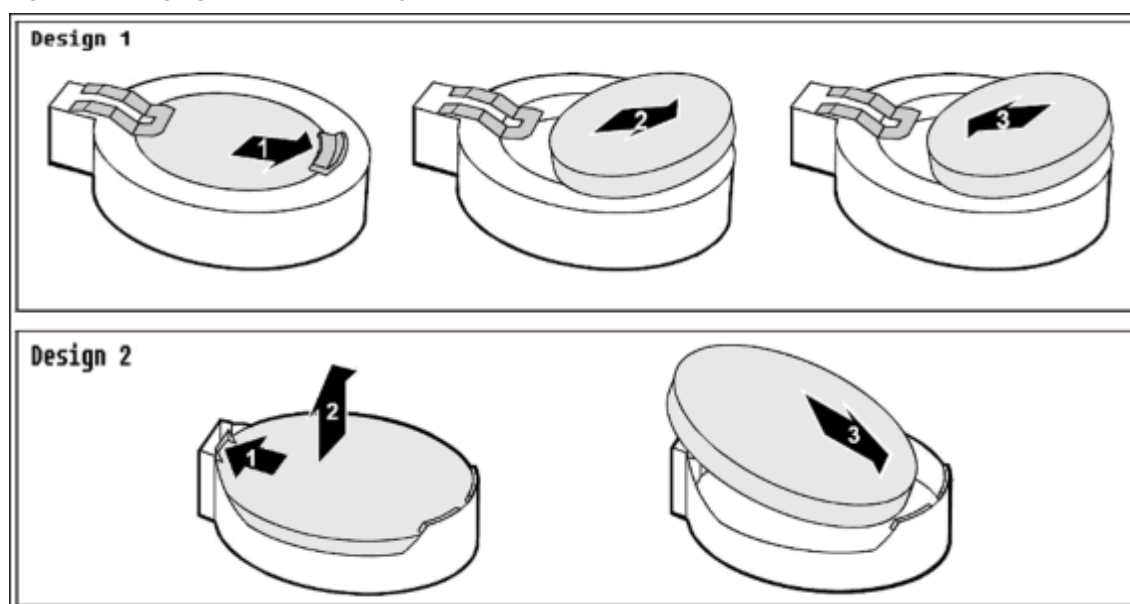
Do not throw lithium batteries into household waste. They must be disposed of in accordance with local regulations concerning hazardous waste.



Ensure that you observe the correct polarity when replacing the lithium battery. The plus pole must be on top.

- Open the housing.
- There are several different lithium battery holder designs; all of them function in a similar way.

Fig. 25: Changing the lithium battery



- Press the catch in the direction of the arrow (1).
 - » The battery jumps out of the holder slightly.
- Remove the battery (2).
- Push the new lithium battery of the identical type into the holder and press it downward until it engages (3).



Restoring BIOS settings is only necessary after battery replacement for older imager HW and SW. All new imagers default BIOS setting are customized for use without any manual configuration.



When the battery is dead, the BIOS is set to factory default.

In addition to activating the system ON/OFF switch, the following may be necessary: - Turn OFF the PC via the power switch on the rear panel of the PC - Turn ON the PC via the rear power switch and press the push button on the front panel of the PC

1.8.2 Final work steps

- Close the housing.
- Perform BIOS setup according to the specific system documentation (only valid for older imager HW and SW).
- Check that the system boots properly.
- Perform the quality assurance procedure: **Image Orientation** in the service platform.
- Make sure that the system is generating images.

1.9 In-Room MRC

1.9.1 Required time and tools

Time

- 120 min

Tools

- Set of standard tools
- TDOC Startup of the In-Room MRC

1.9.2 Final checks

- Run the auto configuration program and perform a visual evaluation according to the start-up of the In-Room MRC, Refer to TDOC start-up of the In-Room MRC

1.9.2.1 Protective conductor test



WARNING

In case of a fault, voltage could be present at the housing of components.

This may lead to severe injuries or death if the resistance of the ground wire is too high.

- » The protective conductor resistance must be $\leq 200 \text{ m}\Omega$. Measured against the LPD protective ground terminal, if a component replacement may affect the ground wire replacement.

For more detailed information, refer to "Protective conductor measurement" in the binder: "Technical Documents"; register: Safety ; chapter: MR-specific safety information.

1.10 Host M470

1.10.1 General

1.10.1.1 Safety Information and/or references to Safety Information

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad (TDK equipment type 8005 from 3M Deutschland GmbH).



WARNING

Please follow the general safety guidelines (→ General Safety Notes / TD00-000.860.01) as well as the MR-specific safety information (→ MR-specific safety information / MR-000.860.01)!

Failure to comply with the safety guidelines may result in death or serious injury.

- » Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Compliance with the information provided is mandatory at all times.



CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

- » Ramp down the magnet or maintain a safe distance from the magnetic field.

Handling of patient data

The data protection guidelines specify the mandatory procedures for handling patient data.

Data carriers, storage devices, or other media (such as the hard disk inside the host PC) must not leave the customer site if there is a chance they still contain patient data.



The service report needs to list the activities performed to ensure the secure handling of patient data.

As a first step, the PC needs to be analyzed. Once a defect in any part of the PC hardware is identified as the cause of the problem, the part replacement follows.

For PCs there are two cases:

1. The PC does not contain any patient data. - The individual product documentation needs to list hard disks without patient data.(→ No patient data on hard disks / Page 37)

2. The PC contains patient data.

The disks need to be returned without patient data.

Please decide how to proceed for your individual case. The recommendation:

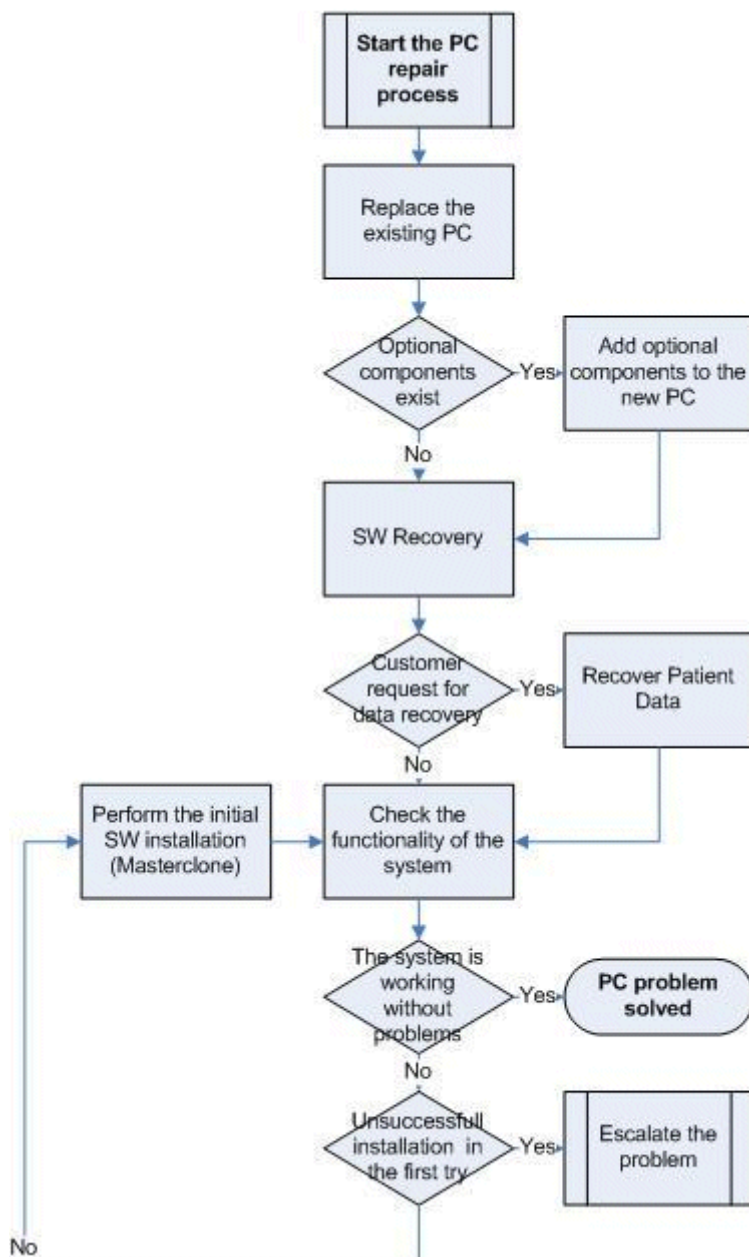
1. Case: Valid if the new PC has the same material number as the defective PC being replaced. The existing disk has to be removed from the host computer that is going to be returned. This disk must be installed in the replacement host computer. At the same time, the new disk that was just removed has to be installed in the defective host PC that is being returned. (→ Swap the disks / Page 38)
2. Delete the patient data with the "blancco" tool. (→ Deleting Patient Data / Page 41)
3. Case: All hard disks potentially containing patient data are removed from the defective PC and remain at the customer site.

No patient data on hard disks



Only continue with the parts replacement procedure outlined in the flow chart below if you have confirmation from the product-specific documentation that the PC contains no patient data, and that no patient information is contained in any logfiles on the PC.

Fig. 26: Replacing a PC without patient data



Swap the disks

The preferred case is replacing the hard disk: This case entails additional activities and this will surely take some time. - This time span may be compensated for by the time required for the restore to be performed. CSML confirms that such an action as opening the housing of the host computer and performing such activities will not negatively influence any warranty rights or claims.

Such a course of action, however, may provide the following benefits:

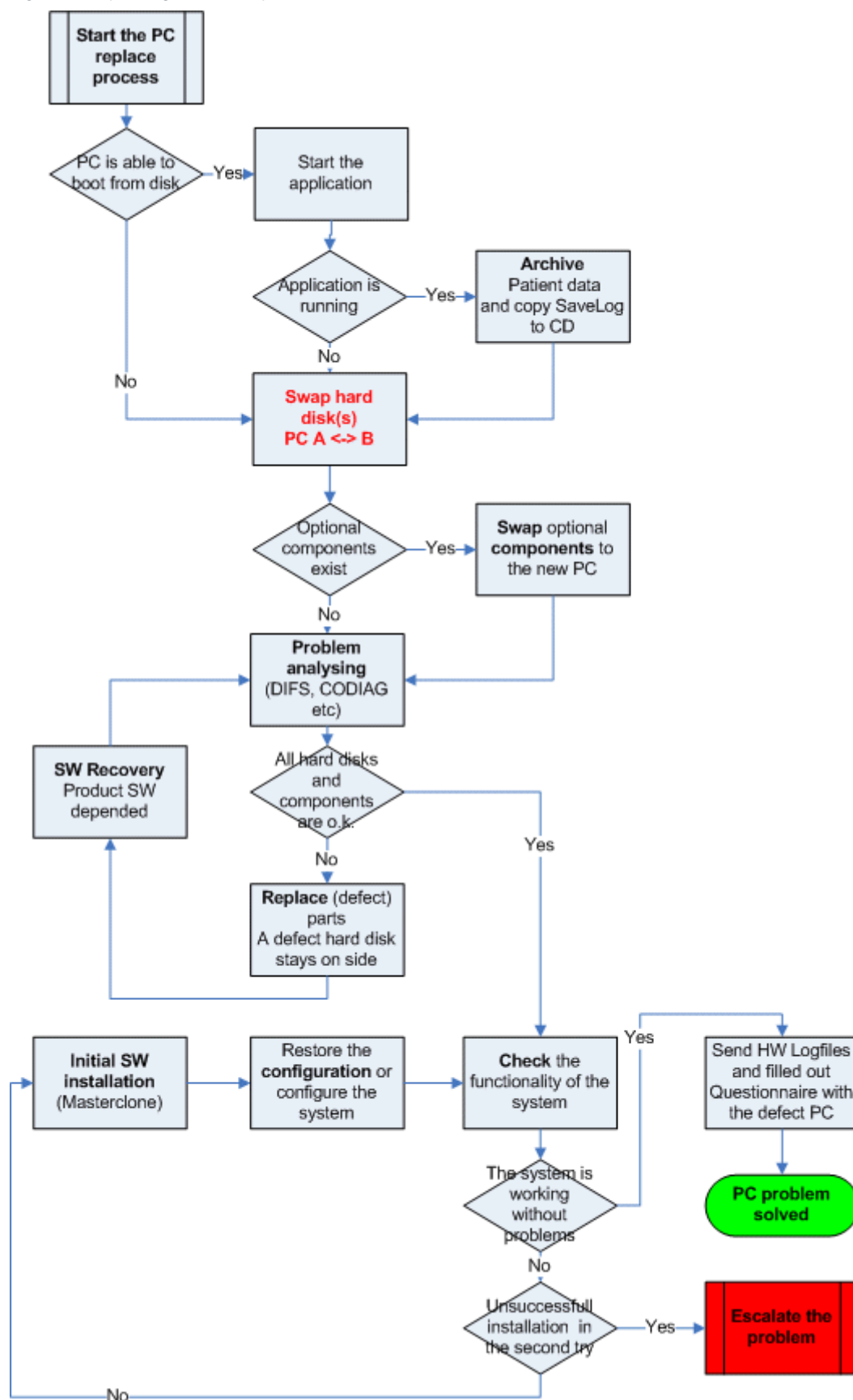
- The returned host computer will not contain any patient data.

- The disk swapped to the new PC will run without need of the rebuild/restore function.
- The customer may even have the benefit of reduced downtime.
- No reconstruction or loading of images from the archive will be necessary (all patient data will be available).

The decision in such cases will be made by the attending CSE or the responsible service manager.

The work steps are shown in the following flowcharts:

Fig. 27: Replacing a PC with patient data



Deleting Patient Data

Since the end of 2013, there is a tool to delete data from hard disk. The tool needs to be used in cases where replacement of the disk from the defective to the new PC is not possible.

The principles of "Return Goods Management" require that all storage devices containing personal patient data, which are to be removed from the site (i.e. spare parts returns), shall have all data wiped clean (securely deleted) or the storage device shall be handed over to the customer when deletion is not possible.

The tool is called "blancco". The tool can be downloaded from the Siemens Intranet.

The certified blancco tool permanently removes data from IT assets such as PCs, servers and data center equipment.

Data erased using blancco cannot be recovered with any existing technology. This helps to have all data wiped clean (securely deleted) and achieve compliance with existing "patient data handling" regulations.

SIEMENS Employees: The tool is available as an iso image from the CS Home page

<https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>

CS Home page -> Tools -> blancco - Erasure Software

For others see e.g.: <http://blancco.com/en/products/>

The time required by the procedure depends on the size of the hard disk.

Warning: before you start using the blancco software, make sure that all files, folders, software applications or any other information that you want to save for later use are backed up on an appropriate media device other than the original hard drive. Data that has been erased from the hard disk by this program cannot be recovered by any existing method.



Selecting the wrong hard disk can result in lose of data with the use of blancco. Be sure that the selected hard disk for deletion is the correct one.

Creating the blancco CD

An ISO image of the blancco CD is available for download from: CS Home page -> Tools -> blancco (CS Home page, <https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>)

1. Download the ISO image.
2. Label the blank CD as "blancco - Erasure Software"
3. Insert the CD into the drive of your service computer.
4. Use a standard burn program (such as Nero) to open the file and burn the CD image to the CD.
5. Remove the CD from the drive.

Using the blanco CD

1. Booting with blanco

1. Insert the "blanco - Erasure Software" CD into the PC with the drive(s) to be deleted

2. Switch off the PC.



If this is an IAS-A PC, disconnect the service backup drive to prevent accidental loss of data.

3. Switch on the computer power to boot the system from the CD (if necessary, change the BIOS boot priority). The blanco Home screen is visible with all identified hard disks of the PC



If necessary, press F2 to change the language and the keyboard layout.

Fig. 28: Blanco language/keyboard layout



[1] F2 to open menu

[2] Language

[3] Keyboard layout

[4] Checkboxes to select hard disks

Deletion with Blancco

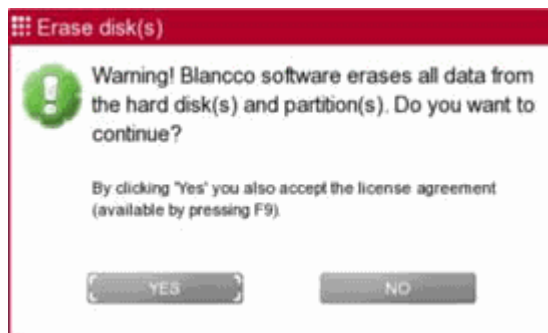
1. Select the hard disks to be deleted.



Blancco can erase up to 16 hard disks simultaneously.

2. Press Erase, a warning message appears.

Fig. 29: Warning message



3. Press Yes to start the erasure process. The erasure process starts and the progress and duration is visible.
4. Click on OK when the process is finished and the Erase successful message appears.
5. Insert a memory stick to save the report.
6. Press Save or F10. The report is saved to the memory stick.
7. Press Ok. The process is finished and the PC can be turned off.

1.10.2 FRU overview

FRU	Time	Instructions	Final checks	Tools
Host	180 min		Perform a measurement, view the images produced. Check with the customer that all options are available	Field service grounding kit (ESD), set of standard tools
Fans	45 min	Fans, ventilation duct, crosspiece	Ensure that the new fan(s) works.	Set of standard tools
DWD-ROM / DVD-RW	45 min	Drives	Burn some test images onto CD-RW or DVD media with the new DVD-RW drive or read Service instructions from a Service DVD from the Service GUI Help > Documentation	Field service grounding kit (ESD), set of standard tools
Power supply	45 min	Power supply	Perform a measurement and view the images produced.	Set of standard tools
Battery	60 min	Lithium battery	Perform BIOS setup according to TDOC Installation Software. Perform a measurement and view the images produced.	Field service grounding kit (ESD), set of standard tools
Dongle (USB)	60 min		Check with the customer that all options are available Perform a measurement and view the images produced.	Service license/Application licenses. The Service/Applications licenses generated in relation to the flexid of the new dongle
In-room MRC	120 min	Refer to the In-room MRC monitor Safety-related tests	Run the auto configuration program and perform a visual evaluation according to the start-up of the In-Room MRC Measure protective conductor resistance for the "In-room MRC"	Set of standard tools and TDOC start-up of the In-room MRC

1.10.3 Host

1.10.3.1 Required time and tools

Time

- 180 min

Tools

- Set of standard tools

1.10.3.2 Preliminary

- If the condition of the "Host" allows access to the "Service platform" (Service Software, e.g., as user administrator), select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master Backup**
- Shutdown and switch off the system.

1.10.3.3 Disassembly



Remember to move the dongle from the old "host" to the new "host"

- Disconnect the main power plug, network drop cable(s), mouse, and keyboard cables.

1.10.3.4 Assembling

- Plug the HW dongle to the new Host.
- Plug in the network drop cable(s), mouse, keyboard cables, monitor, and main power plug.

1.10.3.5 Final checks

- Check if the new "Host" has the correct software version or install software on the new "Host" according to the technical documentation (TDOC) "Install Software" valid for your system.
- Restore "site specific data" according to TDOC "Install Software", valid for your system.
- Start the Local Service, select **Quality Assurance**, perform the Image Orientation "Quality Assurance" step



In case of connection problem with the MARS after replacing the HOST; the firewall settings must be disabled for the internal measurement network.

How to disable the firewall for the internal network:

- Login as user **administrator**
- Open a cmd tool and run **networkcardinstall /I1 /I2 ↵**
- Click **START -> Settings -> Control Panel**

- Change the appearance of the Control Panel to classic view: Click the marked link **Switch to classic view**
- Double click the icon **Windows firewall**
- In the Windows Firewall configuration window select the tab card the icon **Advanced**
- If the Measurement (internal) network checkbox is selected, remove this checkbox
- Close the Windows Firewall configuration window by pressing **OK**
- Re-boot the system and login as user **meduser**

1.10.4 Fans, ventilation duct, crosspiece

1.10.4.1 Front fan

- Open the housing.
- Open the front panel.

Fig. 30: CELSIUS M470 D2778

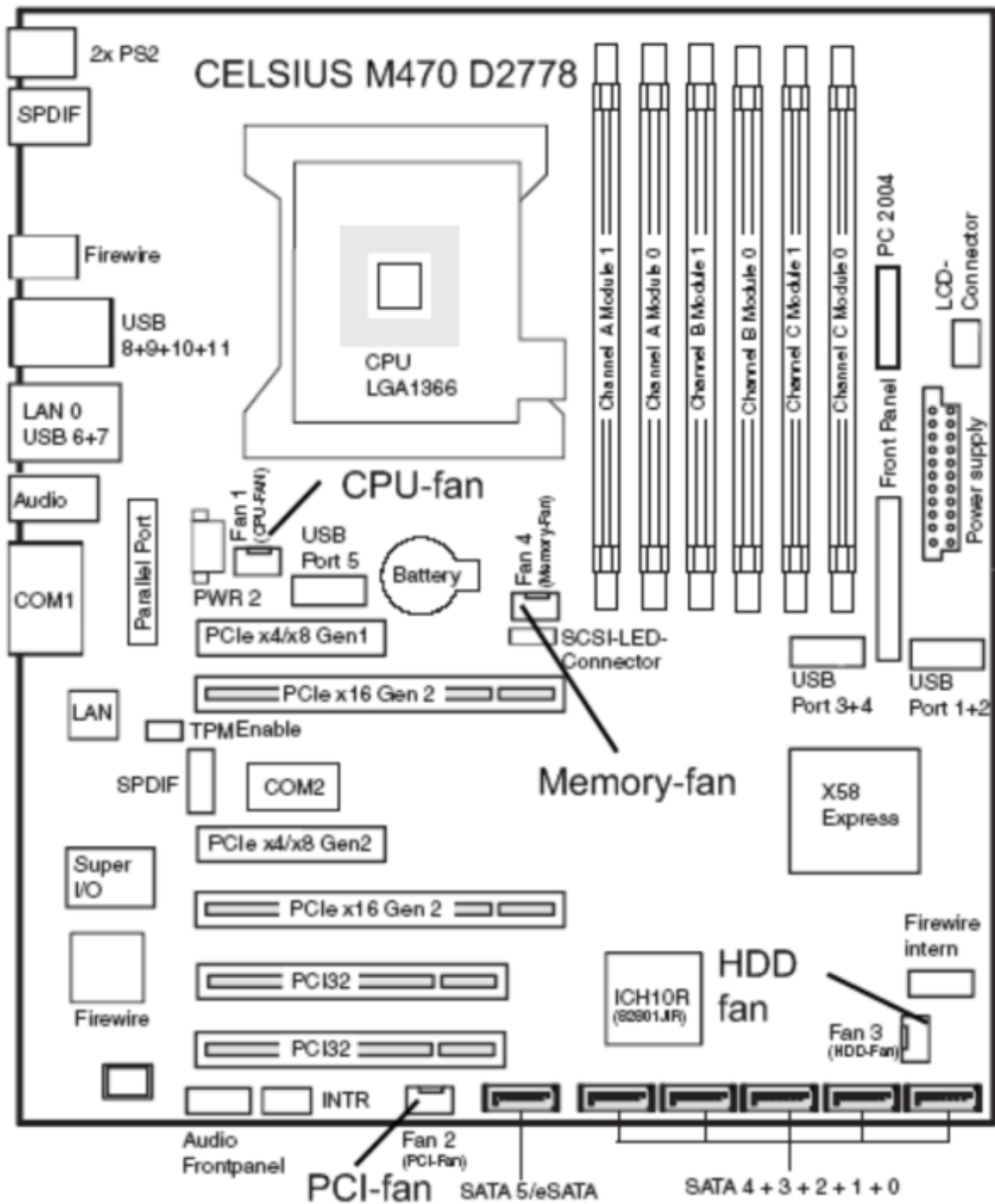
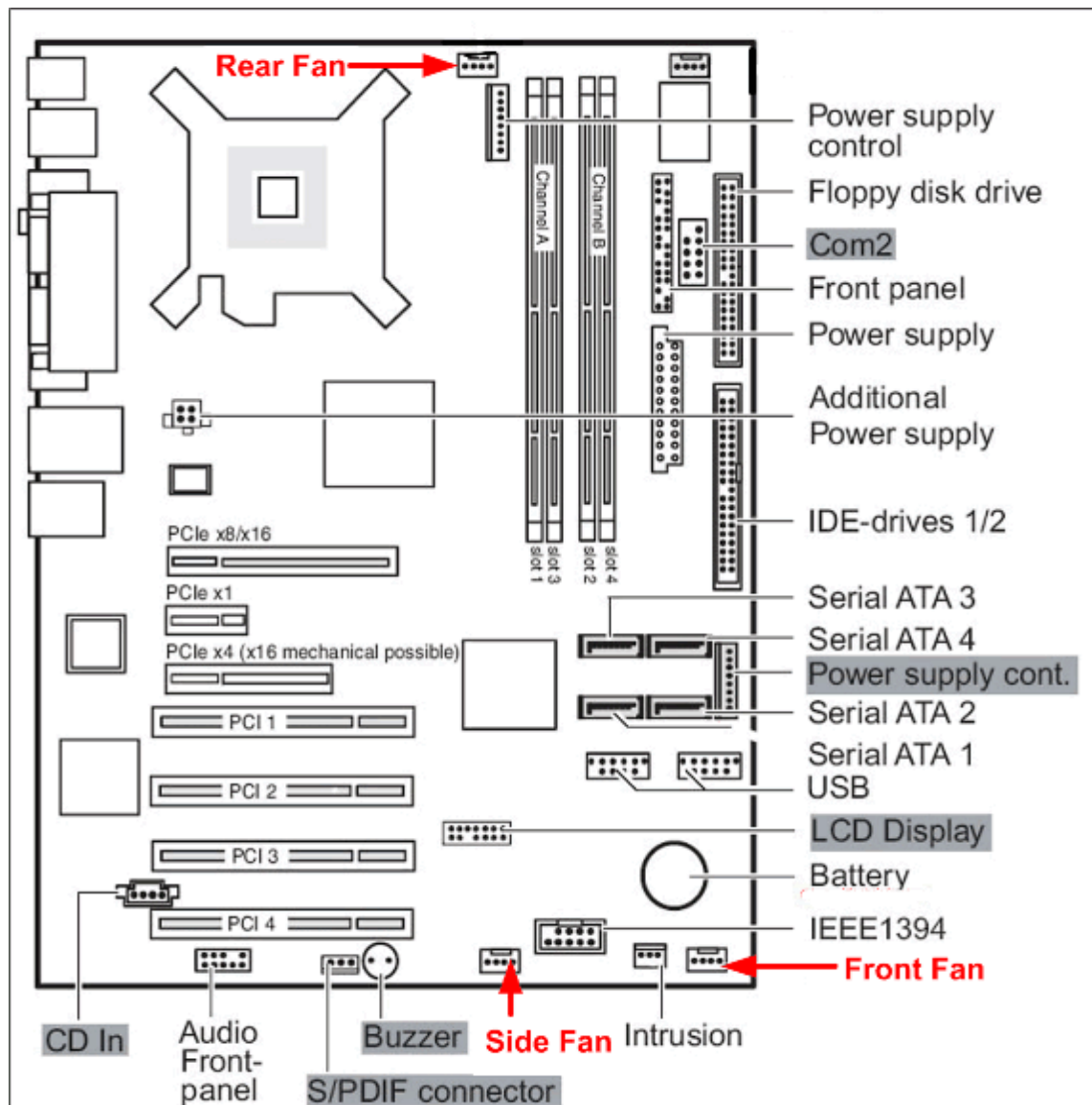
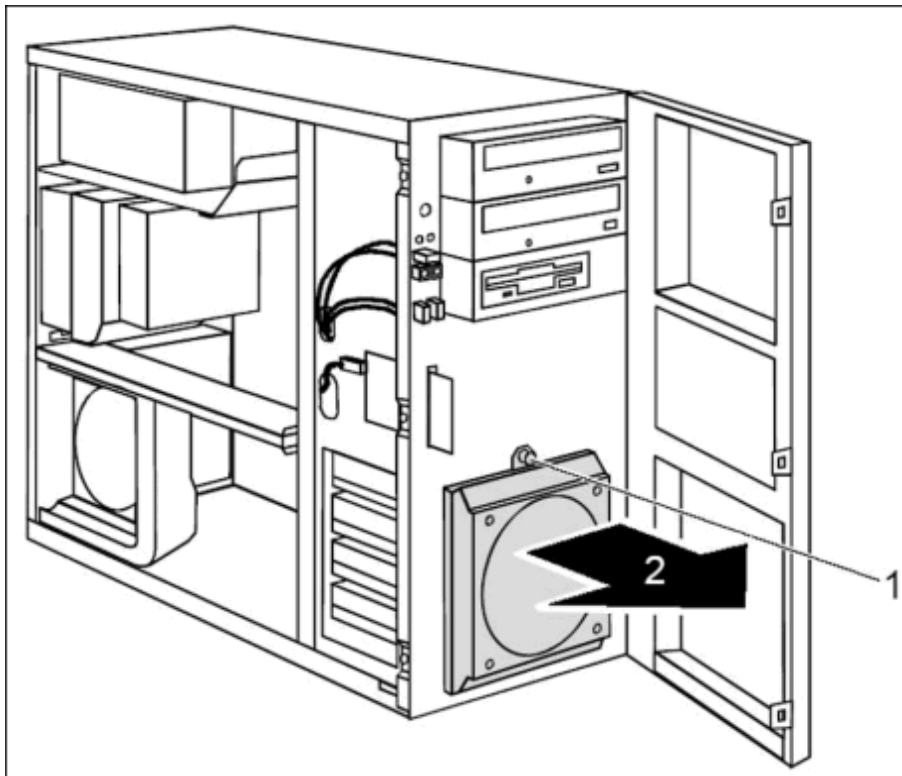


Fig. 31: Fan connectors on the mainboard



Removing the front fan

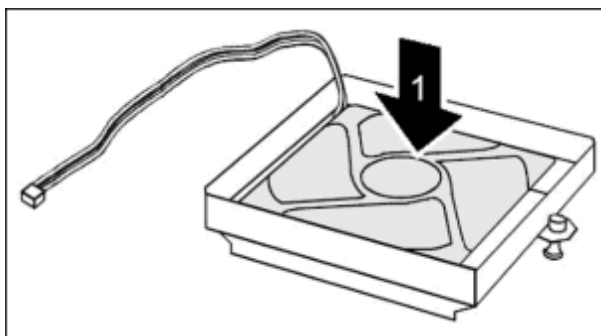
Fig. 32: Removing front fan



- Detach the green expansion rivet (1).
- Ensure that the green expansion rivet (1) is not pulled off the fan.
- Tilt the fan toward the front in the direction of the arrow (2) and remove it from the housing.
- Disconnect the fan cable (refer to (→ Fig. 31 Page 48), "Front fan").
- Pull the fan cable out of the housing.
- Remove the fan.

Securing the front fan on the fan bracket

Fig. 33: Fan bracket

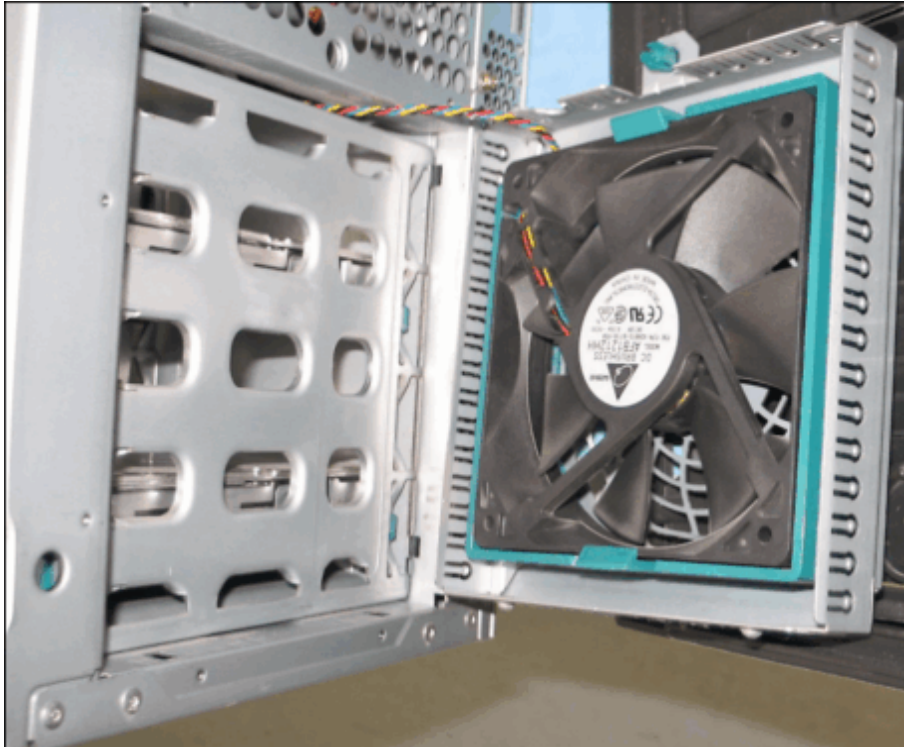


- Position the fan on the fan bracket in the direction of the arrow (1).



When doing so, ensure the proper rotation and air-flow direction. Observe the label of the fan as shown in (→ Fig. 34 Page 50).

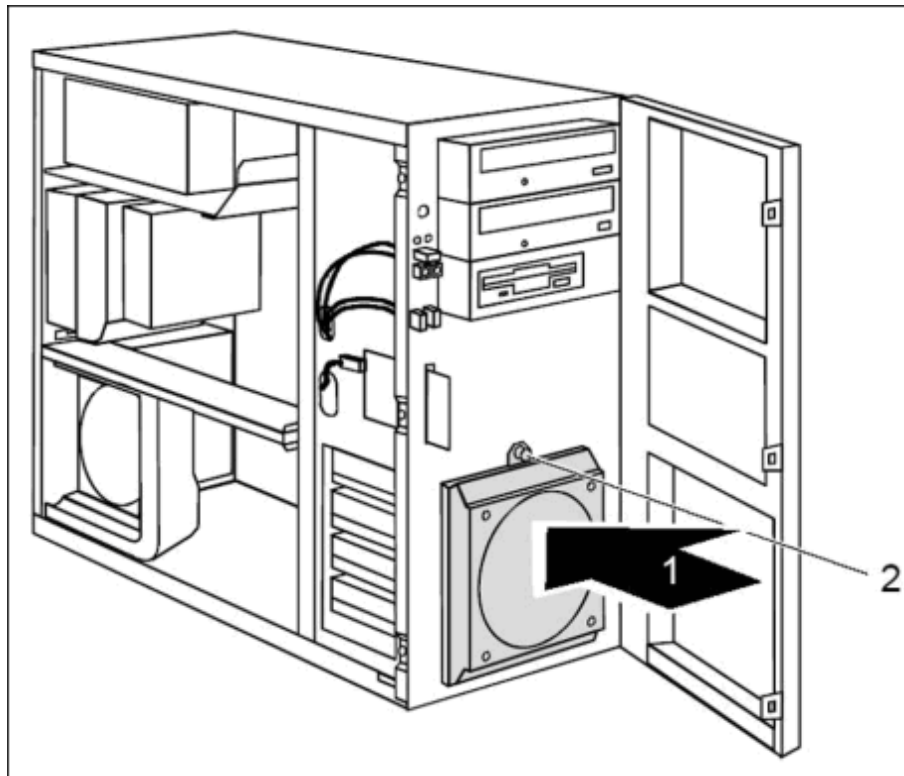
Fig. 34: Air flow of the front fan



- Press the fan firmly into the fan bracket until it is felt to engage.

Installing the front fan

Fig. 35: Installing the front fan



- Connect the fan cable (refer to (→ Fig. 31 Page 48) "Front fan").
- Insert the fan into the housing in the direction of the arrow (1) so that the lower edge of the fan is flush with the bottom of the device.



When doing so, make sure not to pinch the power supply cable.

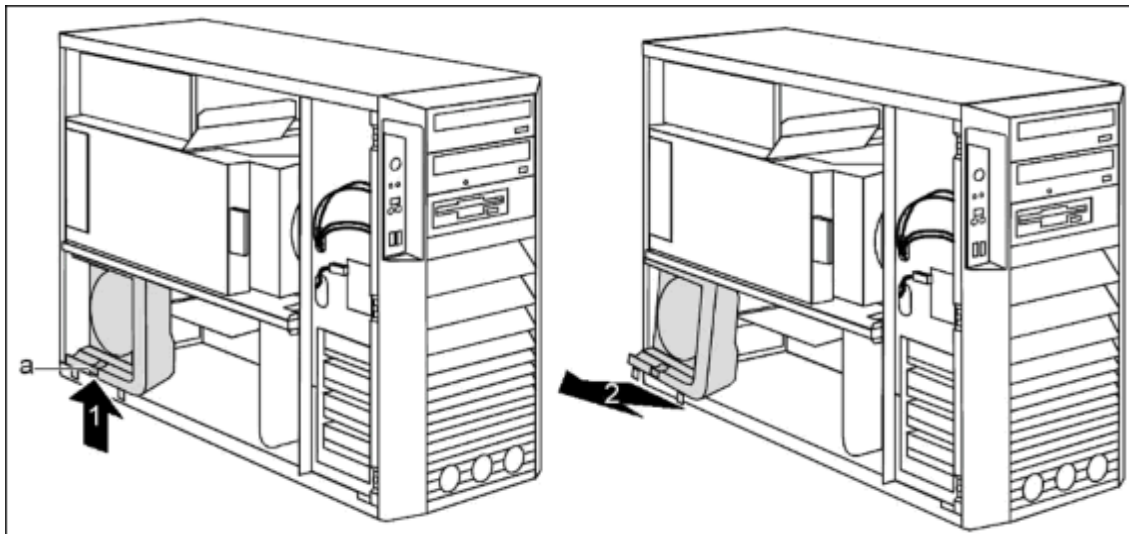
- Slightly pull back the green expansion rivet (2) until the tip of the expansion rivet is no longer spread.
- Press the green expansion rivet (2) into the hole on the housing.
- Secure the fan by pressing the green expansion rivet (2) into the housing until the tip of the expansion rivet is spread again.
- Close the front panel.
- Close the housing.

1.10.4.2 Side fan

You will need to remove the side fan inside the housing in order to install boards or upgrade the main memory.

Removing the side fan

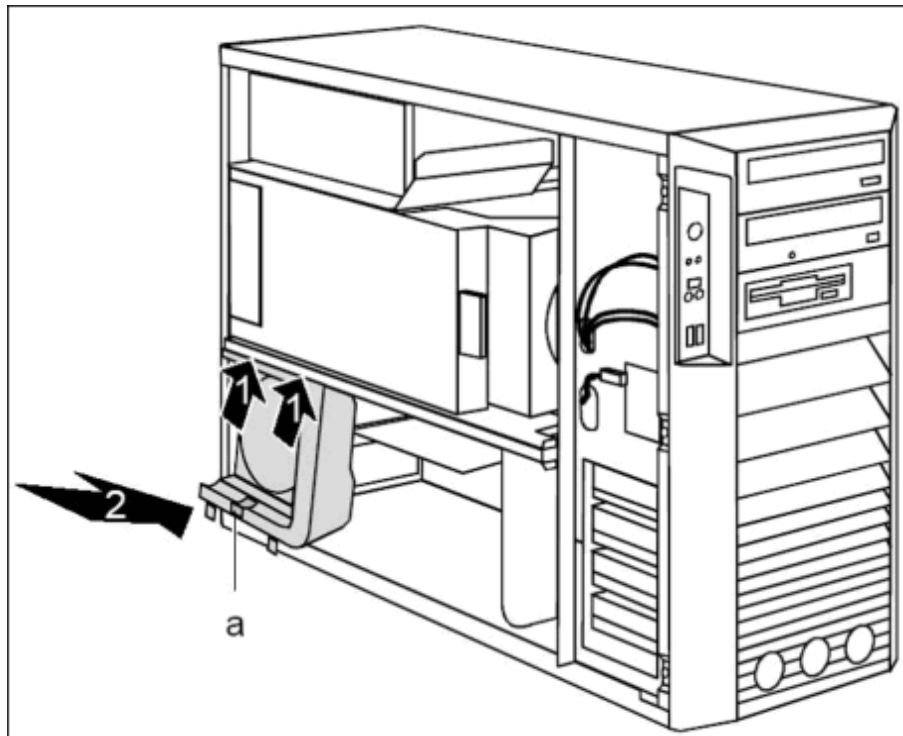
Fig. 36: Removing the side fan



- Open the housing.
- Release the side fan by pressing the locking hook (a) in the direction of the arrow (1).
- Pull the bottom edge of the side fan in the direction of the arrow (2) out of the housing.
- Disconnect the cable of the fan from the main board (refer to (→ Fig. 31 Page 48), "Side fan").
- Remove the side fan from the housing.

Installing the side fan

Fig. 37: Installing the side fan



- Connect the fan cable to the corresponding plug contact on the main board (refer to (→ Fig. 31 Page 48), "Side fan").
- Position the side fan at a slight angle in the direction of the arrow (1) and hook the fan lugs into the notches on the crosspiece.



When doing so, ensure the proper rotation and air-flow direction. Observe the label of the fan as shown in (→ Fig. 38 Page 53).

- Push the side fan into the housing in the direction of the arrow (2).

Fig. 38: Air flow of the side fan



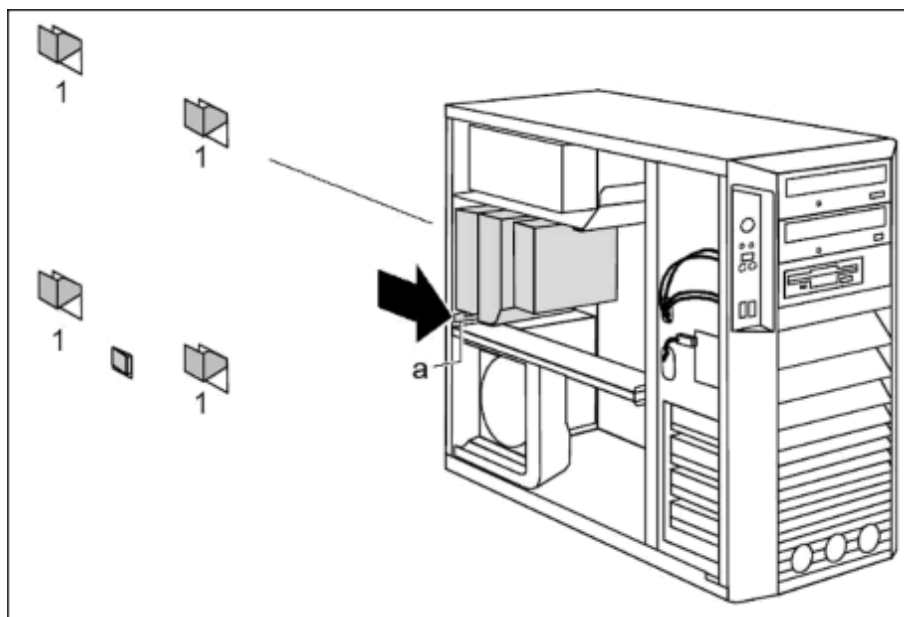
- Lock the side fan by pressing the locking hook (a) downward.
- Close the housing.

1.10.4.3 Ventilation Duct and Rear Fan

Removing the ventilation duct and the rear fan

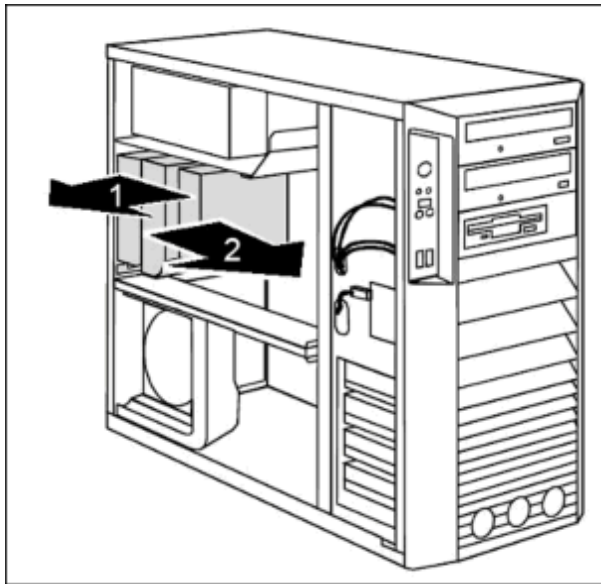
Before you can upgrade the main memory or the processor, or before you can replace the processor or the lithium battery, you must remove the ventilation duct.

Fig. 39: Removing the ventilation duct (1)



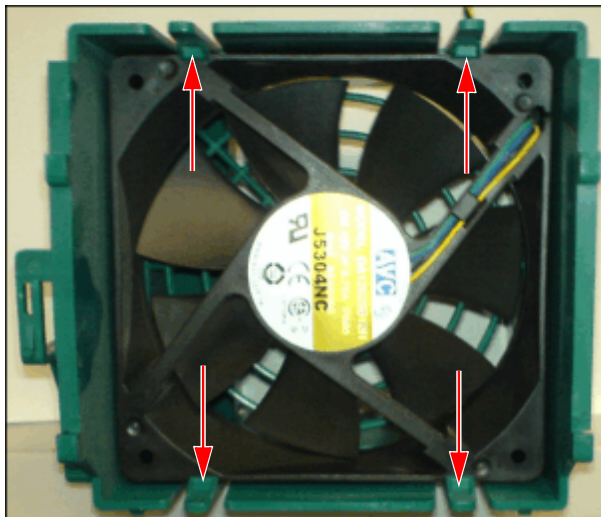
- Open the housing.
- Press the lever (a) in the direction of the arrow until the lever unhooks. The hooks (1) on the back of the housing will release.
- Carefully unhook the hooks (1).

Fig. 40: Removing the ventilation duct (2)



- Pull the ventilation duct and the fan a few millimeters (1) to release the hooks at the back of the housing.
- Remove the ventilation duct and the fan from the housing in the direction of the arrow (2).
- Remove the fan cable from the main board and take note of where it was plugged in (refer to (→ Fig. 31 Page 48), "Rear fan").
- Remove the rear fan from the ventilation duct by pushing the 4 hooks in the direction of the arrows:

Fig. 41: Rear fan (M440/450)



When removing the ventilation duct, be careful not to damage the processor cooler(s) on the mainboard. If necessary, disconnect the cable to the mainboard.

Installing the rear fan and the ventilation duct

- Connect the fan cable to the main board at the connection point you noted before (refer to (→ Fig. 31 Page 48), "Rear fan").

Fig. 42: Air flow of the rear fan



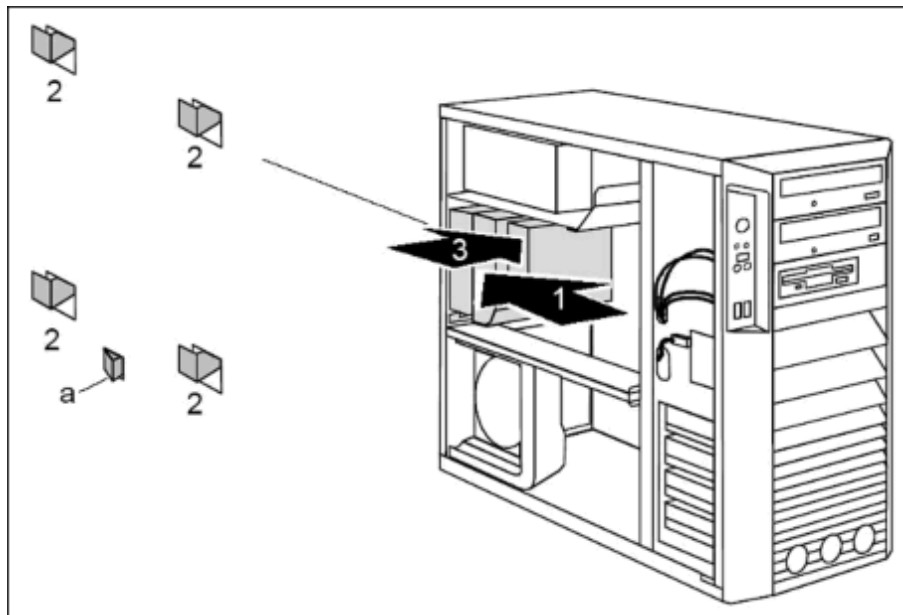
- Mount the rear fan in the ventilation duct (4 hooks).
- Pull the fan cable off the fan before installing the ventilation duct in the device.
- Connect the fan cable to the corresponding connector on the main board (see also the technical manual for the main board).



When routing the fan cable, make sure that it is not kinked or pinched.

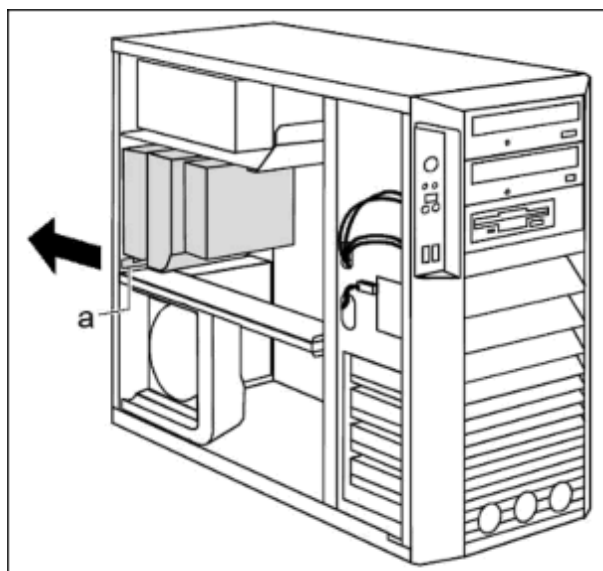
When fitting the ventilation duct, be careful not to damage the processor cooler(s) on the mainboard.

Fig. 43: Installing the ventilation duct (1)



- Place the ventilation duct and the fan in the housing at a slight angle in the direction of the arrow (1). The hooks of the ventilation duct must hook into the openings (2) in the back of the housing.
 - Push cables aside if they get in the way.
 - Press the ventilation duct and the fan into the housing in the direction of the arrow (3) until the ventilation duct engages noticeably.
- The fan is properly engaged when the locking lug (a) protrudes slightly at the back of the housing.
- Connect the fan cable to the fan.

Fig. 44: Installing the ventilation duct (2)



- Press the lever (a) of the ventilation duct in the direction of the arrow until the lever (a) hooks in.

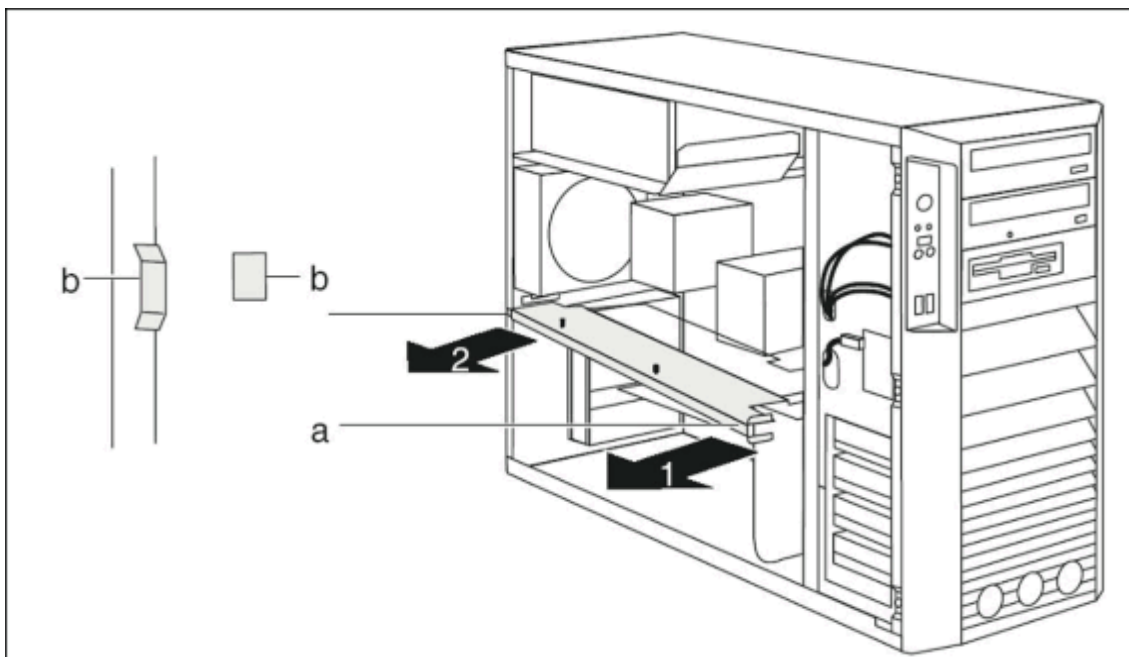
- Close the housing.

1.10.4.4 Crosspiece

Removing the crosspiece

Before replacing the main memory, you will need to remove the crosspiece.

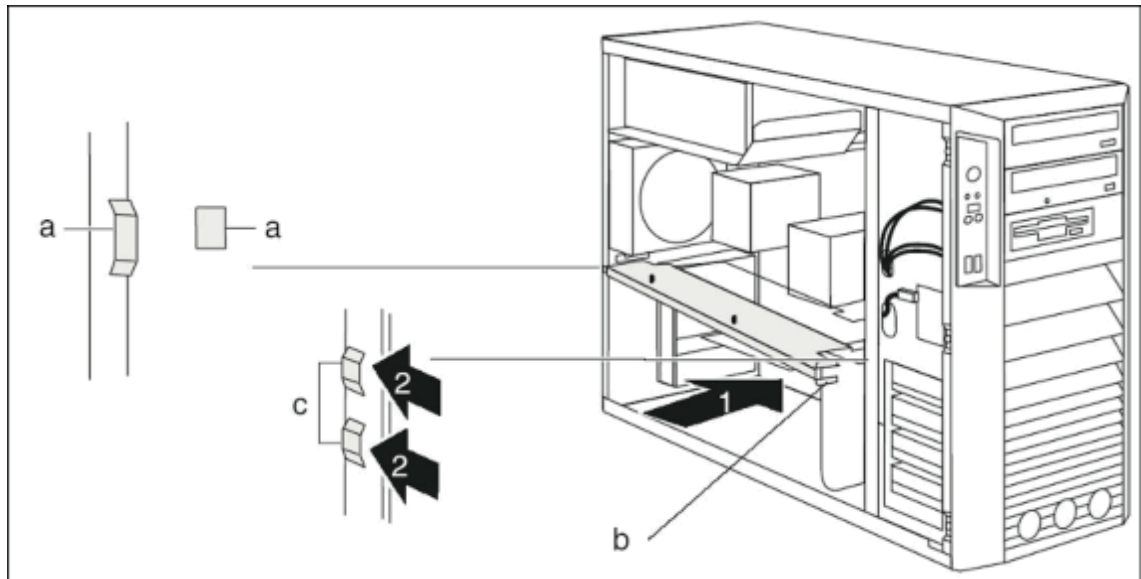
Fig. 45: Removing the crosspiece



- Open the housing.
- Remove the ventilation duct.
- Remove the side fan.
- Grasp the crosspiece by the lug (a) and unhook it from the brackets in the housing.
- Pull the crosspiece slightly out of the housing in the direction of the arrow (1).
- Detach the crosspiece from the brackets (b) on the back of the housing.
- Remove the crosspiece from the housing in the direction of the arrow (2).

Installing the crosspiece

Fig. 46: Installing the crosspiece



- Insert the crosspiece into its brackets (a) on the back of the housing at a slight angle.



Make sure that the Plexiglas plate of the crosspiece is not pinched between the memory slots.

- Press the crosspiece into the housing in the direction of the arrow (1) until the lugs (b) of the crosspiece hook into the brackets (c) in the housing.
- Install the side fan.
- Install the ventilation duct.
- Close the housing.

1.10.5 Power Supply

1.10.5.1 Preparatory work steps

- Open the housing.
- Remove the ventilation duct.
- Remove the crosspiece.

Fig. 47: M470 D2778 power connectors

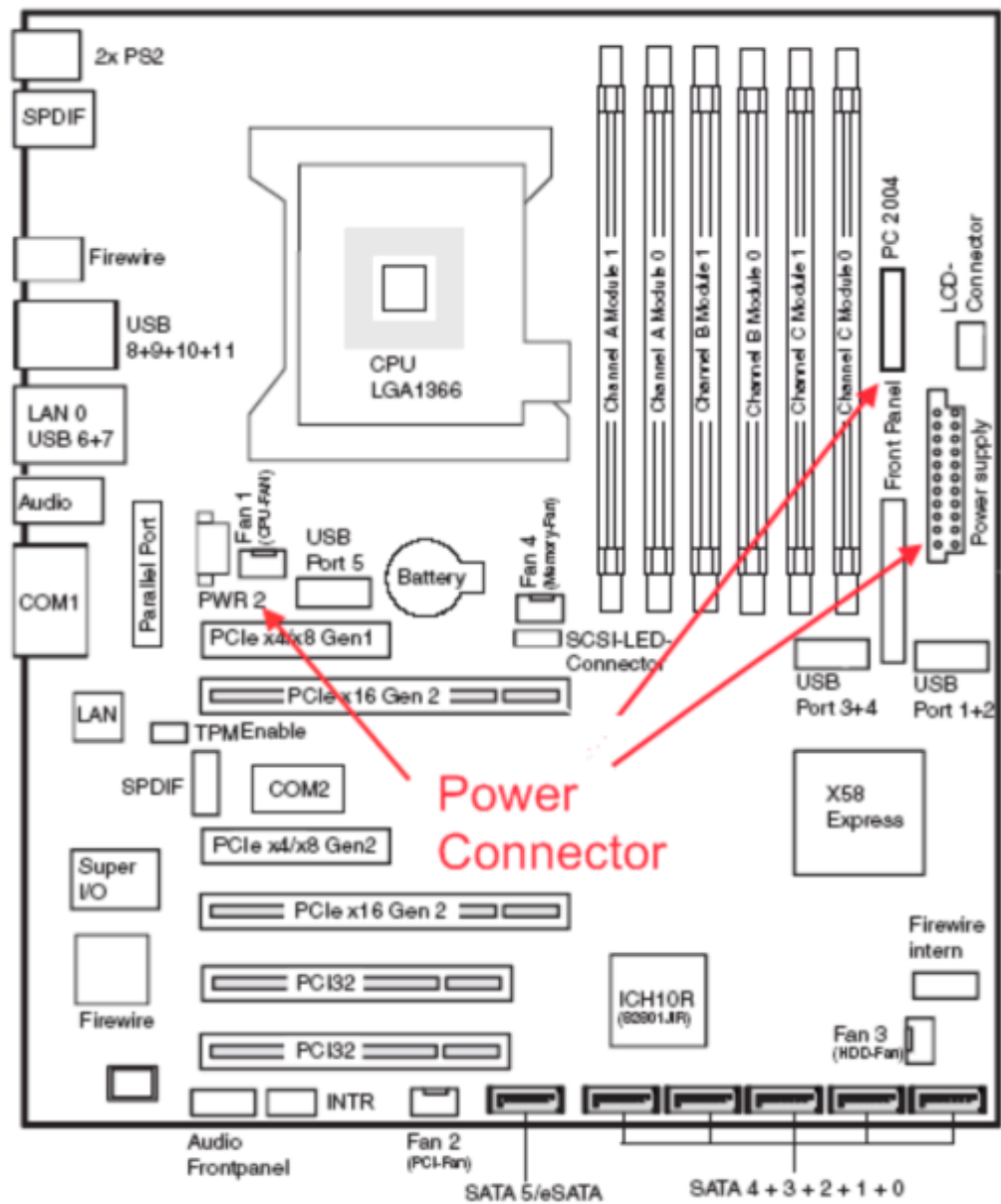
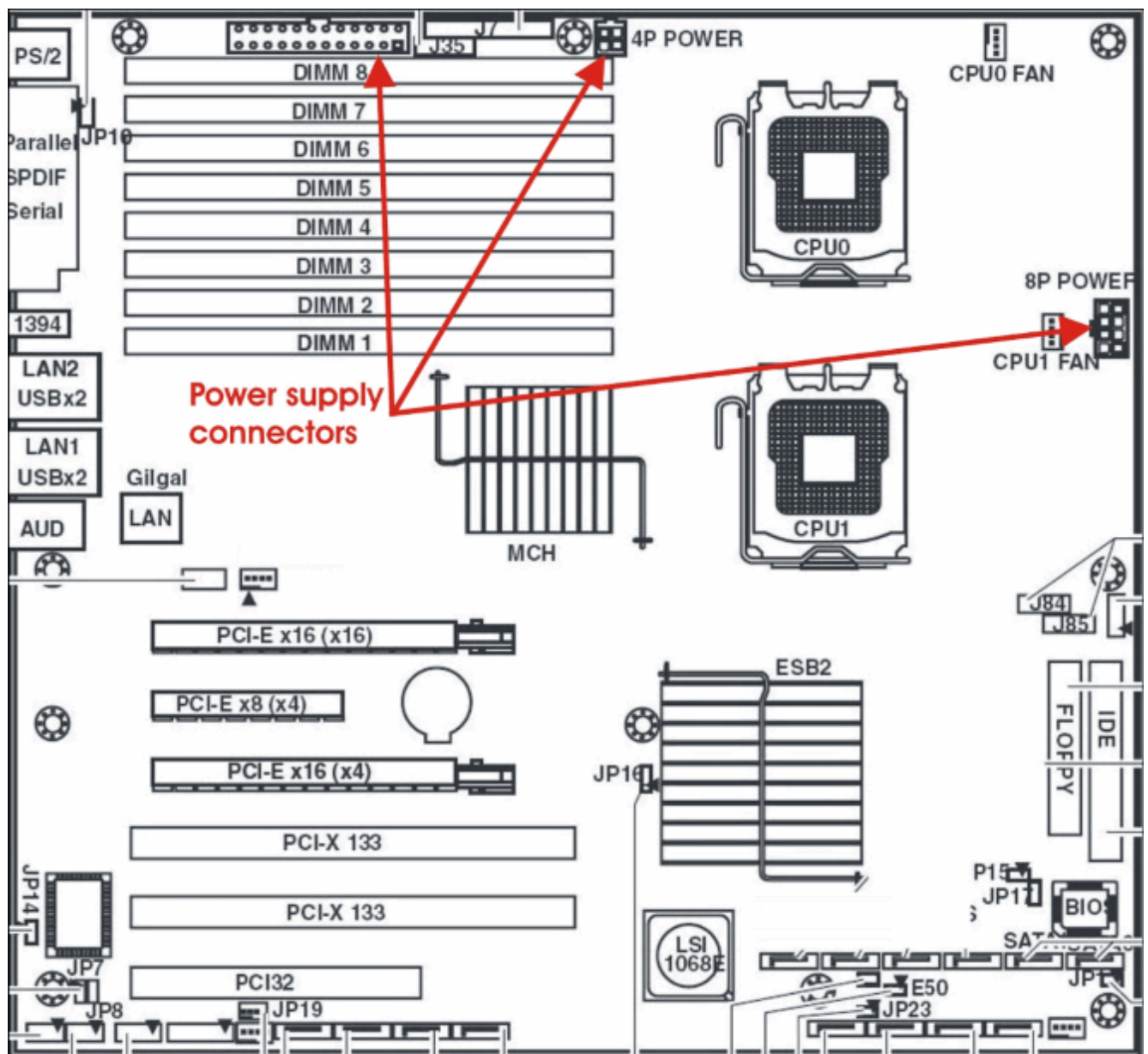
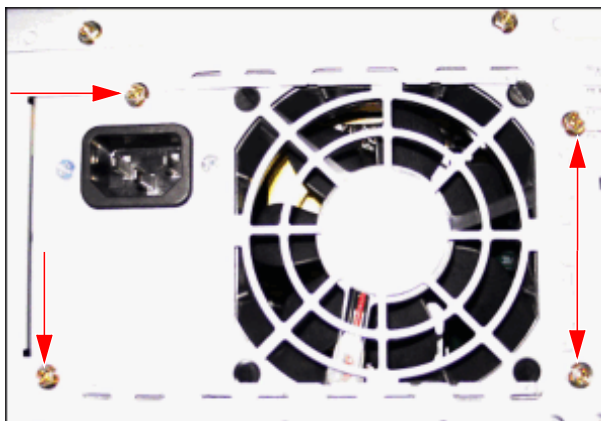


Fig. 48: Mainboard / power supply



1.10.5.2 Changing the power supply

Fig. 49: Power Supply (1).jpg



- Make notes about the power connectors (main board (refer to (→ Fig. 31 Page 48), drives and graphics board) and the ducting of the power supply, and disconnect all of them.
- Remove the 4 screws (arrows) on the back of the power supply.
- Pull the power supply out from the case laterally.

Fig. 50: Pillar



- Remove the pillar (2 screws) and mount it on the new power supply.
- Move the new power supply into the housing and attach it with the 4 screws on the back.
- Connect all power connectors and attach the cables as they were before.

1.10.5.3 Final Steps

- Install the crosspiece.
- Install the ventilation duct.

- Close the housing.

1.10.6 Drives

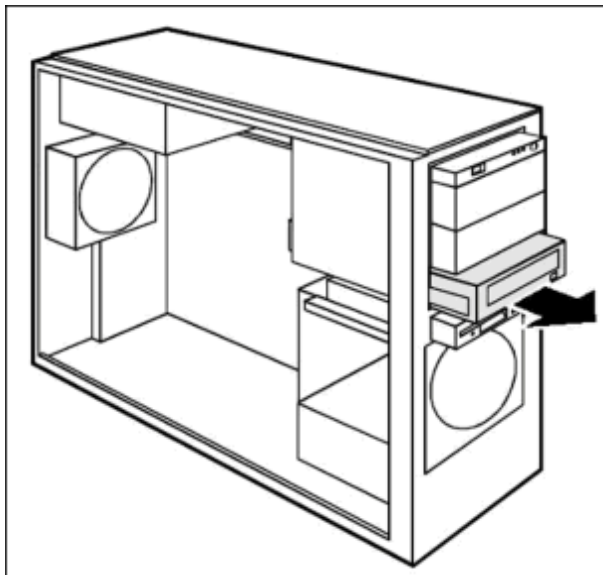
1.10.6.1 Preparatory Work Steps

- Open the housing.
- Remove the front panel.

1.10.6.2 CD/DVD drive

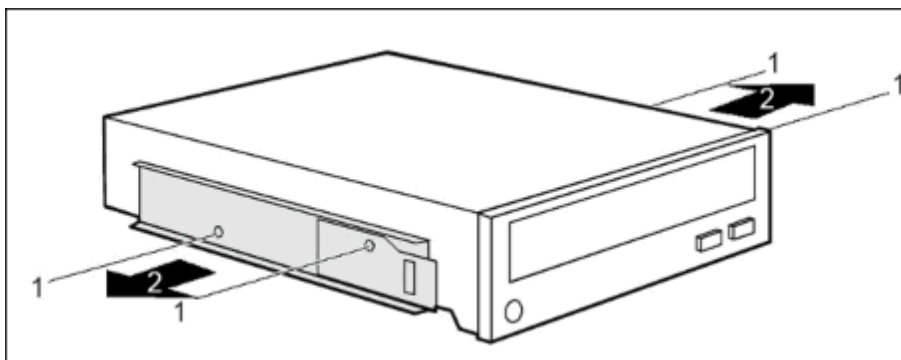
Removing a CD/DVD drive

Fig. 51: Removing a CD/DVD drive (1)



- Pull the data and power supply connectors from the desired drive.
- Pull the drive out of the housing.
- If necessary, make the required settings on the remaining hard disk drive.

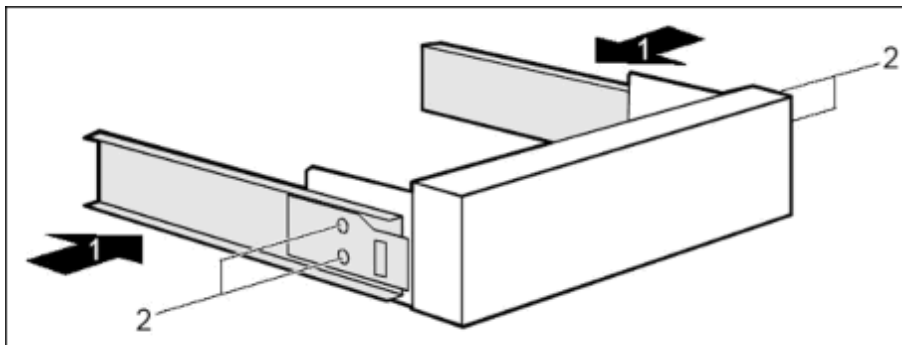
Fig. 52: Removing a CD/DVD drive (2)



- Loosen the screws (1).

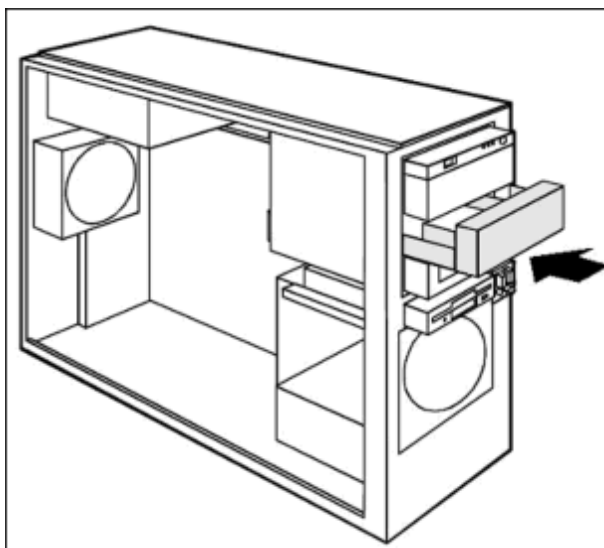
- Remove the rails from the drive (2).

Fig. 53: Removing a CD/DVD drive (3)



- Lay the rails onto the blanking plate in such a way that their holes align with the screw holes (1).
- Fasten the rails with the screws (2).

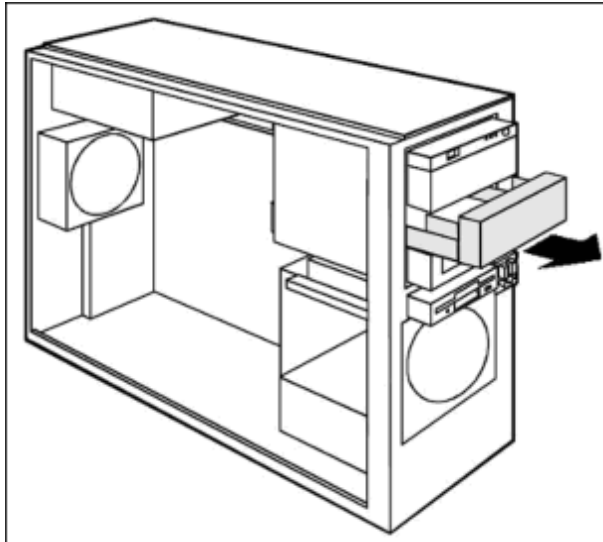
Fig. 54: Removing a CD/DVD drive (4)



- Slide the empty slide-in module into the housing.

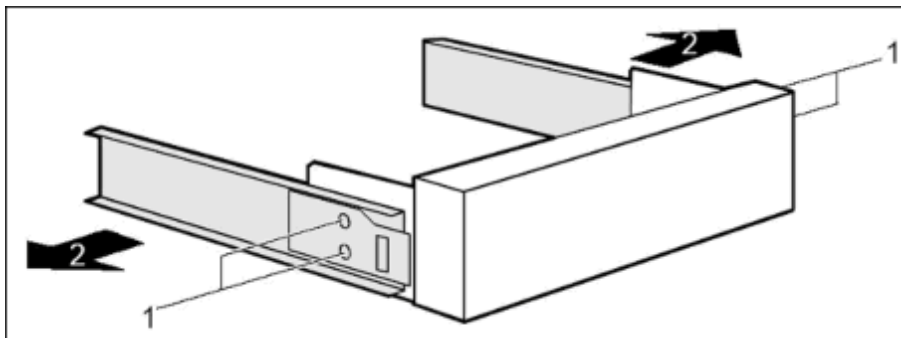
Installing a CD/DVD drive

Fig. 55: Installing a CD/DVD drive (1)



- Pull the empty slide-in module out of the housing.

Fig. 56: Installing a CD/DVD drive (2)



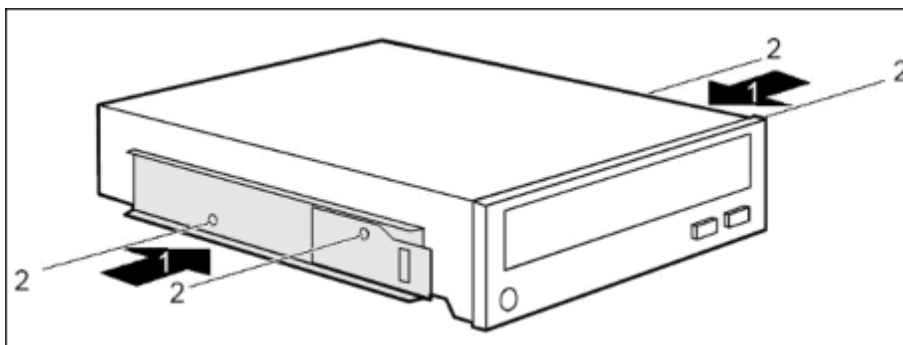
- Loosen the screws (1).
- Remove the rails from the cover (2).



Do not throw away the cover of the empty slide-in module. For cooling, protection against fire, and in order to comply with EMC regulations, you must refit the empty slide-in module if you remove the drive again later.

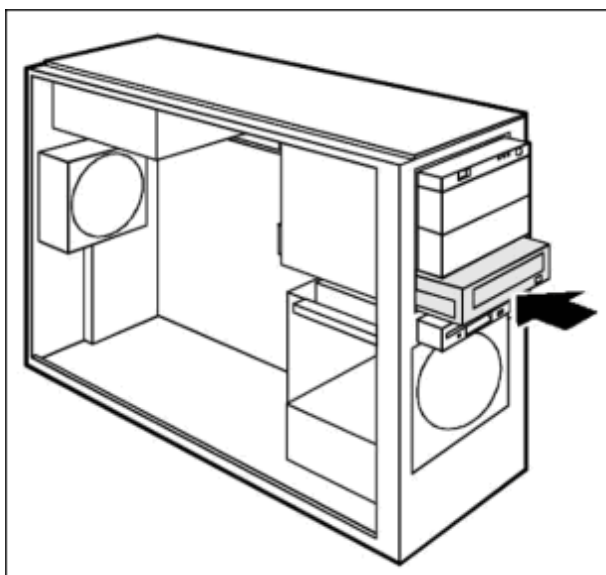
- Take the new drive out of its packaging.
- Adjust the required settings on the drive (if necessary, on already-installed drives as well).

Fig. 57: Installing a CD/DVD drive (3)



- Lay the rails onto the drive in such a way that their holes align with the screw holes (1).
- Fasten the rails with the screws (2).

Fig. 58: Installing a CD/DVD drive (4)



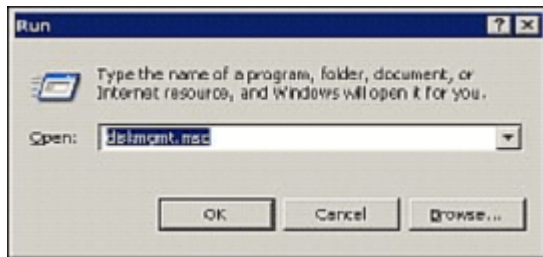
- Slide the accessible drive into the housing.
- Plug the data and power supply connectors into the drive. Make sure the polarity is correct.

Changing the drive letter

- Log in as administrator
- Use the Explorer to check that the CD-writer and DVD-drive still have the proper drive letters. These are:
 - DVD-drive -> R
 - CD-writer -> S

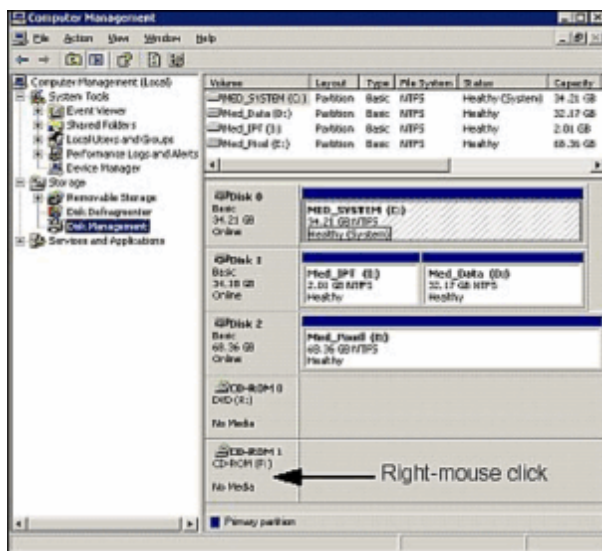
- If the drive letters need to be changed ->
Start the disk manager by selecting START --> RUN and entering diskmgmt.msc.

Fig. 59: Run diskmgmt.msc



- The disk manager pops up.

Fig. 60: Disk manager



- Check that the drive letters for the CD writer and DVD-drive are correct using the "Windows file explorer" (DVD-burner drive letter "S" and DVD-ROM drive letter "R"). (If either of the drive letters is wrong, change the drive letter with the disk manager as follows: Click **Start; Run; diskmgmt.msc** ↵ ; **right click** the drive to be corrected; select **Change Driveletters and Path...**, select the proper drive letter in the drop-down menu, confirm the popup with **Yes**, repeat this sequence until you achieve the correct drive letter assignment)

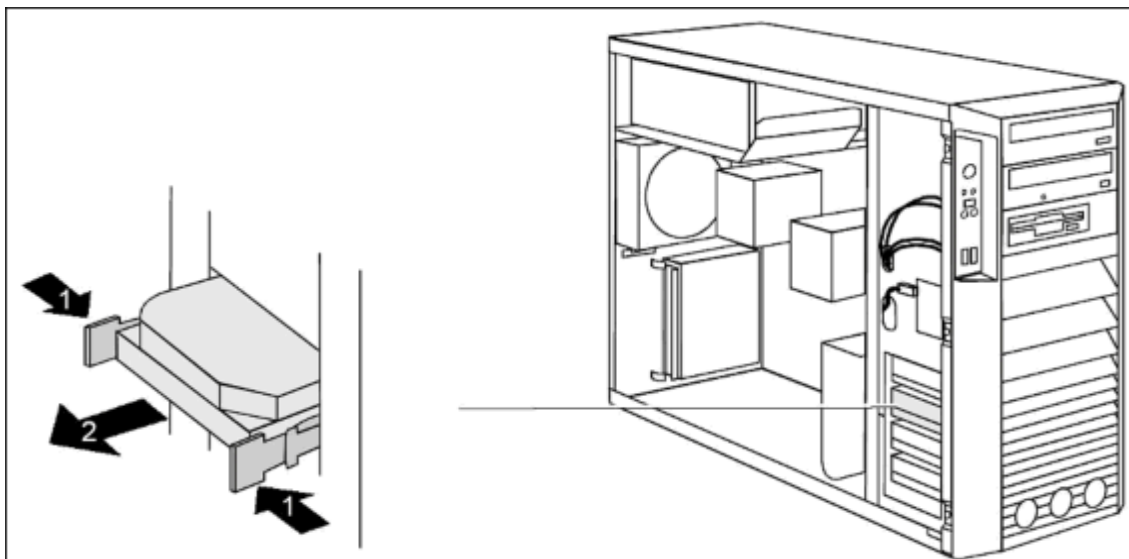
1.10.6.3 Hard Disk Drive



Hard disk replacement is not supported for MR hosts.

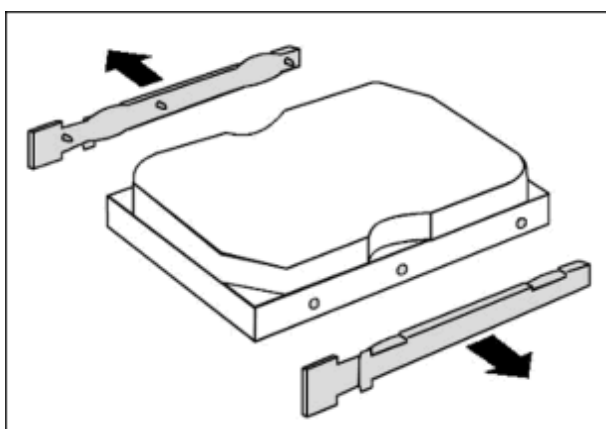
Removing a hard disk drive

Fig. 61: Removing a hard disk drive (1)



- Disconnect the data and power supply cables of the hard disk drive.
- Press the EasyChange rails in the direction of the arrows (1).
- Pull the hard disk drive out of the carrier (2). Set the jumpers as they were before.

Fig. 62: Removing a hard disk drive (2)



- Pull the EasyChange rails off the hard disk drive.
- If necessary, make the required settings on the remaining hard disk drive.

1.10.6.4 Final work steps

- Close the front panel.
- Close the housing.

1.10.7 Lithium Battery



CAUTION

The BIOS update may result in a loss of patient data.

- » Check the specific system documentation for details concerning the BIOS installation.
- » If necessary, save patient data before starting the BIOS installation.
- » On some systems, a BIOS installation requires a complete SW installation

1.10.7.1 Changing the lithium battery

In order to permanently save the system information, a lithium battery is installed to provide the CMOS memory with a current. A corresponding error message notifies the user when the charge is too low or the battery is dead. The lithium battery must then be replaced.



CAUTION

Incorrect replacement of the lithium battery may lead to a risk of explosion.

- » The lithium battery may be replaced only with an identical battery or with a type recommended by the manufacturer, following the instructions below.



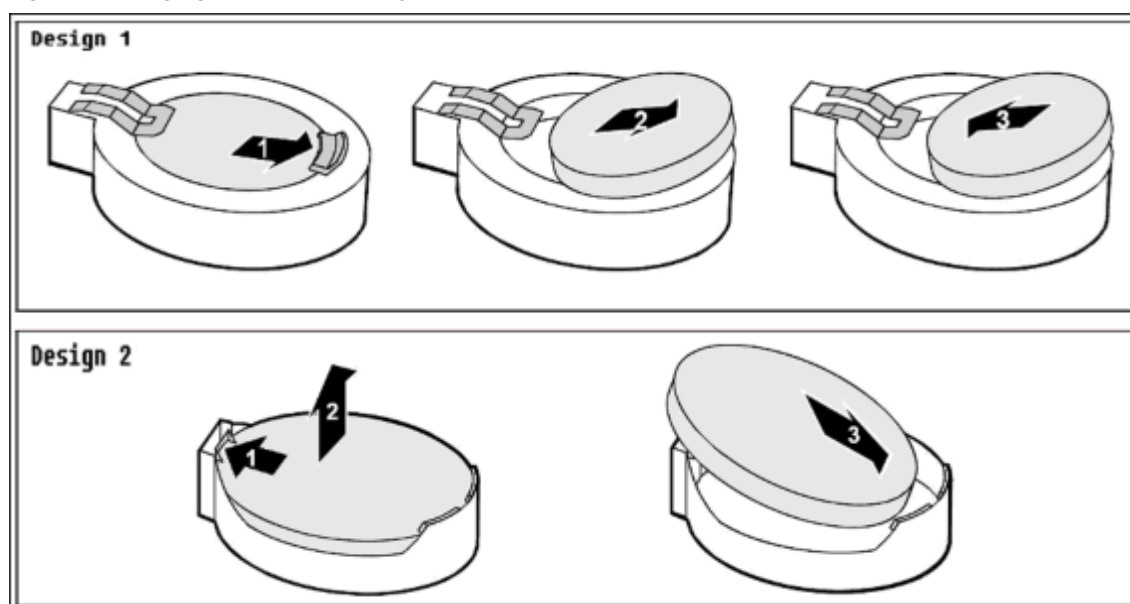
Do not throw lithium batteries into household waste. They must be disposed of in accordance with local regulations concerning hazardous waste.



Ensure that you observe the correct polarity when replacing the lithium battery. The plus pole must be on top.

- Open the housing.
- There are several different lithium battery holder designs; all of them function in a similar way.

Fig. 63: Changing the lithium battery



- Press the catch in the direction of the arrow (1).
 - » The battery jumps out of the holder slightly.
- Remove the battery (2).
- Push the new lithium battery of the identical type into the holder and press it downward until it engages (3).



Restoring BIOS settings is only necessary after battery replacement for older imager HW and SW. All new imagers default BIOS setting are customized for use without any manual configuration.



When the battery is dead, the BIOS is set to factory default.

In addition to activating the system ON/OFF switch, the following may be necessary: - Turn OFF the PC via the power switch on the rear panel of the PC - Turn ON the PC via the rear power switch and press the push button on the front panel of the PC

1.10.7.2 Final work steps

- Close the housing.
- Perform BIOS setup according to the specific system documentation (only valid for older imager HW and SW).
- Check that the system boots properly.
- Perform the quality assurance procedure: **Image Orientation** in the service platform.
- Make sure that the system is generating images.

1.10.8 Dongle

1.10.8.1 Required time and tools

Time

- 60 min

Tools

- Licence file with all customer licenses for the new DONGLE
- Valid service license for the new DONGLE

1.10.8.2 Preliminary

If the service license and all customer options are invalid, the dongle is most likely defective.



Place dongle orders only through TDHSC1 in Erlangen, a dongle defect must be directly escalated to the TDHSC1 in Erlangen, phone number:+49 9191-188080

1.10.8.3 Disassembly

Ensure that the Host is switched off; replace the USB dongle in back of the Host

1.10.8.4 Assembling

- Switch on the system

1.10.8.5 Final checks

- Login as user **administrator**, start the Internet Explorer, start Local Service, select **Configuration**, Service, **Licensing**, Click radio button **Browse to select the license file**, navigate to the media with your new license file, click **Save**.
- Re-boot the system, start the MR application
- Check with the customer that all options are available
- Start the Local service, select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master Backup**

1.10.9 In-Room MRC

1.10.9.1 Required time and tools

Time

- 120 min

Tools

- Set of standard tools

- TDOC Startup of the In-Room MRC

1.10.9.2 Final checks

- Run the auto configuration program and perform a visual evaluation according to the start-up of the In-Room MRC, Refer to TDOC start-up of the In-Room MRC

Protective conductor test



WARNING

In case of a fault, voltage could be present at the housing of components.

This may lead to severe injuries or death if the resistance of the ground wire is too high.

- » The protective conductor resistance must be $\leq 200 \text{ m}\Omega$. Measured against the LPD protective ground terminal, if a component replacement may affect the ground wire replacement.

For more detailed information, refer to "Protective conductor measurement" in the binder: "Technical Documents"; register: Safety ; chapter: MR-specific safety information.

1.11 Host HP Z400

1.11.1 General

1.11.1.1 Safety Information and/or references to Safety Information

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad (TDK equipment type 8005 from 3M Deutschland GmbH).



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. ground pin of the mains plug) and the protective earth wire connected to the "main component" (e.g. screw of the housing).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.



WARNING

Please follow the general safety guidelines (→ General Safety Notes / TD00-000.860.01) as well as the MR-specific safety information (→ MR-specific safety information / MR-000.860.01)!

Failure to comply with the safety guidelines may result in death or serious injury.

- » Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Compliance with the information provided is mandatory at all times.



CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

- » Ramp down the magnet or maintain a safe distance from the magnetic field.

1.11.1.2 Handling of patient data

The data protection guidelines specify the mandatory procedures for handling patient data.

Data carriers, storage devices, or other media (such as the hard disk inside the host PC) must not leave the customer site if there is a chance they still contain patient data.



The service report needs to list the activities performed to ensure the secure handling of patient data.

As a first step, the PC needs to be analyzed. Once a defect in any part of the PC hardware is identified as the cause of the problem, the part replacement follows.

For PCs there are two cases:

1. The PC does not contain any patient data. - The individual product documentation needs to list hard disks without patient data. (→ No patient data on hard disks / Page 74)
2. The PC contains patient data.
The disks need to be returned without patient data.

Please decide how to proceed for your individual case. The recommendation:

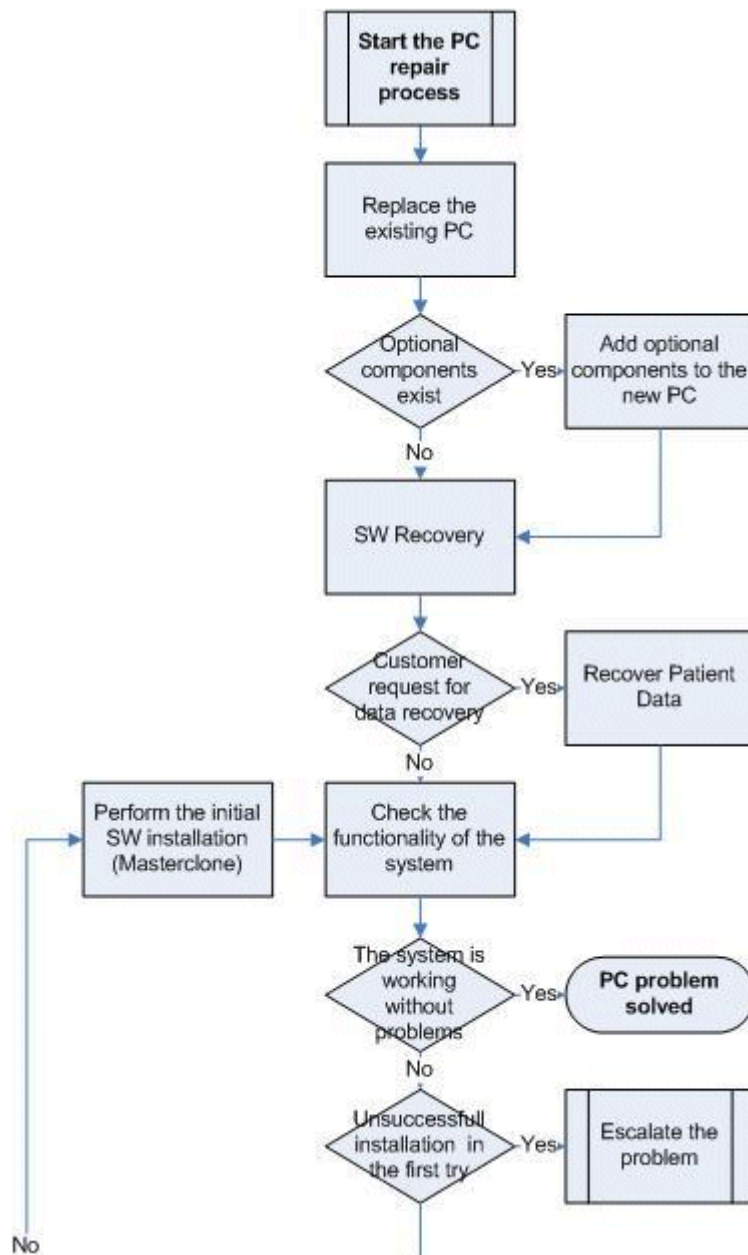
1. Case: Valid if the new PC has the same material number as the defective PC being replaced. The existing disk has to be removed from the host computer that is going to be returned. This disk must be installed in the replacement host computer. At the same time, the new disk that was just removed has to be installed in the defective host PC that is being returned. (→ Swap the disks / Page 75)
2. Delete the patient data with the "blancco" tool. (→ Deleting Patient Data / Page 78)
3. Case: All hard disks potentially containing patient data are removed from the defective PC and remain at the customer site.

No patient data on hard disks



Only continue with the parts replacement procedure outlined in the flow chart below if you have confirmation from the product-specific documentation that the PC contains no patient data, and that no patient information is contained in any logfiles on the PC.

Fig. 64: Replacing a PC without patient data



Swap the disks

The preferred case is replacing the hard disk: This case entails additional activities and this will surely take some time. - This time span may be compensated for by the time required for the restore to be performed. CSML confirms that such an action as opening the housing of the host computer and performing such activities will not negatively influence any warranty rights or claims.

Such a course of action, however, may provide the following benefits:

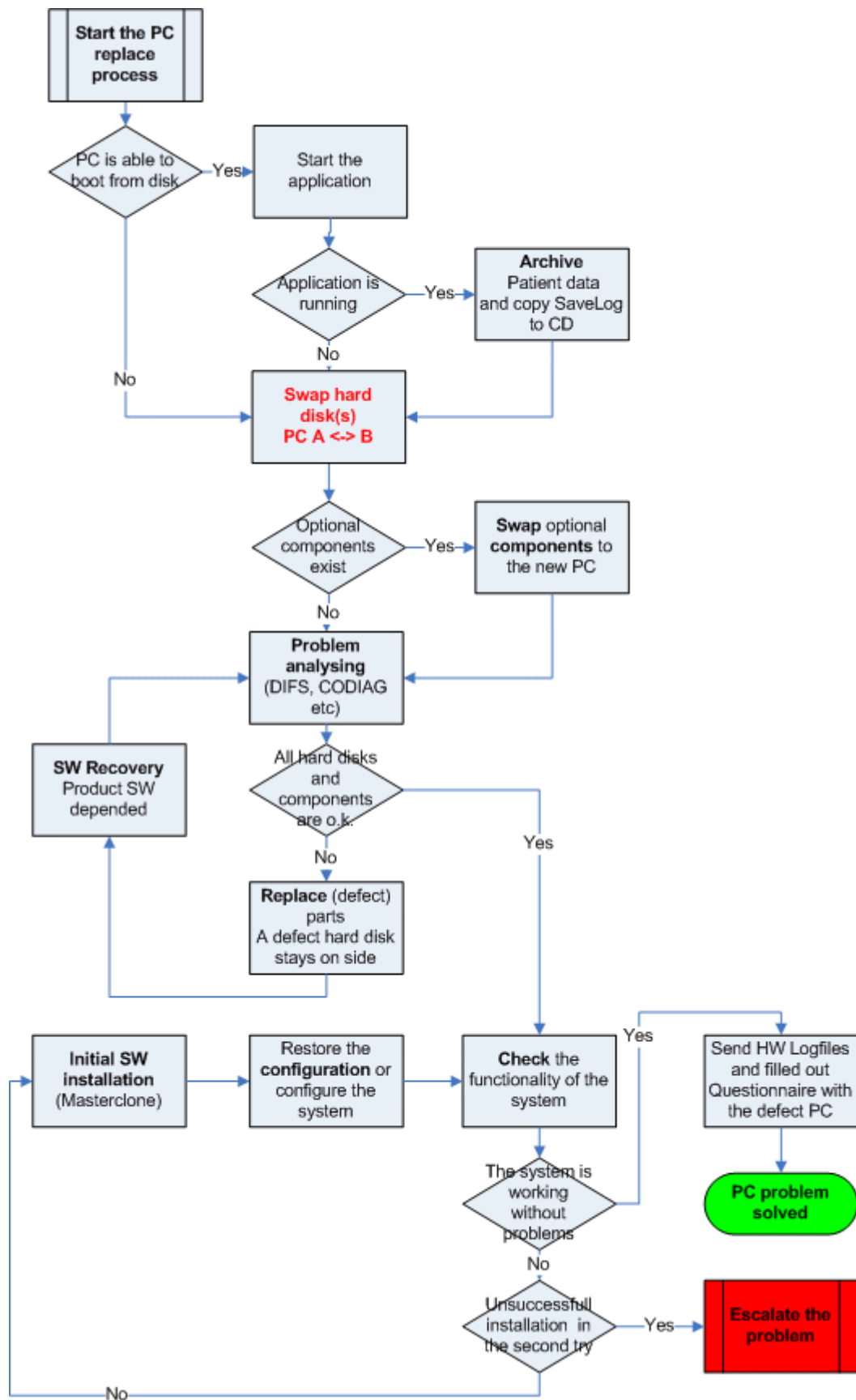
- The returned host computer will not contain any patient data.

- The disk swapped to the new PC will run without need of the rebuild/restore function.
- The customer may even have the benefit of reduced downtime.
- No reconstruction or loading of images from the archive will be necessary (all patient data will be available).

The decision in such cases will be made by the attending CSE or the responsible service manager.

The work steps are shown in the following flowcharts:

Fig. 65: Replacing a PC with patient data



Deleting Patient Data

Since the end of 2013, there is a tool to delete data from hard disk. The tool needs to be used in cases where replacement of the disk from the defective to the new PC is not possible.

The principles of "Return Goods Management" require that all storage devices containing personal patient data, which are to be removed from the site (i.e. spare parts returns), shall have all data wiped clean (securely deleted) or the storage device shall be handed over to the customer when deletion is not possible.

The tool is called "blancco". The tool can be downloaded from the Siemens Intranet.

The certified blancco tool permanently removes data from IT assets such as PCs, servers and data center equipment.

Data erased using blancco cannot be recovered with any existing technology. This helps to have all data wiped clean (securely deleted) and achieve compliance with existing "patient data handling" regulations.

SIEMENS Employees: The tool is available as an iso image from the CS Home page

<https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>

CS Home page -> Tools -> blancco - Erasure Software

For others see e.g.: <http://blancco.com/en/products/>

The time required by the procedure depends on the size of the hard disk.

Warning: before you start using the blancco software, make sure that all files, folders, software applications or any other information that you want to save for later use are backed up on an appropriate media device other than the original hard drive. Data that has been erased from the hard disk by this program cannot be recovered by any existing method.



Selecting the wrong hard disk can result in lose of data with the use of blancco. Be sure that the selected hard disk for deletion is the correct one.

Creating the blancco CD

An ISO image of the blancco CD is available for download from: CS Home page -> Tools -> blancco (CS Home page, <https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>)

1. Download the ISO image.
2. Label the blank CD as "blancco - Erasure Software"
3. Insert the CD into the drive of your service computer.
4. Use a standard burn program (such as Nero) to open the file and burn the CD image to the CD.
5. Remove the CD from the drive.

Using the blanco CD

1. Booting with blanco

1. Insert the "blanco - Erasure Software" CD into the PC with the drive(s) to be deleted

2. Switch off the PC.



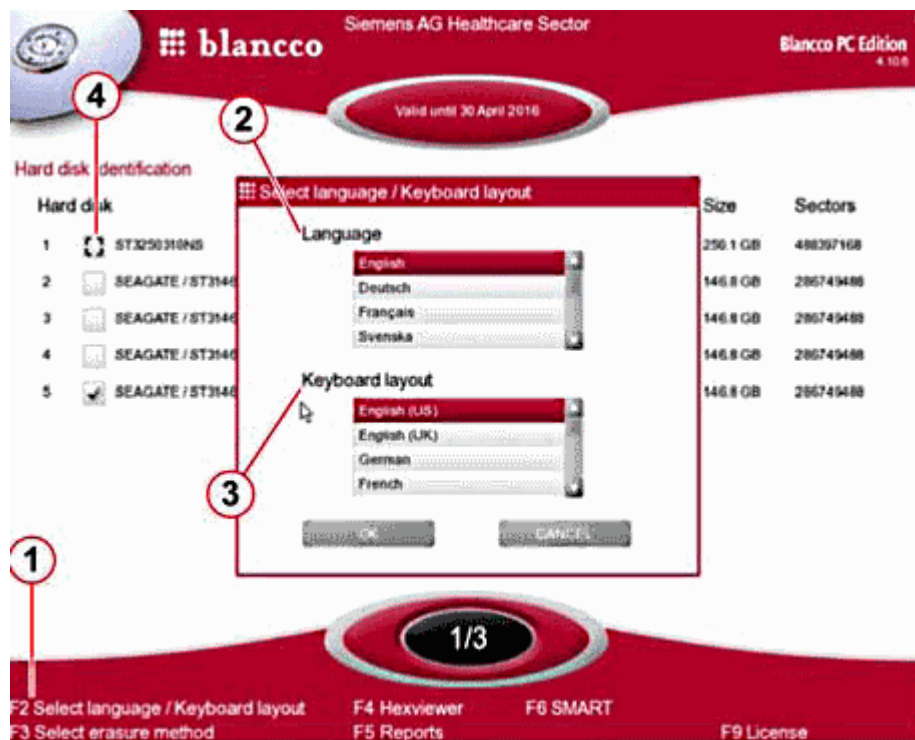
If this is an IAS-A PC, disconnect the service backup drive to prevent accidental loss of data.

3. Switch on the computer power to boot the system from the CD (if necessary, change the BIOS boot priority). The blanco Home screen is visible with all identified hard disks of the PC



If necessary, press F2 to change the language and the keyboard layout.

Fig. 66: Blanco language/keyboard layout



- [1] F2 to open menu
- [2] Language
- [3] Keyboard layout
- [4] Checkboxes to select hard disks

Deletion with blanco

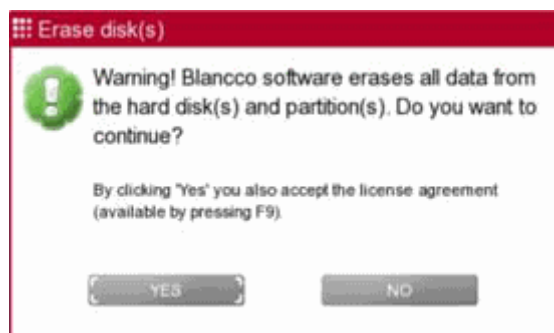
1. Select the hard disks to be deleted.



blanco can erase up to 16 hard disks simultaneously.

2. Press Erase, a warning message appears.

Fig. 67: Warning message



3. Press Yes to start the erasure process. The erasure process starts and the progress and duration is visible.
4. Click on OK when the process is finished and the Erase successful message appears.
5. Insert a memory stick to save the report.
6. 6. Press Save or F10. The report is saved to the memory stick.
7. Press Ok. The process is finished and the PC can be turned off.

1.11.2 FRU overview

FRU	Time	Instructions	Final checks	Tools
Host	180 min		Perform a measurement, view the images produced. Check with the customer that all options are available	Field service grounding kit (ESD), set of standard tools
DWD-ROM / DVD-RW	45 min	Drives	Burn some test images onto CD-RW or DVD media with the new DVD-RW drive or read Service instructions from a Service DVD from the Service GUI Help > Documentation	Field service grounding kit (ESD), set of standard tools
Battery	60 min	Lithium battery	Perform BIOS setup according to TDOC Installation Software. Perform a measurement and view the images produced.	Field service grounding kit (ESD), set of standard tools
Dongle (USB)	60 min		Check with the customer that all options are available Perform a measurement and view the images produced.	Service license/Application licenses. The Service/Applications licenses generated in relation to the flexid of the new dongle

1.11.3 Host

1.11.3.1 Required time and tools

Time

- 180 min

Tools

- Set of standard tools

1.11.3.2 Preliminary

- If the condition of the "Host" allows access to the "Service platform" (Service Software, e.g., as user administrator), select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master Backup**
- Shutdown and switch off the system.

1.11.3.3 Disassembly



Remember to move the dongle from the old "host" to the new "host"

- Disconnect the main power plug, network drop cable(s), mouse, and keyboard cables.

1.11.3.4 Assembling

- Plug the HW dongle to the new Host.
- Plug in the network drop cable(s), mouse, keyboard cables, monitor, and main power plug.

1.11.3.5 Final checks

- Check if the new "Host" has the correct software version or install software on the new "Host" according to the technical documentation (TDOC) "Install Software" valid for your system.
- Restore "site specific data" according to TDOC "Install Software", valid for your system.
- Start the Local Service, select **Quality Assurance**, perform the Image Orientation "Quality Assurance" step



In case of connection problem with the MARS after replacing the HOST; the firewall settings must be disabled for the internal measurement network.

How to disable the firewall for the internal network:

- Login as user **administrator**
- Open a cmd tool and run **networkcardinstall /I1 /I2 ↵**
- Click **START -> Settings -> Control Panel**

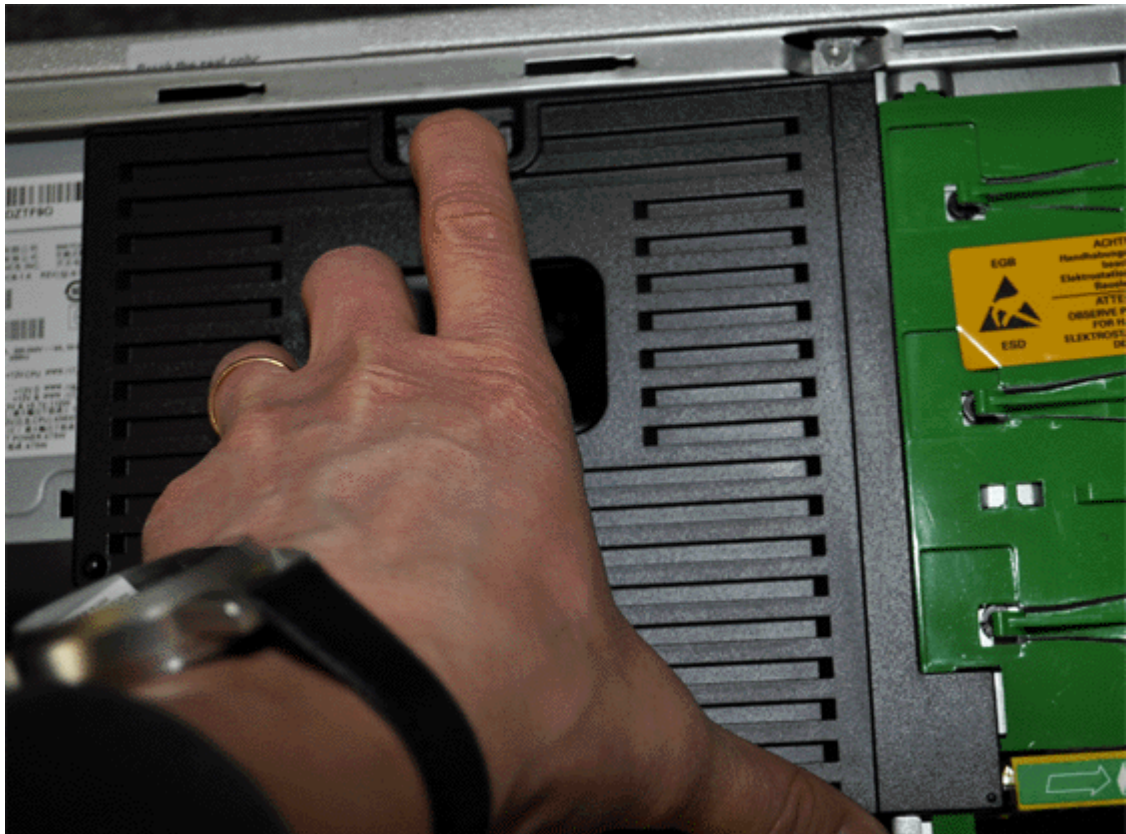
- Change the appearance of the Control Panel to classic view: Click the marked link **Switch to classic view**
- Double click the icon **Windows firewall**
- In the Windows Firewall configuration window select the tab card the icon **Advanced**
- If the Measurement (internal) network checkbox is selected, remove this checkbox
- Close the Windows Firewall configuration window by pressing **OK**
- Re-boot the system and login as user **meduser**

1.11.4 Drives

1.11.4.1 Preparatory Work Steps

- Open the housing.
- Remove the front panel.

Fig. 68: HP Z400 - release the airflow guide



- Remove the airflow guide.

1.11.4.2 CD/DVD drive

Removing a CD/DVD drive

- Pull the data and power supply connectors from the desired drive.

Fig. 69: HP Z400 - release the clip to unlock the DVD drive



- Release the clip and pull the drive out of the housing.
- If necessary, make the required settings on the remaining hard disk drive.
- Slide the empty slide-in module into the housing.

Installing a CD/DVD drive

- Take the new drive out of its packaging.
- Slide the accessible drive into the housing.
- Plug the data and power supply connectors into the drive. Make sure the polarity is correct.

1.11.4.3 Final work steps

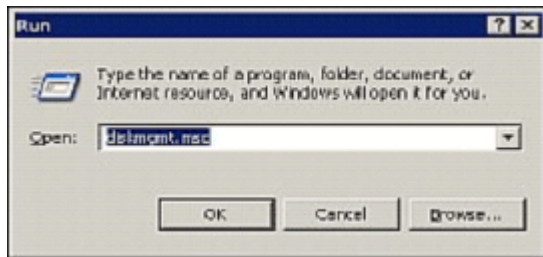
- Insert the airflow guide.
- Close the front panel.
- Close the housing.

Changing the drive letter

- Log in as administrator
- Use the Explorer to check that the CD-writer and DVD-drive still have the proper drive letters. These are:
 - DVD-drive -> R
 - CD-writer -> S

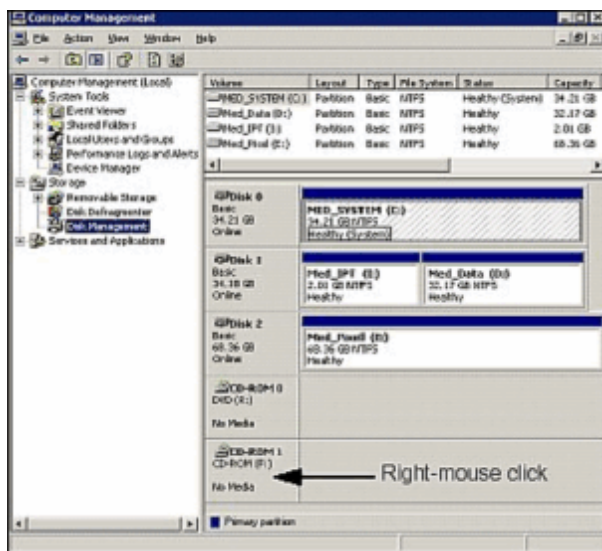
- If the drive letters need to be changed ->
Start the disk manager by selecting START --> RUN and entering diskmgmt.msc.

Fig. 70: Run diskmgmt.msc



- The disk manager pops up.

Fig. 71: Disk manager



- Check that the drive letters for the CD writer and DVD-drive are correct using the "Windows file explorer" (DVD-burner drive letter "S" and DVD-ROM drive letter "R"). (If either of the drive letters is wrong, change the drive letter with the disk manager as follows: Click **Start; Run; diskmgmt.msc** ↵ ; **right click** the drive to be corrected; select **Change Driveletters and Path...**, select the proper drive letter in the drop-down menu, confirm the popup with **Yes**, repeat this sequence until you achieve the correct drive letter assignment)

1.11.5 Lithium Battery



CAUTION

The BIOS update may result in a loss of patient data.

- » Check the specific system documentation for details concerning the BIOS installation.
- » If necessary, save patient data before starting the BIOS installation.
- » On some systems, a BIOS installation requires a complete SW installation

1.11.5.1 Changing the lithium battery

In order to permanently save the system information, a lithium battery is installed to provide the CMOS memory with a current. A corresponding error message notifies the user when the charge is too low or the battery is dead. The lithium battery must then be replaced.



CAUTION

Incorrect replacement of the lithium battery may lead to a risk of explosion.

- » The lithium battery may be replaced only with an identical battery or with a type recommended by the manufacturer, following the instructions below.



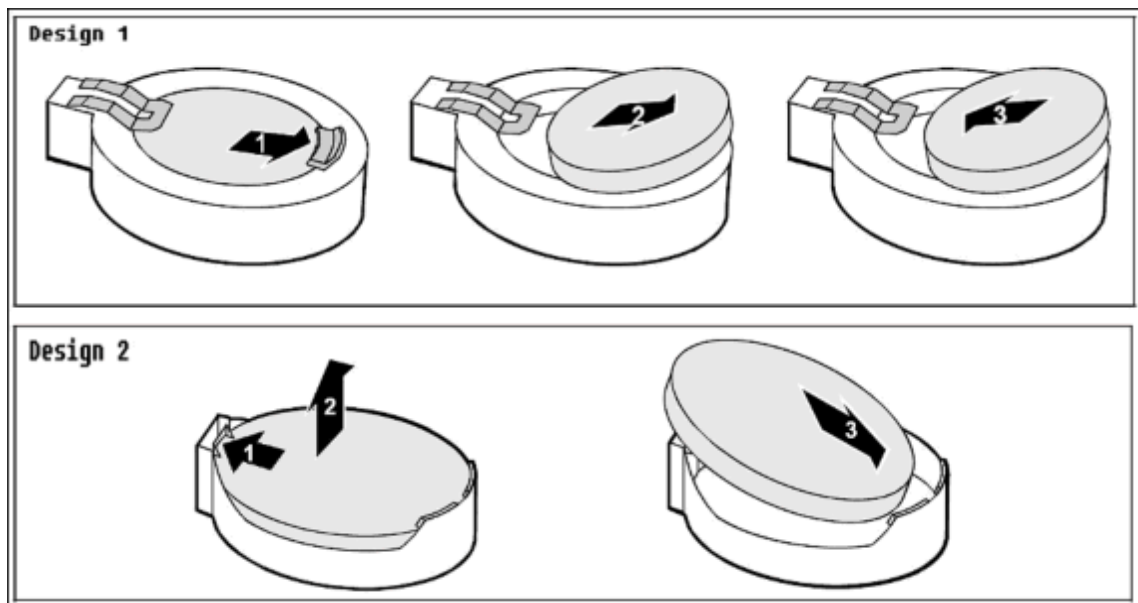
Do not throw lithium batteries into household waste. They must be disposed of in accordance with local regulations concerning hazardous waste.



Ensure that you observe the correct polarity when replacing the lithium battery. The plus pole must be on top.

- Open the housing.
- There are several different lithium battery holder designs; all of them function in a similar way.

Fig. 72: Changing the lithium battery



- Press the catch in the direction of the arrow (1).
 - » The battery jumps out of the holder slightly.
- Remove the battery (2).
- Push the new lithium battery of the identical type into the holder and press it downward until it engages (3).



Restoring BIOS settings is only necessary after battery replacement for older imager HW and SW. All new imagers default BIOS setting are customized for use without any manual configuration.



When the battery is dead, the BIOS is set to factory default.

In addition to activating the system ON/OFF switch, the following may be necessary: - Turn OFF the PC via the power switch on the rear panel of the PC - Turn ON the PC via the rear power switch and press the push button on the front panel of the PC

1.11.5.2 Final work steps

- Close the housing.
- Perform BIOS setup according to the specific system documentation (only valid for older imager HW and SW).
- Check that the system boots properly.
- Perform the quality assurance procedure: **Image Orientation** in the service platform.
- Make sure that the system is generating images.

1.11.6 Dongle

1.11.6.1 Required time and tools

Time

- 60 min

Tools

- Licence file with all customer licenses for the new DONGLE
- Valid service license for the new DONGLE

1.11.6.2 Preliminary

If the service license and all customer options are invalid, the dongle is most likely defective.



Place dongle orders only through TDHSC1 in Erlangen, a dongle defect must be directly escalated to the TDHSC1 in Erlangen, phone number:+49 9191-188080

1.11.6.3 Disassembly

Ensure that the Host is switched off; replace the USB dongle in back of the Host

1.11.6.4 Assembling

- Switch on the system

1.11.6.5 Final checks

- Login as user **administrator**, start the Internet Explorer, start Local Service, select **Configuration**, Service, **Licensing**, Click radio button **Browse to select the license file**, navigate to the media with your new license file, click **Save**.
- Re-boot the system, start the MR application
- Check with the customer that all options are available
- Start the Local service, select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master Backup**

1.12 Host HP Z420

1.12.1 Host Z420

1.12.1.1 General

Safety Information and/or References to Safety Information

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad (TDK equipment type 8005 from 3M Deutschland GmbH).



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. ground pin of the mains plug) and the protective earth wire connected to the "main component" (e.g. screw of the housing).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.



WARNING

Please follow the general safety guidelines (→ General Safety Notes / TD00-000.860.01) as well as the MR-specific safety information (→ MR-specific safety information / MR-000.860.01)!

Failure to comply with the safety guidelines may result in death or serious injury.

- » Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Compliance with the information provided is mandatory at all times.



CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

- » Ramp down the magnet or maintain a safe distance from the magnetic field.

Handling of patient data

The data protection guidelines specify the mandatory procedures for handling patient data.

Data carriers, storage devices, or other media (such as the hard disk inside the host PC) must not leave the customer site if there is a chance they still contain patient data.



The service report needs to list the activities performed to ensure the secure handling of patient data.

As a first step, the PC needs to be analyzed. Once a defect in any part of the PC hardware is identified as the cause of the problem, the part replacement follows.

For PCs there are two cases:

1. The PC does not contain any patient data. - The individual product documentation needs to list hard disks without patient data.(→ No patient data on hard disks / Page 90)
2. The PC contains patient data.
The disks need to be returned without patient data.

Please decide how to proceed for your individual case. The recommendation:

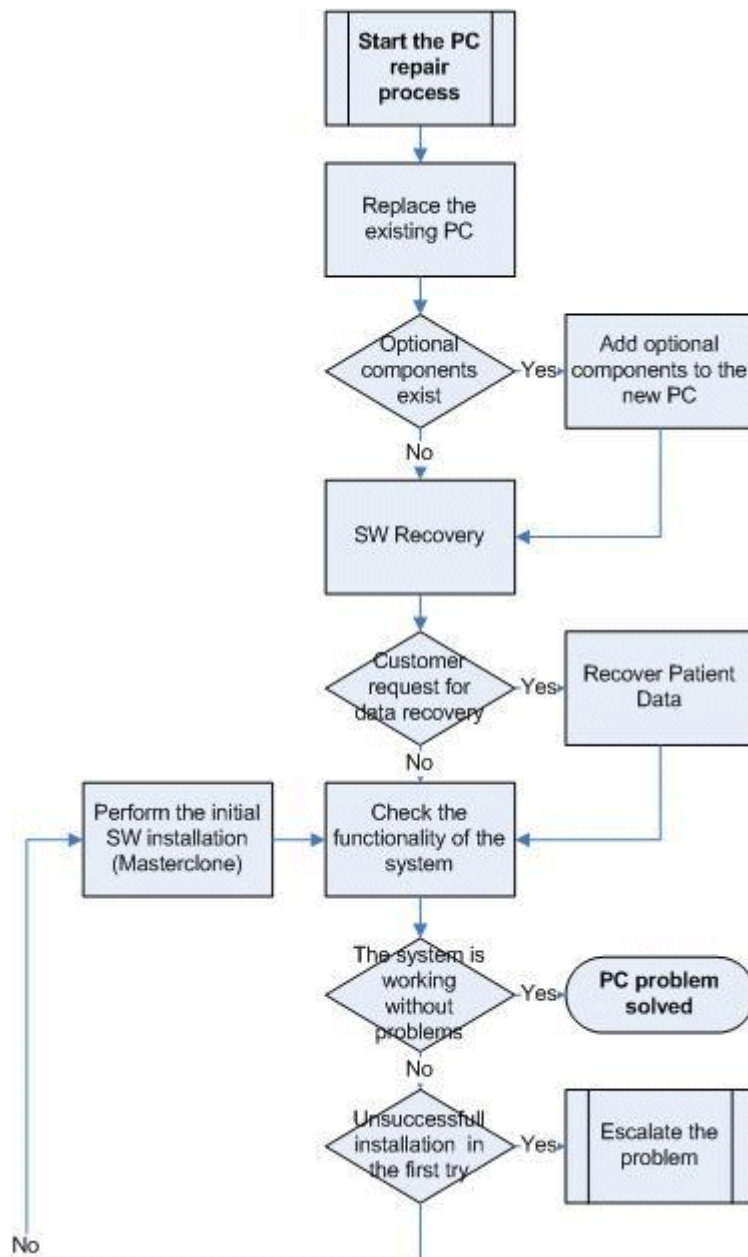
1. Case: Valid if the new PC has the same material number as the defective PC being replaced. The existing disk has to be removed from the host computer that is going to be returned. This disk must be installed in the replacement host computer. At the same time, the new disk that was just removed has to be installed in the defective host PC that is being returned. (→ Swap the disks / Page 91)
2. Delete the patient data with the "blancco" tool. (→ Deleting Patient Data / Page 94)
3. Case: All hard disks potentially containing patient data are removed from the defective PC and remain at the customer site.

No patient data on hard disks



Only continue with the parts replacement procedure outlined in the flow chart below if you have confirmation from the product-specific documentation that the PC contains no patient data, and that no patient information is contained in any logfiles on the PC.

Fig. 73: Replacing a PC without patient data



Swap the disks

The preferred case is replacing the hard disk: This case entails additional activities and this will surely take some time. - This time span may be compensated for by the time required for the restore to be performed. CSML confirms that such an action as opening the housing of the host computer and performing such activities will not negatively influence any warranty rights or claims.

Such a course of action, however, may provide the following benefits:

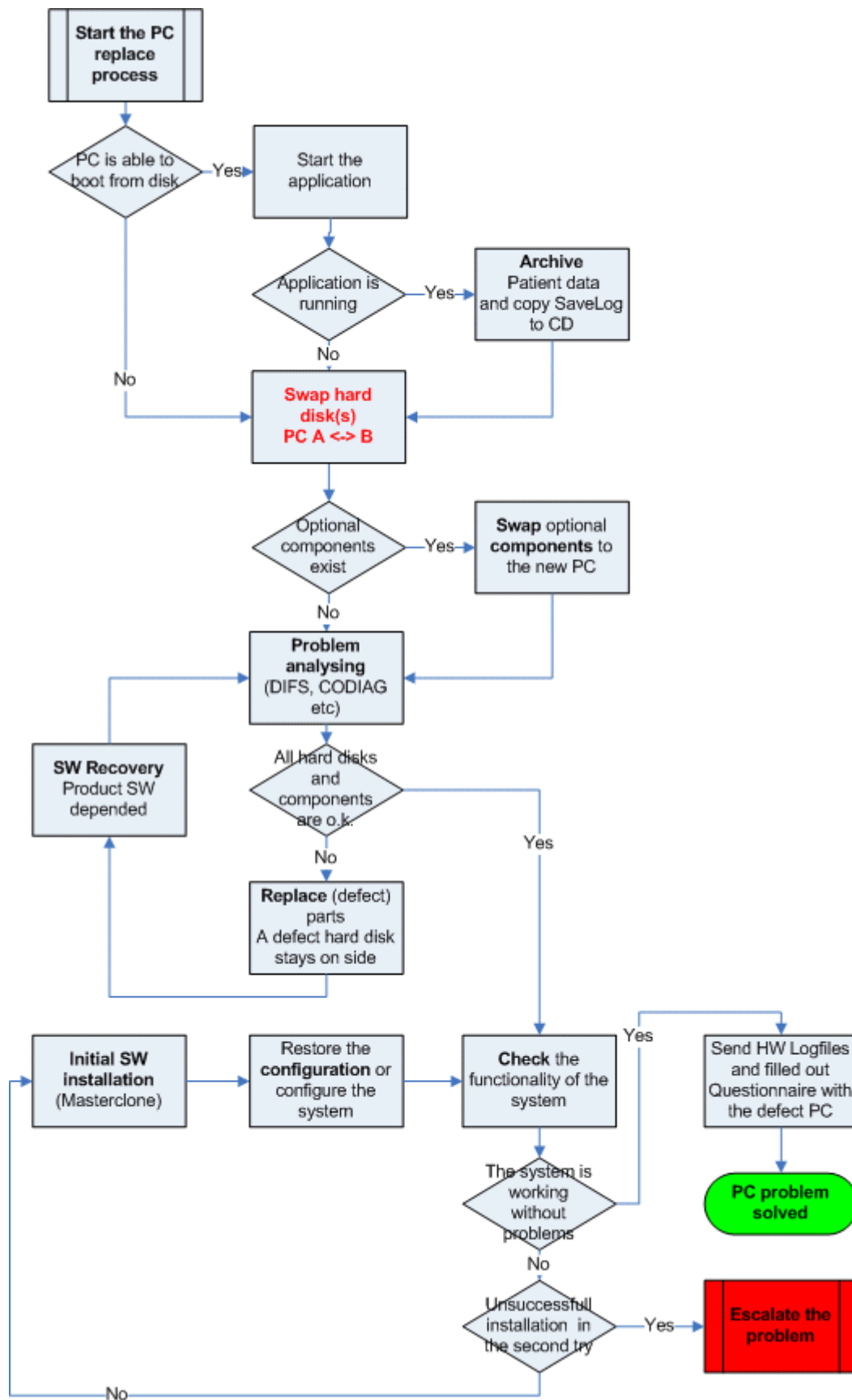
- The returned host computer will not contain any patient data.

- The disk swapped to the new PC will run without need of the rebuild/restore function.
- The customer may even have the benefit of reduced downtime.
- No reconstruction or loading of images from the archive will be necessary (all patient data will be available).

The decision in such cases will be made by the attending CSE or the responsible service manager.

The work steps are shown in the following flowcharts:

Fig. 74: Replacing a PC with patient data



Deleting Patient Data

Since the end of 2013, there is a tool to delete data from hard disk. The tool needs to be used in cases where replacement of the disk from the defective to the new PC is not possible.

The principles of "Return Goods Management" require that all storage devices containing personal patient data, which are to be removed from the site (i.e. spare parts returns), shall have all data wiped clean (securely deleted) or the storage device shall be handed over to the customer when deletion is not possible.

The tool is called "blancco". The tool can be downloaded from the Siemens Intranet.

The certified blancco tool permanently removes data from IT assets such as PCs, servers and data center equipment.

Data erased using blancco cannot be recovered with any existing technology. This helps to have all data wiped clean (securely deleted) and achieve compliance with existing "patient data handling" regulations.

SIEMENS Employees: The tool is available as an iso image from the CS Home page

<https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>

CS Home page -> Tools -> blancco - Erasure Software

For others see e.g.: <http://blancco.com/en/products/>

The time required by the procedure depends on the size of the hard disk.

Warning: before you start using the blancco software, make sure that all files, folders, software applications or any other information that you want to save for later use are backed up on an appropriate media device other than the original hard drive. Data that has been erased from the hard disk by this program cannot be recovered by any existing method.



Selecting the wrong hard disk can result in lose of data with the use of blancco. Be sure that the selected hard disk for deletion is the correct one.

Creating the blancco CD

An ISO image of the blancco CD is available for download from: CS Home page -> Tools -> blancco (CS Home page, <https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>)

1. Download the ISO image.
2. Label the blank CD as "blancco - Erasure Software"
3. Insert the CD into the drive of your service computer.
4. Use a standard burn program (such as Nero) to open the file and burn the CD image to the CD.
5. Remove the CD from the drive.

Using the blanco CD

1. Booting with blanco

1. Insert the "blanco - Erasure Software" CD into the PC with the drive(s) to be deleted

2. Switch off the PC.



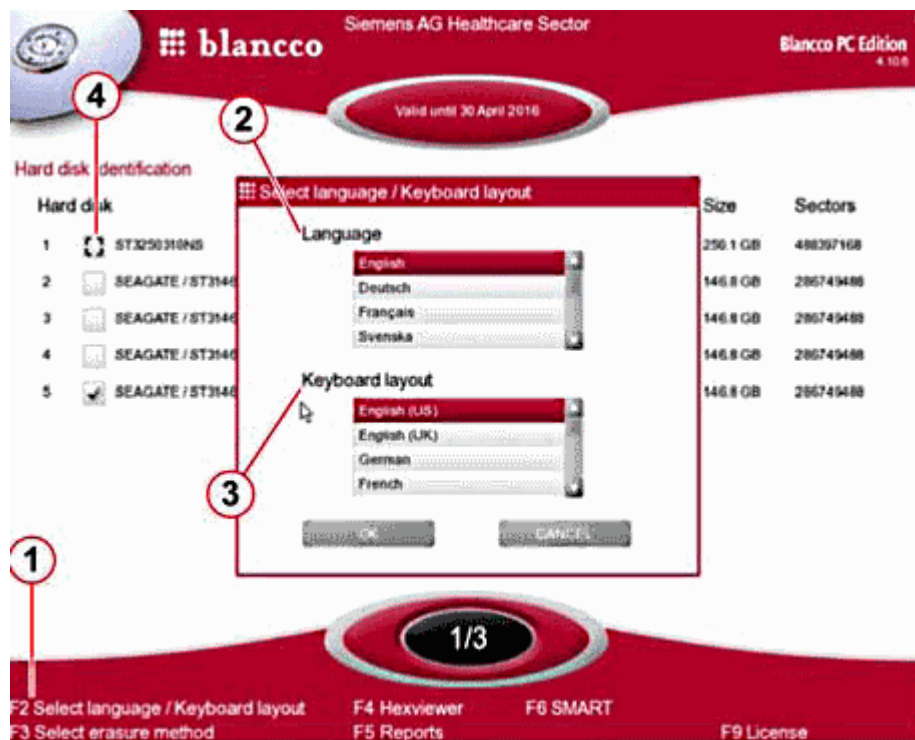
If this is an IAS-A PC, disconnect the service backup drive to prevent accidental loss of data.

3. Switch on the computer power to boot the system from the CD (if necessary, change the BIOS boot priority). The blanco Home screen is visible with all identified hard disks of the PC



If necessary, press F2 to change the language and the keyboard layout.

Fig. 75: Blanco language/keyboard layout



[1] F2 to open menu

[2] Language

[3] Keyboard layout

[4] Checkboxes to select hard disks

Deletion with blanco

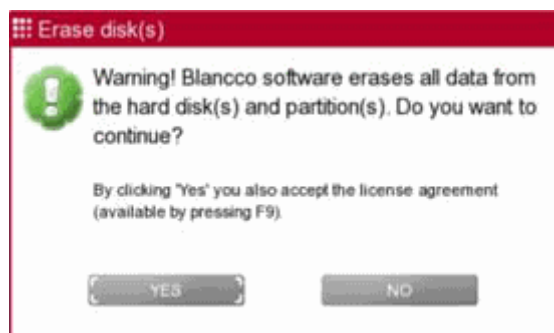
1. Select the hard disks to be deleted.



blanco can erase up to 16 hard disks simultaneously.

2. Press Erase, a warning message appears.

Fig. 76: Warning message



3. Press Yes to start the erasure process. The erasure process starts and the progress and duration is visible.
4. Click on OK when the process is finished and the Erase successful message appears.
5. Insert a memory stick to save the report.
6. 6. Press Save or F10. The report is saved to the memory stick.
7. Press Ok. The process is finished and the PC can be turned off.

1.12.1.2 FRU overview

FRU	Time	Instructions	Final checks	Tools
Host	200 min	Handling of patient data Host	Perform a measurement, view the images produced. Check with the customer that all options are available	Field service grounding kit (ESD), set of standard tools
DWD-ROM / DVD-RW	45 min	Drives	Burn some test images onto CD-RW or DVD media with the new DVD-RW drive or read Service instructions from a Service DVD from the Service GUI Help > Documentation	Field service grounding kit (ESD), set of standard tools
Power supply	45 min	Power supply	Perform a measurement and view the images produced.	Set of standard tools
Battery	60 min	Lithium battery	Perform BIOS setup according to TDOC Installation Software. Perform a measurement and view the images produced.	Field service grounding kit (ESD), set of standard tools
Dongle (USB)	60 min		Check with the customer that all options are available Perform a measurement and view the images produced.	Service license/Application licenses. The Service/Applications licenses generated in relation to the flexid of the new dongle

1.12.1.3 Host

Required time and tools

Time

- 180 min

Tools

- Set of standard tools

Preliminary

- If the condition of the "Host" allows access to the "Service platform" (service software, e.g., as user administrator), select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master backup**
- Shut down and switch off the system.

Disassembly



Remember to move the dongle from the old "host" to the new "host"

- Disconnect the main power plug, network drop cable(s), mouse, and keyboard cables.

Assembling

- Plug the HW dongle into the new host.
- Plug in the network drop cable(s), mouse, keyboard cables, monitor, and main power plug.

Final checks

- Check to see if the new "Host" has the correct software version or install software on the new "Host" according to the technical documentation (TDOC) "Install Software" valid for your system.
- Restore "site-specific data" according to TDOC "Install Software", valid for your system.
- Start the Local Service, select **Quality Assurance**, perform the Image Orientation "Quality Assurance" step



In case of connection problem with MARS after replacing the HOST; the firewall settings must be disabled for the internal measurement network.

How to disable the firewall for the internal network:

- Log in as user **administrator**

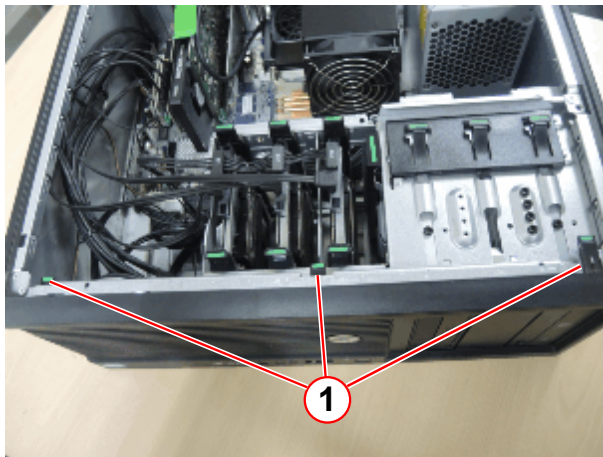
- Open a cmd tool and run **networkcardinstall /I1 /I2 ↵**
- Click **START -> Settings -> Control Panel**
- Change the appearance of the Control Panel to classic view: Click the marked link **Switch to classic view**
- Double click the **Windows firewall** icon
- In the Windows Firewall configuration window, select the **Advanced** tab card.
- If the measurement (internal) network checkbox is selected, remove this checkbox
- Close the Windows Firewall configuration window by pressing **OK**
- Reboot the system and login as **meduser**

1.12.1.4 Drives

Preparatory Work Steps

- Open the housing.
- Remove the front panel.

Fig. 77: HP Z420 - remove the front panel



(1) Front panel fastener

CD/DVD drive

Removing a CD/DVD drive

- Pull the data and power supply connectors from the desired drive.

- Release the clip and pull the drive out of the housing.

Fig. 78: HP Z420 - release the clip to unlock the DVD drive



- If necessary, make the required settings on the remaining hard disk drive.
- Slide the empty slide-in module into the housing.

Installing a CD/DVD drive

- Take the new drive out of its packaging.
- Slide the accessible drive into the housing.
- Plug the data and power supply connectors into the drive. Make sure the polarity is correct.

Final work steps

- Close the front panel.
- Close the housing.

Changing the drive letter

- Log in as administrator
- Use the Explorer to check that the CD-writer and DVD-drive still have the proper drive letters. These are:
 - DVD-drive -> R
 - DVD-writer -> S
- If the drive letters need to be changed ->

Start the disk manager by selecting START --> RUN and entering diskmgmt.msc.
- The disk manager pops up.
- Check that the drive letters for the CD writer and DVD-drive are correct using the "Windows file explorer" (DVD-burner drive letter "S" and DVD-ROM drive letter "R"). (If either of the drive letters is wrong, change the drive letter with the disk manager as follows: Click **Start; Run; diskmgmt.msc** ↵; **right click** the drive to be corrected; select **Change Driveletters and Path...**, select the proper drive letter in the drop-down

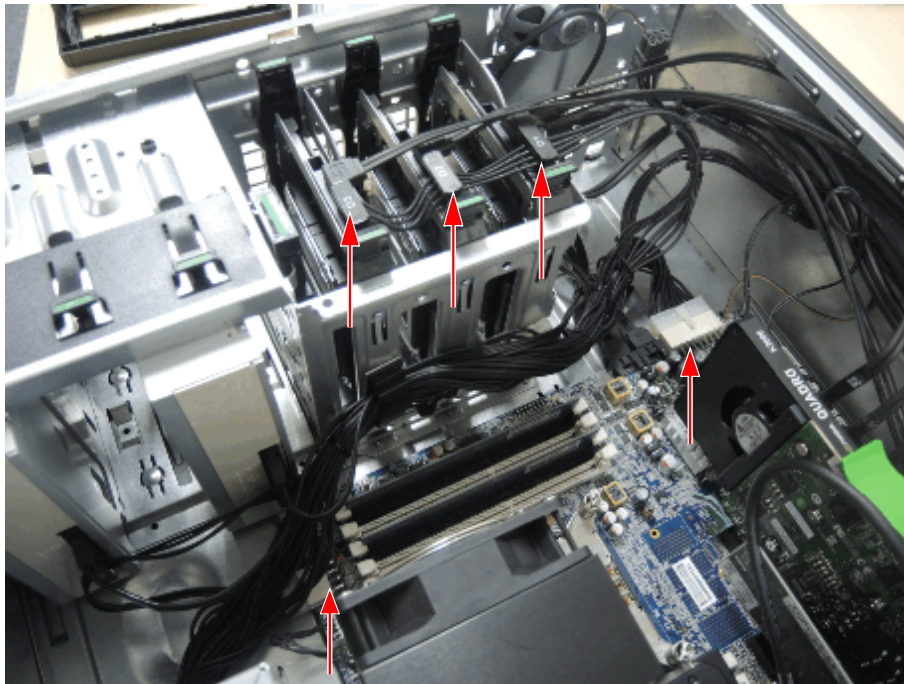
menu, confirm the popup with **Yes**, repeat this sequence until you achieve the correct drive letter assignment)

1.12.1.5 Power Supply

Preparatory Work Steps

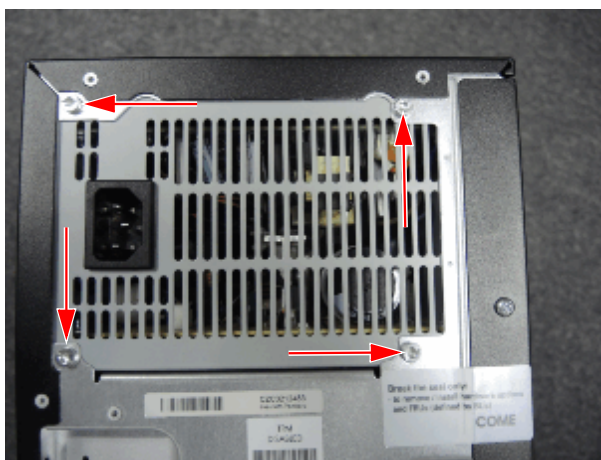
- Open the housing.
- Remove the power connectors.

Fig. 79: Power supply connectors



Changing the Power Supply

Fig. 80:



- Make notes about the power connectors (mainboard, drives and graphics board) and the ducting of the power supply, and disconnect all of them.

- Remove the 4 screws (arrows) on the back of the power supply.
- Laterally pull the power supply out from the case.
- Move the new power supply into the housing and attach it with the 4 screws on the back.
- Connect all power connectors and attach the cables as they were before.

Final Steps

- Install the crosspiece.
- Install the ventilation duct.
- Close the housing.

1.12.1.6 Lithium Battery



CAUTION

The BIOS update may result in a loss of patient data.

- » Check the specific system documentation for details concerning the BIOS installation.
- » If necessary, save patient data before starting the BIOS installation.
- » On some systems, a BIOS installation requires a complete SW installation

Changing the Lithium Battery

In order to permanently save the system information, a lithium battery is installed to provide the CMOS memory with a current. A corresponding error message notifies the user when the charge is too low or the battery is dead. The lithium battery must then be replaced.



CAUTION

Incorrect replacement of the lithium battery may lead to a risk of explosion.

- » The lithium battery may be replaced only with an identical battery or with a type recommended by the manufacturer, following the instructions below.



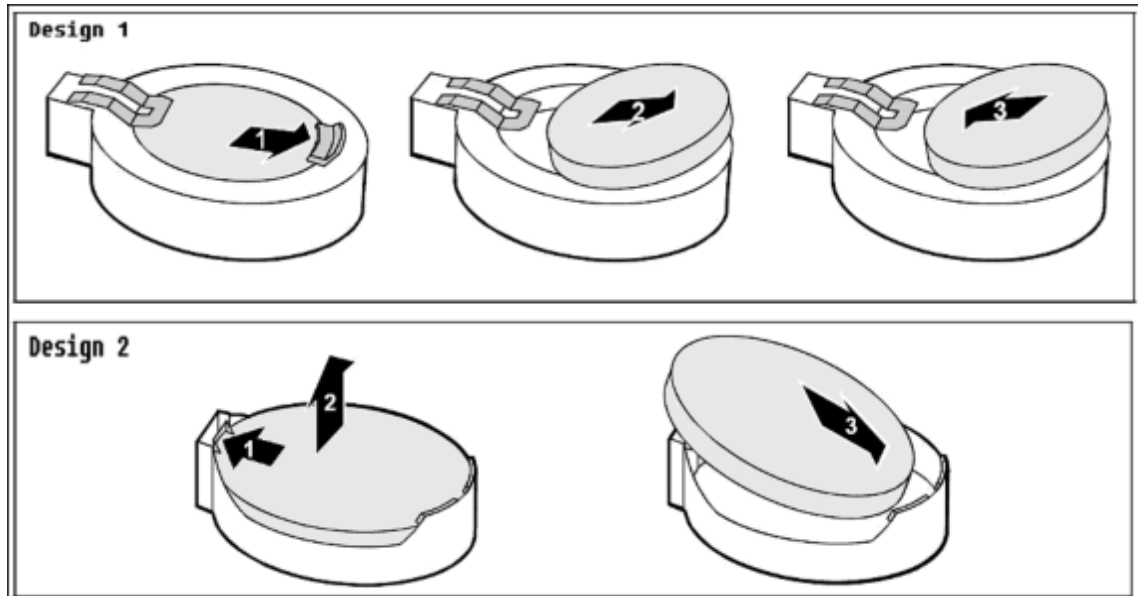
Do not throw lithium batteries into household waste. They must be disposed of in accordance with local regulations concerning hazardous waste.



Ensure that you observe the correct polarity when replacing the lithium battery. The plus pole must be on top.

- Open the housing.
- There are several different lithium battery holder designs; all of them function in a similar way.

Fig. 81: Changing the lithium battery



- Press the catch in the direction of the arrow (1).
 - » The battery slightly jumps out of the holder.
- Remove the battery (2).
- Push the new lithium battery of identical type into the holder and press it downward until it engages (3).
- **Restore BIOS settings** (necessary after battery replacement)



When the battery is dead, the BIOS is set to the factory default.

In addition to activating the system ON/OFF switch, the following may be necessary: - Turn OFF the PC via the power switch on the rear panel of the PC - Turn ON the PC via the rear power switch and press the push button on the front panel of the PC

Final Work Steps

- Close the housing.
- Perform BIOS setup according to the specific system documentation.
- Check to ensure that the system boots properly.

1.12.1.7 Dongle

Tools and Time required

Time

- 60 min

Tools

- Licence file with all customer licenses for the new DONGLE
- Valid service license for the new dongle.



Dongle could only be used on USB2.0 port. Please notice the difference on the USB port.

Disassembly

Ensure that the Host is switched off; replace the USB dongle in back of the Host

Assembling

- Switch on the system.

Final checks

- Log in as user **administrator**, start the Internet Explorer, start Local Service, select **Configuration**, Service, **Licensing**, Click radio button **Browse to select the license file**, navigate to the media with your new license file, click **Save**.
- Reboot the system, then start the MR application.
- Check with the customer to ensure that all options are available
- Start Local Service, select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master backup**

1.13 Host HP Z440

1.13.1 General

1.13.1.1 Safety Information and/or References to Safety Information

 Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad (TDK equipment type 8005 from 3M Deutschland GmbH).



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. ground pin of the mains plug) and the protective earth wire connected to the "main component" (e.g. screw of the housing).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.



WARNING

Please follow the general safety guidelines (→ General Safety Notes / TD00-000.860.01) as well as the MR-specific safety information (→ MR-specific safety information / MR-000.860.01)!

Failure to comply with the safety guidelines may result in death or serious injury.

- » Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Compliance with the information provided is mandatory at all times.



CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

- » Ramp down the magnet or maintain a safe distance from the magnetic field.

1.13.1.2 Handling patient data

The data protection guidelines specify the mandatory procedures for handling patient data.

Data carriers, storage devices, or other media (such as the hard disk inside the host PC) must not leave the customer site if there is a chance that they still contain patient data.



The service report needs to list the activities performed to ensure secure handling of patient data.

As a first step, the PC needs to be analyzed. Once a defect in any part of the PC hardware is identified as the cause of the problem, part replacement follows.

For PCs two cases apply:

1. The PC does not contain any patient data. The individual product documentation needs to list hard disks without patient data.
2. The PC contains patient data.

The disks need to be returned without patient data.

Please decide how to proceed regarding your individual case. Recommendations:

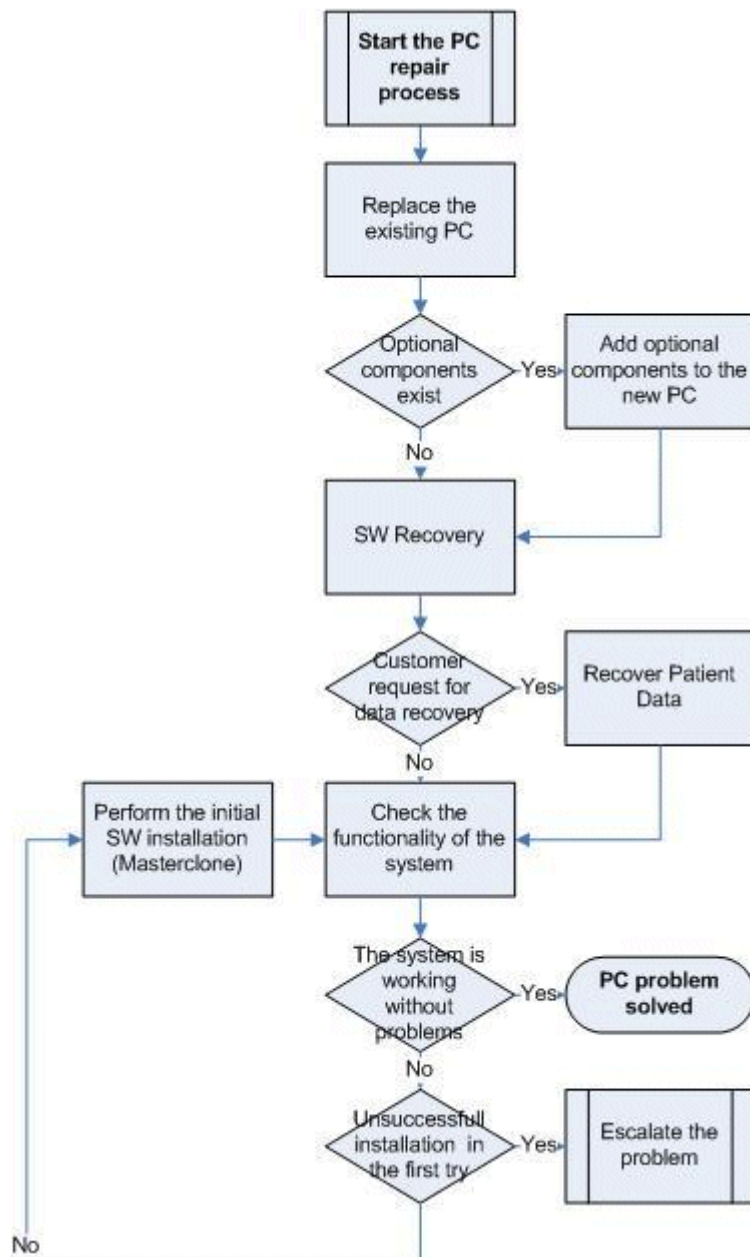
1. Delete the patient data with the "blancco" tool.
2. Case: All hard disks potentially containing patient data are removed from the defective PC and remain at the customer site.

No Patient Data on Hard Disks



Only continue with the parts replacement procedure outlined in the flow chart below if you have confirmation from the product-specific documentation that the PC contains no patient data and no patient information is contained in any log files on the PC.

Fig. 83: Replacing a PC without patient data



Deleting Patient Data

Since the end of 2013 there has been a tool to delete data from the hard disk. The tool needs to be used in cases where replacement of the disk from the defective to the new PC is not possible.

The principles of "Return Goods Management" require that all storage devices containing personal patient data, which are to be removed from the site (i.e. spare parts returns), will have all data wiped clean (securely deleted) or the storage device will be handed over to the customer when deletion is not possible.

The tool is called "blancco". The tool can be downloaded from the Siemens Intranet.

The certified blancco tool permanently removes data from IT assets such as PCs, servers, and data center equipment.

Data erased using blancco cannot be recovered with any existing technology. This helps to have all data wiped clean (securely deleted) and achieve compliance with existing "patient data handling" regulations.

SIEMENS Employees: The tool is available as an iso image from the CS homepage

<https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>

CS Home page -> Tools -> blancco - Erasure Software

For other, see e.g.: <http://blancco.com/en/products/>

The time required by the procedure depends on the size of the hard disk.

Warning: before you start using the blancco software, make sure that all files, folders, software applications or any other information that you want to save for later use are backed up on an appropriate media device other than the original hard drive. Data that has been erased from the hard disk by this program cannot be recovered by any existing method.



Selecting the wrong hard disk can result in the loss of data when using blancco. Be sure that the hard disk selected for deletion is the correct one.

Creating the blancco CD

An ISO image of the blancco CD is available for download from: CS Home page -> Tools -> blancco (CS Home page, <https://intranet.healthcare.siemens.com/cms/CS/en/Tools/Pages/Default.aspx>)

1. Download the ISO image.
2. Label the blank CD as "blancco - Erasure Software"
3. Insert the CD into the drive of your service computer.
4. Use a standard burn program (such as Nero) to open the file and burn the CD image to the CD.
5. Remove the CD from the drive.

Using the blancco CD

1. Booting with blancco
 1. Insert the "blancco - Erasure Software" CD into the PC with the drive(s) to be deleted
2. Switch off the PC.



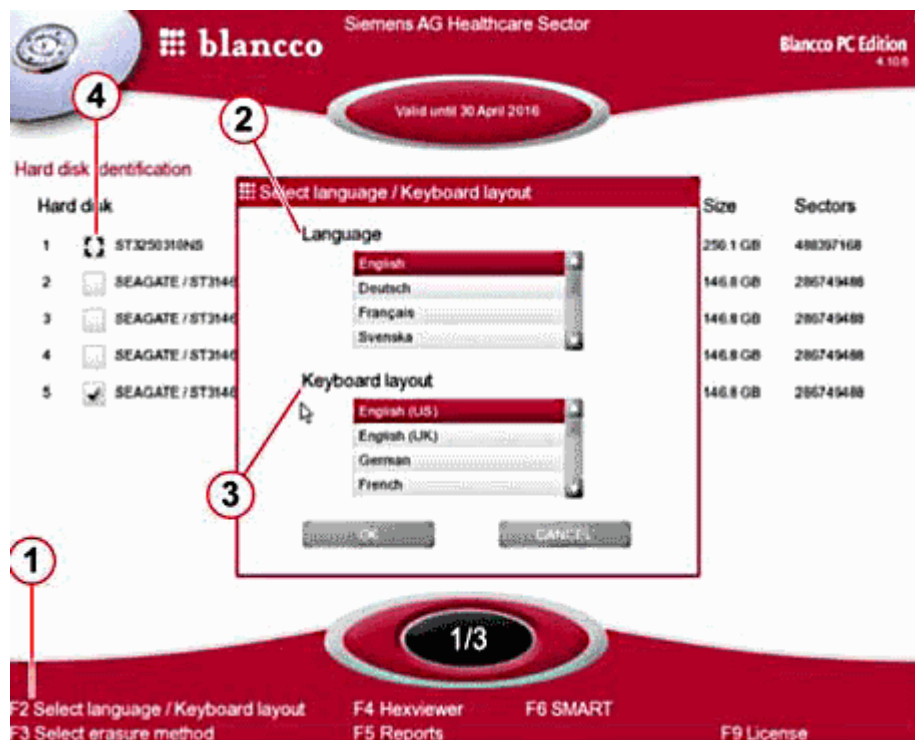
If this is an IAS-A PC, disconnect the service backup drive to prevent accidental loss of data.

- Switch on the computer power to boot the system from the CD (if necessary, change the BIOS boot priority). The blanco Home screen is visible with all identified hard disks of the PC



If necessary, press F2 to change the language and the keyboard layout.

Fig. 84: Blanco language/keyboard layout



- [1] F2 to open menu
- [2] Language
- [3] Keyboard layout
- [4] Checkboxes to select hard disks

Deletion with blanco

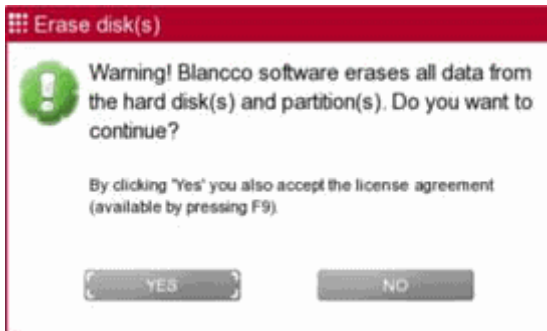
- Select the hard disks to be deleted.



blanco can erase up to 16 hard disks simultaneously.

2. Press Erase, a warning message appears.

Fig. 85: Warning message



3. Press Yes to start the erasure process. The erasure process starts and the progress and duration is visible.
4. Click OK when the process is finished and the Erase successful message appears.
5. Insert a memory stick to save the report.
6. Press Save or F10. The report is saved to the memory stick.
7. Press Ok. The process is finished and the PC can be switched off.

1.13.2 FRU overview

FRU	Time	Instructions	Final checks	Tools
Host	200 min	Handling of patient data Host	Perform a measurement, view the images produced. Check with the customer that all options are available	Field service grounding kit (ESD), set of standard tools
DWD-ROM / DVD-RW	45 min	Drives	Burn some test images onto CD-RW or DVD media with the new DVD-RW drive or read Service instructions from a Service DVD from the Service GUI Help > Documentation	Field service grounding kit (ESD), set of standard tools
Power supply	45 min	Power supply	Perform a measurement and view the images produced.	Set of standard tools
Battery	60 min	Lithium battery	Perform BIOS setup according to TDOC Installation Software. Perform a measurement and view the images produced.	Field service grounding kit (ESD), set of standard tools
Dongle (USB)	60 min		Check with the customer that all options are available Perform a measurement and view the images produced.	Service license/Application licenses. The Service/Applications licenses generated in relation to the flexid of the new dongle

1.13.3 Host

1.13.3.1 Required time and tools

Time

- 180 min

Tools

- Set of standard tools

1.13.3.2 Preliminary

- If the condition of the "Host" allows access to the "Service platform" (service software, e.g., as user administrator), select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master backup**
- Shut down and switch off the system.

1.13.3.3 Disassembly



Remember to move the dongle from the old "host" to the new "host"

- Disconnect the main power plug, network drop cable(s), mouse, and keyboard cables.

1.13.3.4 Assembling

- Plug the HW dongle into the new host.
- Plug in the network drop cable(s), mouse, keyboard cables, monitor, and main power plug.

1.13.3.5 Final checks

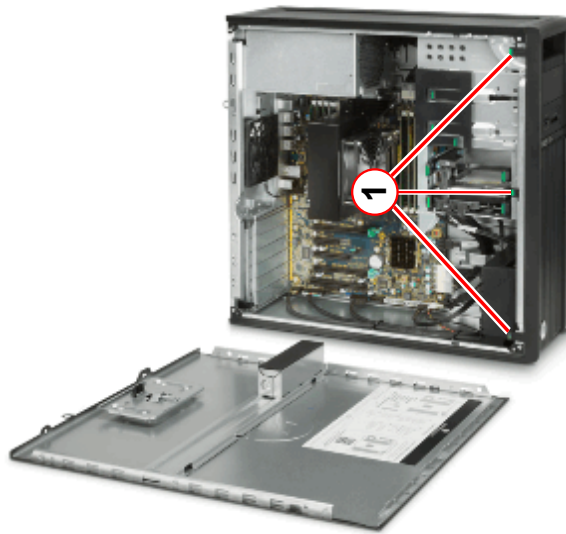
- Check to see if the new "Host" has the correct software version or install software on the new "Host" according to the technical documentation (TDOC) "Install Software" valid for your system.
- Restore "site-specific data" according to TDOC "Install Software", valid for your system.
- Start the Local Service, select **Quality Assurance**, perform the Image Orientation "Quality Assurance" step

1.13.4 Drives

1.13.4.1 Preparatory Work Steps

- Open the housing.
- Remove the front panel.

Fig. 86: HP Z440 - remove the front panel



(1) Front panel fastener

1.13.4.2 CD/DVD drive

Removing a CD/DVD drive

- Pull the data and power supply connectors from the desired drive.
- Release the clip and pull the drive out of the housing.

Fig. 87: HP Z440 - release the clip to unlock the DVD drive



- If necessary, make the required settings on the remaining hard disk drive.
- Slide the empty slide-in module into the housing.

Installing a CD/DVD drive

- Take the new drive out of its packaging.

- Slide the accessible drive into the housing.
- Plug the data and power supply connectors into the drive. Make sure the polarity is correct.

1.13.4.3 Final work steps

- Close the front panel.
- Close the housing.

Changing the drive letter

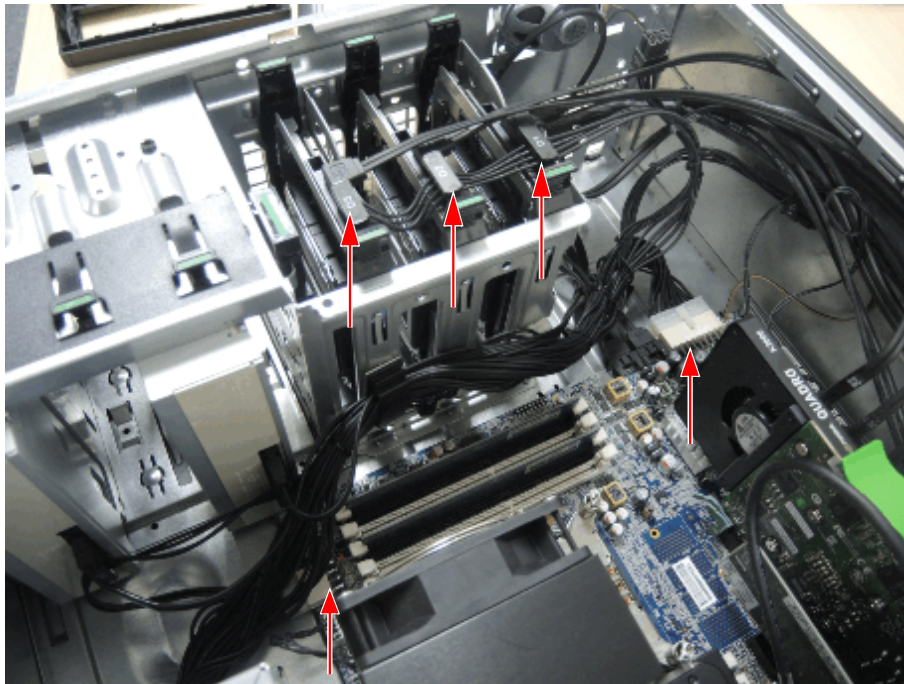
- Log in as administrator
- Use the Explorer to check that the CD-writer and DVD-drive still have the proper drive letters. These are:
 - DVD-drive -> R
 - DVD-writer -> S
- If the drive letters need to be changed ->
Start the disk manager by selecting START --> RUN and entering diskmgmt.msc.
- The disk manager pops up.
- Check that the drive letters for the CD writer and DVD-drive are correct using the "Windows file explorer" (DVD-burner drive letter "S" and DVD-ROM drive letter "R"). (If either of the drive letters is wrong, change the drive letter with the disk manager as follows: Click **Start; Run; diskmgmt.msc** ↵ ; **right click** the drive to be corrected; select **Change Driveletters and Path...**, select the proper drive letter in the drop-down menu, confirm the popup with **Yes**, repeat this sequence until you achieve the correct drive letter assignment)

1.13.5 Power Supply

1.13.5.1 Preparatory Work Steps

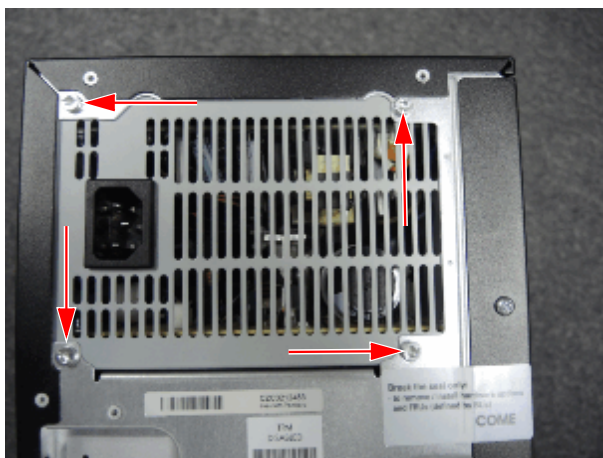
- Open the housing.
- Remove the power connectors.

Fig. 88: Power supply connectors



1.13.5.2 Changing the Power Supply

Fig. 89: Power supply



- Make notes about the power connectors (mainboard, drives and graphics board) and the ducting of the power supply, and disconnect all of them.
- Remove the 4 screws (arrows) on the back of the power supply.
- Laterally pull the power supply out from the case.
- Move the new power supply into the housing and attach it with the 4 screws on the back.
- Connect all power connectors and attach the cables as they were before.

1.13.5.3 Final Steps

- Install the crosspiece.

- Install the ventilation duct.
- Close the housing.

1.13.6 Lithium Battery



CAUTION

The BIOS update may result in a loss of patient data.

- » Check the specific system documentation for details concerning the BIOS installation.
- » If necessary, save patient data before starting the BIOS installation.
- » On some systems, a BIOS installation requires a complete SW installation

1.13.6.1 Changing the Lithium Battery

In order to permanently save the system information, a lithium battery is installed to provide the CMOS memory with a current. A corresponding error message notifies the user when the charge is too low or the battery is dead. The lithium battery must then be replaced.



CAUTION

Incorrect replacement of the lithium battery may lead to a risk of explosion.

- » The lithium battery may be replaced only with an identical battery or with a type recommended by the manufacturer, following the instructions below.



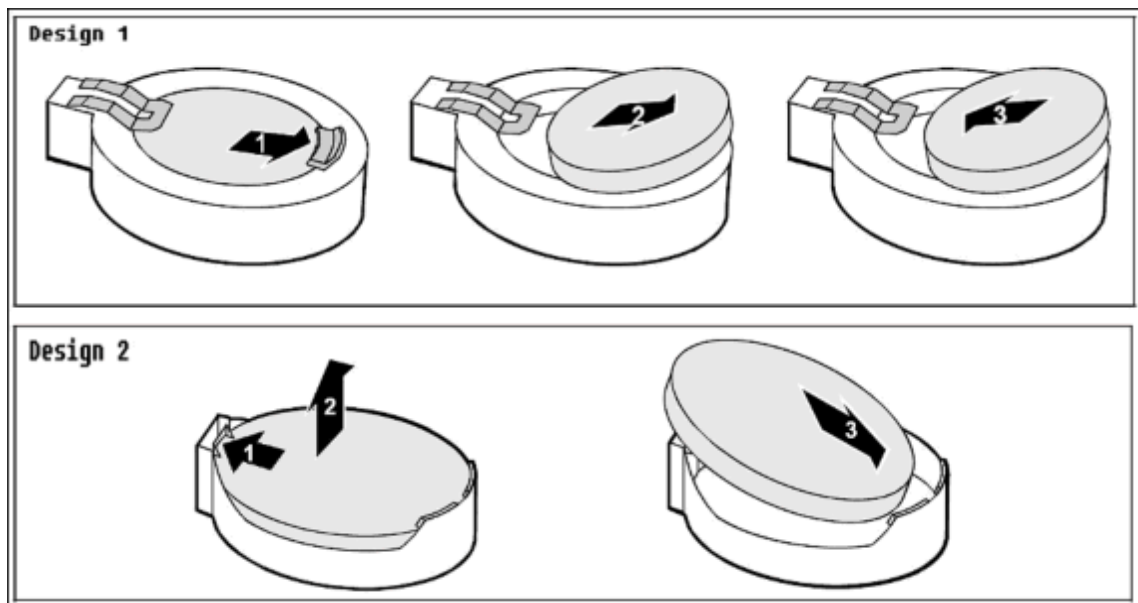
Do not throw lithium batteries into household waste. They must be disposed of in accordance with local regulations concerning hazardous waste.



Ensure that you observe the correct polarity when replacing the lithium battery. The plus pole must be on top.

- Open the housing.
- There are several different lithium battery holder designs; all of them function in a similar way.

Fig. 90: Changing the lithium battery



- Press the catch in the direction of the arrow (1).
 - » The battery slightly jumps out of the holder.
- Remove the battery (2).
- Push the new lithium battery of identical type into the holder and press it downward until it engages (3).
- **Restore BIOS settings** (necessary after battery replacement)



When the battery is dead, the BIOS is set to the factory default.

In addition to activating the system ON/OFF switch, the following may be necessary: - Turn OFF the PC via the power switch on the rear panel of the PC - Turn ON the PC via the rear power switch and press the push button on the front panel of the PC

1.13.6.2 Final Work Steps

- Close the housing.
- Perform BIOS setup according to the specific system documentation.
- Check to ensure that the system boots properly.

1.13.7 Dongle

1.13.7.1 Tools and Time required

Time

- 60 min

Tools

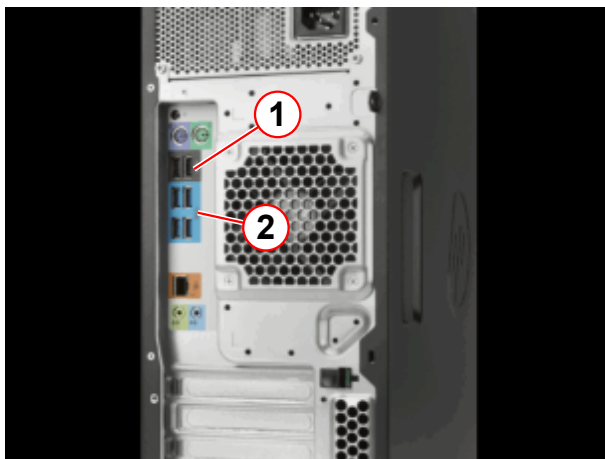
- Licence file with all customer licenses for the new DONGLE

- Valid service license for the new dongle.



Dongle could only be used on USB2.0 port. Please notice the difference on the USB port.

Fig. 91: Back of the Z440 host



- (1) 2.0 USB ports
- (2) 3.0 USB ports

1.13.7.2 Preliminary

If the service license and all customer options are invalid, the dongle is most likely defective.



Place dongle orders only through the Licensing Center in Erlangen. A dongle defect must be directly escalated to Licensing Center in Erlangen, phone number +49 9191-188080

1.13.7.3 Disassembly

Ensure that the Host is switched off; replace the USB dongle in back of the Host

1.13.7.4 Assembling

- Switch on the system.

1.13.7.5 Final checks

- Log in as user **administrator**, start the Internet Explorer, start Local Service, select **Configuration**, Service, **Licensing**, Click radio button **Browse to select the license file**, navigate to the media with your new license file, click **Save**.
- Reboot the system, then start the MR application.
- Check with the customer to ensure that all options are available

- Start Local Service, select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master backup**

2.1 Imager FRU Overview

FRU	Time	Instructions	Final checks	Tools
Imager	70 min.	(→ Imager / Page 123)	Perform the “Image Orientation” measurement. Service Software, Quality Assurance, General Quality Assurance, Image Orientation Perform a measurement, view the images produced.	Set of standard tools
PCI RX boards	70 min.	(→ PCI RX Boards / Page 125)	Perform the “Image Orientation” measurement. Service Software, Quality Assurance, General Quality Assurance, Image Orientation Perform a measurement and view the images produced.	Field service grounding kit (ESD), set of standard tools (The imager is heavy, approx. 20 kg)
Lithium Battery	70 min.	(→ Lithium Battery / Page 128)	Perform “Image Orientation” measurement. Service Software, Quality Assurance, General Quality Assurance, Image Orientation Perform a measurement and view the images produced.	Field service grounding kit (ESD), set of standard tools

2.2 Imager

The imager is heavy (approx. 20 kg).

2.2.1 Required Time and Tools

Time

- 70 min.

Tools

- Set of standard tools

2.2.2 Preliminary Steps

- Shutdown and switch off the system.

2.2.3 Disassembly



Ensure that the cables remains intact (at the back of the ACC).

Fig. 92: Imager step 7



- Disconnect the main power plug, network drop cable, fiber optic links, data receiver cable, and power remote switch cable.
- Remove the 4x Allen screws (within/below the side handles), pull/slide the "Imager" forward with care.
- Check that the number of PCI-RX boards is the same in the new "Imager" as the original "Imager".

⚠ Adhere to ESD regulations and use the appropriate tool set: anti-static wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

2.2.4 Assembling

- Insert the new "Imager", mount the 4x Allen screws (within/below the side handles).
- Plug in all connectors/fiber fiber optic links at the front of the Imager

- Re-assemble the screws for the Imager container to the cabinet



Ensure that the cables remains intact (at the back of the ACC).

2.2.5 Final Steps

- Switch on the system
- Install BIOS for systems with VB15, VB17, and VB19A. BIOS update to Step 7 Imager is not required.
- Start Local Service, select **Quality Assurance**, Perform the "Quality Assurance" step Image Orientation
- Perform a measurement, view the images produced

2.3 PCI RX Boards

⚠ Adhere to all ESD regulations and use the appropriate tool kit: Antistatic wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

2.3.1 Tools and Time required

Time

- 70 min

Tools

- Set of standard tools

2.3.2 Preliminary

- Shut down and switch off the system.

2.3.3 Disassembly

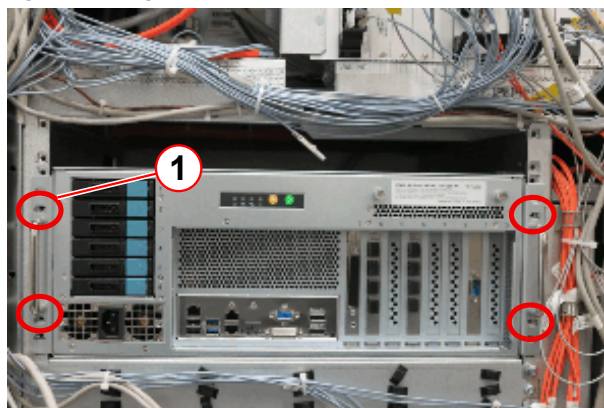
Fig. 93: Imager step 7



The imager is heavy (approx. 20 kg).

- Unplug the main power connector, network drop cable, fiber optic links, data receiver cable, and power remote switch cable.

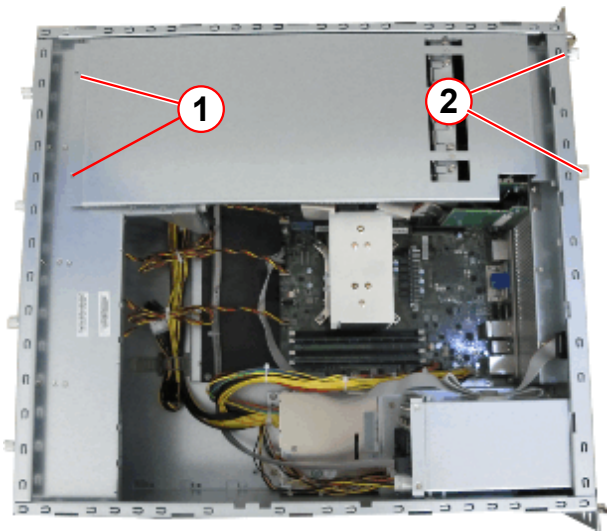
Fig. 94: Imager



- Remove the 4 Allen screws (1), pull/slide the "Imager" forward with care - **remember the "Imager" is heavy!**

(1) Screws

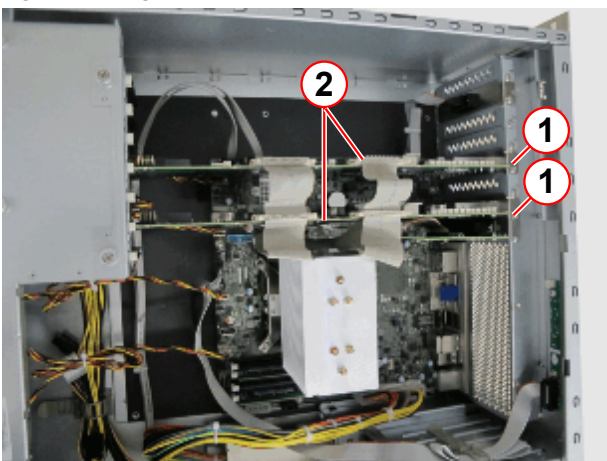
Fig. 95: Imager open



(1) Phillips screws
(2) Screws

- Remove the top cover from the Imager, loosen and remove the 2 Phillips screws (1) and loosen the two screws (2)
Pull and lift the top cover.

Fig. 96: Imager inside



(1) Screws
(2) PCI boards

- Unplug the flat cables.
- Unlock and lift up the PCI-RX board to be replaced, and insert the new PCI-RX board.
- Reassemble the “Imager”.
- Perform the final checks.

Tab. 1 Imager (PCI slots)

Label	Use
Slot 2	PCI_RX16 board
Slot 3	PCI_RX16_8 board or PCI_RX16 board

2.3.4 Assembly

- Reconnect the cables for the imager.

2.3.5 Final Steps

- Re-assemble the screws for the imager

2.3.6 Final Checks

- Switch on the system
- Start the Local Service, select **Quality Assurance**, Perform the “Quality Assurance” step Image Orientation
- Perform a measurement and view the images produced.

2.4 Lithium Battery



CAUTION

The BIOS update may result in a loss of patient data.

- » Check the specific system documentation for details concerning the BIOS installation.
- » If necessary, save patient data before starting the BIOS installation.
- » On some systems, a BIOS installation requires a complete SW installation



The battery in Imager Step 5 will not be changed with software version VB19B because BIOS switches to a setting that is not compatible with VB19B and BIOS reinstallation is not possible with VB19B. Therefore a replacement in Imager Step 7 is necessary.

2.4.1 Changing the Lithium Battery

In order to permanently save system information, a lithium battery is installed to provide the CMOS memory with a current. A corresponding error message notifies the user when the charge is too low or the battery is dead. The lithium battery must then be replaced.



CAUTION

Incorrect replacement of the lithium battery may lead to a risk of explosion.

- » The lithium battery may be replaced only with an identical battery or with a type recommended by the manufacturer, and by following the instructions below.



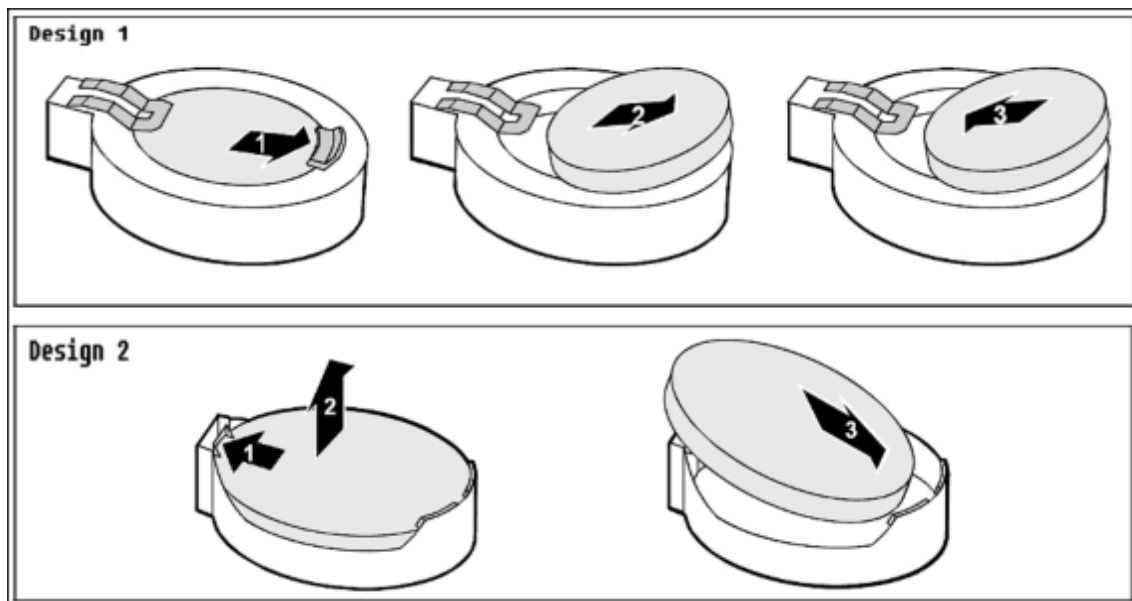
Do not throw lithium batteries into household waste. They must be disposed of in accordance with local regulations concerning hazardous waste.



Ensure that you observe the correct polarity when replacing the lithium battery. The plus pole must be on top.

- Open the housing.
- There are several different lithium battery holder designs; all of them function in a similar way.

Fig. 97: Changing the lithium battery



- Press the catch in the direction of the arrow (1).
 - » The battery jumps out of the holder slightly.
- Remove the battery (2).
- Push the new lithium battery of the identical type into the holder and press it downward until it engages (3).



Restoring BIOS settings is only necessary after battery replacement for older imager HW and SW. All new imager default BIOS settings are customized for use without any manual configuration.



When the battery is dead, the BIOS is set to the factory default.

In addition to activating the system ON/OFF switch, the following may be necessary: - Turn OFF the PC via the power switch on the rear panel of the PC - Turn ON the PC via the rear power switch and press the push button on the front panel of the PC

2.4.2 Final Work Steps

- Close the housing.
- Perform the BIOS setup according to the specific system documentation (only valid for imager working with VB15, VB17, and VB19A).
- Check to ensure that the system boots properly.
- Perform the quality assurance procedure: **Image Orientation** in the service platform.
- Make sure that the system is generating images.

3.1 BIOS CD update and installation

3.1.1 General

This chapter describes how to install the current BIOS CD.

3.1.2 Prerequisites

3.1.2.1 For downloaded files

Material no.	Name
03 07 86 31	Blank "write once read many" (WORM) CD media TDK CDR 74 or equivalent.
n/a	SPC equipped with CD-R drive (CD burner)
n/a	Up-to-date virus scanner on SPC
n/a	Burning software such as Nero or equivalent on SPC

3.1.3 BIOS Update on Hosts



CAUTION

The BIOS update may result in a loss of patient data.

- » Check the specific system documentation for details concerning the BIOS installation. If necessary, save patient data before starting the BIOS installation. On some systems a BIOS installation requires a complete SW installation



WARNING

Interrupting the BIOS update or not adhering to the following steps may lead to irreparable damage to the PC.

This might require replacement of the complete PC.

- » Do not turn off or disconnect power from the PC during the BIOS update!



The following message will be displayed during the BIOS update (→ Fig. 98 Page 131)). Please do **not** confuse this with the end of the BIOS update, do **not** remove any media or reboot the computer!

Remove flash diskette from drive. Cold reboot in 5 seconds ...

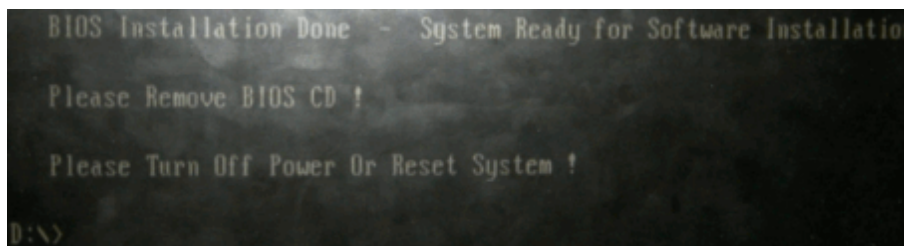
Please wait until the following message is displayed (→ Fig. 99 Page 131)). Now, the update and all necessary settings are made.

BIOS Installation Done - System Ready for Software Installation

Fig. 98: Message during BIOS update (to be ignored)



Fig. 99: Message after complete BIOS update



The BIOS update optimizes the settings of NUMARIS/4 for your host; it sets the NUMARIS/4 BIOS password and boot device order

ACTION	Host
During the boot-up process, press F2 or F10 or F12 (depending on host type) repeatedly until the BIOS set-up appears.	☐
Check the date/time and correct if necessary.	☐
Only for R630: Go to Security and check if the entry "Start from IDE CD-ROM" is set to Enabled.	☐
Go to Boot -> Boot Device Priority and move the CD-ROM Drive/Optical Drive to the top by selecting the "CD-ROM Drives" entry and pressing the + key. (This may be done slightly differently on your host.)	☐

ACTION	Host
■ Insert the BIOS-CD into the CD/DVD-ROM drive (R: !).	☐
■ Save and exit BIOS.	☐
■ The BIOS CD is booted and the BIOS updates run automatically; no user action is necessary. Ignore any message to reboot the PC, also see (→ Fig. 98 Page 131)!	☐
■ The BIOS update ends when the following information is displayed BIOS Installation Done - System Ready for Software Installation Please Remove the BIOS CD ! Please Turn Off Power or the Reset System ! C:\>	☐
■ Remove the BIOS-CD.	☐
■ Press the key combination <CTRL> + <ALT> + (on a German keyboard use <STRG> + <ALT> + <ENTF>) to reboot.	☐

3.1.4 Final checks



Check if hotfixes and service packs are available for this software version according to the various update instructions.

- Perform the “Quality Assurance” procedure: “Image Orientation” in the service platform.
- Make sure that the system is generating images.

- » NEW: Replacement of fans, power supply, and battery.

Version 03

- Chapter 03:
 - Editorial changes

Version 04

- Chapter 03:
 - Chapter restructured.
 - Added information for Remote Monitor support for z400/M470.

Version 05

- Hint: "Protective earth connection test" was added.
- Chapter Host:
 - Editorial changes

Version 06

- Chapter Host:
 - Updated section "Lithium Battery": Added warning and harmonized replacement description.
- Chapter Imager:
 - Added section "Lithium Battery".
- Chapter TFT monitor:
 - Added section "Digital TFT monitor DSC 1910-D".
- Chapter BIOS CD:
 - Added warnings about consequences in case of failure during BIOS update. Harmonized BIOS update procedures.
 - Editorial changes.

Version 07

- Chapter Host
 - Added M470 (→ Host M470 / Page 36)
 - Added Z400 (→ Host HP Z400 / Page 73)
 - Added Z420 (→ Host HP Z420 / Page 89)
 - Added, handling of patient data (→ Host / Page 5)

Version 08

- Minor changes

Version 09

- Chapter Monitor deleted.

Version 10

- Chapter "Host"
 - Z440 added (→ Host HP Z440 / Page 107)
- Chapter "Imager"
 - (→ Imager / Page 122) reworked.

There are no Hazard IDs in this document.

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